The Formula-Footprint Barrier
for Label-Induced Hardness Predicates

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Abstract

Let a batched generator map each source input $x \in \{0,1\}^n$ to M_n short strings, and let an external Boolean labeling of those strings induce a source Boolean function through a fixed row aggregator. The formula-footprint barrier says that any completion-sound certificate that this induced function has circuit complexity greater than s_n must expose many generated labels of both polarities. We study this barrier in the standard nondeterministic decision-tree sense of inputwise 1-certificates. The input variables may be labels of distinct generated strings, independent labels of generated occurrences, or quotient labels after sound equality identifications.

The main result is a formula-label footprint lower bound. A sorting-network dictionary evaluates a batch of M variable keys against K hardwired exceptional keys in size

$$\widetilde{O}\big((M+K)(\ell+\log(M+K))\big).$$

Consequently, every nondeterministic decision-tree 1-certificate for hardness above s_n must contain, separately, a large set of generated labels fixed to 0 and a large set fixed to 1. In the regime $s_n = n^{1+\varepsilon-o(1)}$, $M_n = n^{q+o(1)}$, and generator size $n^{\rho+o(1)}$ with $\max\{q,\rho\} < 1+\varepsilon$, each polarity has size $n^{1+\varepsilon-o(1)}$.

This lower bound is close to best possible in the standard identity truth-table carrier. A circuit-counting test-set argument shows that a generic function of circuit complexity above s has a completion-sound certificate of size $O(s\log(s+n))$, with both polarities represented. Thus the formula-footprint theorem is not merely a necessary condition; in the baseline model it captures the right certificate scale up to polylogarithmic factors.

Two supporting structural lemmas record what changes under different label granularities. In the private occurrence model, sparse-row patching shows that a sound certificate must force $\Omega(s_n/n)$ rows to each output value. If $\mathrm{force}_b(A_M)$ is the minimum number of coordinates needed by any partial assignment forcing an aggregator A_M to b, then every certificate queries

$$\Omega\Big((\mathrm{force}_0(A_{M_n}) + \mathrm{force}_1(A_{M_n}))\frac{s_n}{n}\Big)$$

occurrences. For parity, $\mathrm{force}_0 = \mathrm{force}_1 = M_n$, so every forced row is complete and the query lower bound is $\Omega(M_n s_n/n)$.

For parity after sound equality identifications, the residual completion space is an affine image, and soundness requires every completion to stay at Hamming distance $\Omega(s_n/n)$ from both constant functions. For nonlinear semantic relations, support locality forces an ordinary relation-membership certificate on each constrained tuple; in particular, an unsatisfiable SAT self-reduction split $y = y_0 \vee y_1$ must expose parent and both children. For an acyclic OR self-reduction tree after internal nodes are eliminated, the corresponding frontier relation has an exact global certificate: root 0 forces the root and all frontier leaves, while root 1 needs the root and one true leaf. The same exact certificate holds for finite acyclic OR DAG frontiers after repeated reachable frontier formulas are soundly quotiented: sharing changes raw leaf occurrences into distinct reachable frontier classes, but not the residual OR certificate. More generally, any exposed finite acyclic Boolean frontier circuit has residual support cost equal to the root coordinate plus ordinary pointwise certificate complexity for the induced frontier

function. Finally, bounded-error randomized verifiers inherit the same lower bounds for the union of labels queried on accepting executions: that union is itself a deterministic certificate whenever the acceptance probability on the hard labeling exceeds the soundness bound for easy completions. A displayed-atom extraction theorem applies the formula-footprint bound to any proof interface whose accepted proof explicitly displays a faithful, completion-sound set of formula labels. Restricted resolution theorems apply this to explicit label assumptions, to the used unit core of full-assignment refutations, to the support of explicit local label clauses, and to auxiliary encodings that are support-local under all completions. For extension proofs, the paper gives both a declared dependency-cone normalizer and a canonical syntactic-cone normalizer extracting the same support directly from the extension-definition graph; the final backward-slice corollary records the corresponding proof-object budget, a slice-explicit extension-resolution corollary turns that budget into a restricted ER-style size lower bound, a displayed acyclic ER corollary shows that ordinary displayed extension clauses already charge the same boundary, an expansion-bounded compressed ER transfer states the exact decompression budget needed for compressed representations, a macro-compressed ER corollary turns ordinary macro reuse into an explicit reuse-product lower bound, a support-declared macro ER corollary charges any sound declared support budget, and a support-audited LRAT corollary packages the same obligation for checker-visible clausal traces. A provenance-typed LRAT corollary shows that a trusted variable-role map plus block-local clause provenance supplies such a support audit, and a namespace-typed LRAT corollary shows that, with reserved formula-label variables, the LRAT antecedent closure of the empty clause already identifies the used label support when the used auxiliary clauses have a hard witness. A strict-prefix Qrep replay proposition adds the quotient-compression version of the same rule: representative reuse is transparent only when an already charged same-theory certificate forest can replay it. A trace-complete DPLL decision-log corollary gives the first DPLL-style proof-object instantiation of the hidden-adaptive audit, and an antecedent-complete CDCL corollary adds learned clauses when their acyclic antecedent DAGs expose all local support used by the replay. A CDCL antecedent-erasure trichotomy records the matching boundary: if learned-clause antecedents and role provenance are erased, a support extractor must add equivalent provenance, pay conservative support, or be unsound on a same-skeleton lift. A RUP-complete CDCL corollary gives the positive non-plain bridge: checker-visible RUP propagation certificates over a known live clause database, together with trusted label namespace or role parsing and retained deleted-clause support, recover the support audit without requiring an explicit learned-clause antecedent DAG. A canonical-namespace RUP-CDCL corollary then makes the namespace route checkable from a deterministic encoding manifest with a reserved label-variable interval and live-database reconstruction. A support-audit necessity separator then shows why the LRAT audit is a real invariant rather than a harmless annotation: checked clausal unsatisfiability alone can coexist with an unsound small support declaration. A support-erasure separator shows the provenance version of the same boundary: the same untyped LRAT skeleton may have an auxiliary lift with empty formula-label support and a label lift requiring all labels. A plain LRAT extraction trichotomy packages the raw-trace route: add provenance, pay role-agnostic support, or be unsound on a same-skeleton semantic lift. A support-local normalization transfer theorem then gives the general proof-size consequence for any proof system that admits such a normalizer with controlled support budget. These are certificate-complexity statements. They do not prove $P \neq NP$, do not prove SAT $\notin$ P/poly, do not construct an explicit hard SAT-labeled carrier, and do not imply a lower bound for Frege, Extended Frege, IPS, raw RUP/DRUP without a manifest or support parsing, arbitrary CDCL without provenance or a certified RUP support audit, or arbitrary compressed proof systems without supplying the required normalization theorem. The carrier-independent audit upgrade criterion isolates the remaining top-paper proof-system target: a proof-system-intrinsic extractor, not a carrier-specific manifest or hidden role annotation, would instantiate the transfer theorem and yield the same footprint lower bound with its audit budget. A self-auditing RUP-CDCL instantiation shows that this target is nonvacuous: if the proof syntax itself contains the checked label decoder, live-database reconstruction, RUP propagation witnesses, and retained deleted-clause support, then the audit is computed from the proof artifact itself. An append-only

self-auditing variant removes the retained deleted-clause support field by making the clause database monotone. A prefix-typed append-only variant removes the proof-internal decoder as well when formula-label variables are a public syntactic class of the proof system. These remain typed certificate-format theorems, not claims about raw DIMACS/RUP/DRUP syntax.

1 Introduction

Hardness magnification asks whether a lower bound only slightly beyond an accessible frontier can imply a major complexity separation. The transfer is useful only when its quantitative losses are controlled. The proof of the source lower bound then becomes the central issue: which kinds of local evidence could certify that the source function is hard, and what information must such evidence expose?

This paper isolates a certificate-complexity question that arises naturally in this setting. A batched generator produces a list of short objects from a source input. An external Boolean labeling assigns a bit to each object, and a fixed aggregator combines the labels in each source row. The labeling therefore induces an n-input Boolean function. We ask for nondeterministic decision-tree certificates of the predicate

"the induced source function has circuit complexity greater than s_n."

The key point is that this is an ordinary Boolean property of the labels. A sound local transcript is exactly a 1-certificate: every completion of the unexposed labels must still induce a hard function.

The contribution is deliberately narrower than an unrestricted proof-system lower bound or an explicit SAT lower bound. The main technical point is the *formula-footprint barrier*: distinct generated formulas share a single static dictionary, so a small two-polarity formula footprint gives a cheap completion circuit. The row-forcing and quotient-distance results are structural tests for local proof interfaces; they are not claimed as lower bounds for arbitrary proof systems. The SAT parity carrier is used only as a conditional calibration: if an explicit source lower bound were supplied, local completion-sound certificates would have to pay the costs identified here.

We analyze three granularities of label variables.

- In the *formula-label model*, equal generated strings share one variable. A certificate may exploit all collisions in the generator range.
- In the *occurrence-label model*, every generated position (x, j) has an independent variable. This relaxation is stronger adversarially and makes row structure explicit.
- In the *quotient-label model* for parity, sound equality links identify labels before the row parities are formed. The residual completion space is an affine image under a binary incidence map.

Across these models, the same proof strategy appears in different forms:

> A sound certificate cannot admit a completion whose induced source function has a cheap circuit.

For formula labels, a cheap completion is nearly constant on the generated formula universe. For occurrence labels, it is constant outside a small set of forced rows. For equality quotients, it is an affine completion close to a constant source truth table.

1.1 Conceptual spine: one obstruction in three interfaces

The results are meant to be read as one cohered theorem package, not as a fragmented ensemble of nearby observations or disconnected payoffs. The organizing obstruction is *completion soundness*: after a local proof, verifier, or certificate exposes some labels, all unexposed labels remain adversarially completable. A certificate is strong only if no such completion makes the induced source function easy.

The sharp result carrying the paper is the formula-footprint theorem. Before this theorem, a local hardness certificate for a generator-induced labeling could be viewed as an arbitrary sparse partial truth table. After the theorem, a small two-polarity footprint has a concrete consequence: the unexposed labels can be set to a constant, the exposed exceptions can be evaluated by a shared batched dictionary, and the resulting source function has a circuit of size nearly linear in the batch size plus the footprint. Thus, near the magnification threshold, every completion-sound formula-local certificate, equivalently every nondeterministic decision-tree 1-certificate for this hardness predicate, must expose $n^{1+\varepsilon-o(1)}$ generated formulas of each polarity.

The row-forcing and quotient-distance theorems are not competing main results. They explain how the same obstruction changes when the local interface changes. Private occurrence access turns completion soundness into many forced source rows; parity then makes each forced row complete. Equality links turn completion soundness into an affine residual-distance condition. Nonlinear semantic relations, such as one-step SAT self-reduction, are not freely available under arbitrary completion: Theorem 6.6 shows that a support-local audit must contain an ordinary membership certificate for each relation instance, Theorem 6.7 gives the exact residual certificate for acyclic OR self-reduction trees after eliminating internal nodes, Theorem 6.8 extends that exact certificate to finite acyclic OR DAG frontiers after sound quotienting of repeated reachable frontier formulas, and Theorem 6.10 gives the general exposed acyclic Boolean frontier version by reducing support-locality to ordinary pointwise certificate complexity of the induced frontier function. The identity-carrier sharpness theorem shows that the formula footprint scale is essentially optimal in the baseline truth-table model. The displayed-atom theorem, the restricted resolution theorems, the support-local extension-resolution instantiation, the dependency-cone normalizer, canonical syntactic-cone normalizer, final backward-slice lower bound, slice-explicit extension-resolution lower bound, displayed acyclic ER length lower bound, expansion-bounded compressed ER transfer, macro-compressed ER reuse tradeoff, support-declared macro ER normalizer, support-audited LRAT trace normalizer, provenance-typed LRAT support audit, namespace-typed LRAT antecedent-closure normalizer, strict-prefix Qrep replay normalizer, trace-complete DPLL decision-log normalizer, antecedent-complete CDCL learned-clause normalizer, CDCL antecedent-erasure trichotomy, RUP-complete CDCL support normalizer, canonical-namespace RUP-CDCL standardizer, self-auditing RUP-CDCL instantiation, append-only self-auditing RUP-CDCL instantiation, prefix-typed append-only RUP-CDCL instantiation, support-audit necessity separator, support-erasure indistinguishability separator, plain LRAT extraction trichotomy, cone-budget corollary, bounded-depth tradeoff, and the transfer theorem in Theorems 4.10, 4.13, 4.15, 4.16, 4.18, 4.20, 4.23, 4.25, 4.27, 4.30, 4.32, 4.34, 4.36, 4.38, 4.40, 4.42, 4.44, 4.46, 4.48, 4.50, 4.52, 4.53, 4.55, 4.57 to 4.59, 4.61 and 6.15 to 6.19 are the proof-complexity bridge for this package: a proof-system application must extract, with controlled overhead, one of these completion-sound local objects or expose a syntactic dependency-cone, typed provenance or namespace package, strict-prefix replay dictionary, trace-complete DPLL controller log, antecedent-complete CDCL learned-clause DAG, RUP-complete propagation-trace support audit, canonical label-interval encoding manifest, self-auditing label decoder, append-only RUP audit, prefix-typed label namespace, non-plain CDCL semantic invariant, or sound support-declaration budget for its auxiliary encoding.

The carrier-independent audit upgrade criterion is the same proof obligation in admission-rule form. It says exactly what a stronger recognizable proof-system payoff must contribute: the support audit must be forced by the proof-system artifact itself, not supplied by a carrier-specific manifest or hidden role advice. This keeps the proof-system route attached to the flagship obstruction rather than becoming a fragmented list of proof-format examples. Theorem 4.57 supplies the corresponding positive instantiation for an enriched but checkable RUP-CDCL certificate format: the label decoder and support audit are part of the proof object, so the audit is intrinsic to the proof system rather than imported from a side manifest. Theorem 4.58 removes one annotation layer in the deletion-free case: no retained deleted-clause support records are needed when the live database is append-only. Theorem 4.59 removes another: no proof-internal decoder is needed when the proof system has a public formula-label variable namespace.

This changes the proof obligation. The paper does not say that a major lower bound follows from locality alone. It says that a local route to such a lower bound must either pay the formula-footprint, row-forcing, or quotient-distance cost, or explain why the proof interface escapes those completion models.

1.2 The formula-footprint barrier theorem

Let

$$\mathbf{G}_n(x) = \big(G_{n,1}(x), \dots, G_{n,M_n}(x)\big)$$

be a single multi-output circuit of size g_n, with each output string of length ℓ_n. Let U_n be the set of distinct generated strings. For a label table $\lambda : U_n \to \{0,1\}$ and an aggregator A_{M_n}, define

$$H^{\lambda}_{n,A}(x) = A_{M_n}\big(\lambda(G_{n,1}(x)), \dots, \lambda(G_{n,M_n}(x))\big).$$

The formula-label hardness predicate is 1 precisely when $\mathrm{C}(H^{\lambda}_{n,A}) > s_n$.

Suppose a certificate fixes K_b distinct generated formulas to the value b. Complete every other formula to the opposite value $1-b$. The resulting label table differs from the constant $1-b$ on exactly those K_b formulas. The naive circuit compares every generated query against every exception and costs $\Theta(M_n K_b \ell_n)$. The useful observation is that the dictionary is shared by the entire batch. Sorting the M_n variable query records together with the K_b hardwired records gives a circuit of size

$$O\Big((M_n + K_b)(\ell_n + \log(M_n + K_b)) \log^2(M_n + K_b)\Big).$$

Thus a sound certificate must satisfy, separately for $b = 0$ and $b = 1$,

$$g_n + \widetilde{O}\big((M_n + K_b)\ell_n\big) > s_n.$$

In particular, if $s_n = n^{a-o(1)}$, $M_n = n^{q+o(1)}$, $g_n = n^{\rho+o(1)}$, $\ell_n = n^{o(1)}$, and $a > \max\{q, \rho\}$, then

$$K_0, K_1 \geq n^{a-o(1)}.$$

This is an inputwise 1-certificate lower bound, not an average-case or naturalness statement. It respects every equality among repeated generated formulas.

1.3 The row-forcing theorem

Now give each occurrence (x, j) an independent label $z(x, j)$. A partial assignment on row x *forces* output b if every completion of the unqueried positions has aggregator value b. Let R_b be the set of rows forced to b.

If a row is not forced to b, it has a completion to $1-b$. Complete every such row independently in this way. The resulting source function differs from the constant $1-b$ only on R_b, and equality tests patch those rows with $O(n|R_b|)$ gates. Therefore every hardness certificate above s_n must satisfy

$$|R_0|, |R_1| = \Omega(s_n/n).$$

Let $\mathrm{force}_b(A_M)$ be the minimum size of a partial assignment that forces $A_M = b$. Every row in R_b contains at least $\mathrm{force}_b(A_M)$ queried coordinates, giving

$$|Q| = \Omega\left((\mathrm{force}_0(A_{M_n}) + \mathrm{force}_1(A_{M_n}))\frac{s_n}{n}\right).$$

For parity, both forcing numbers equal M_n. Hence all forced rows are complete and

$$|Q| = \Omega(M_n s_n/n).$$

At $M_n = n^{q+o(1)}$ and $s_n = n^{1+\varepsilon-o(1)}$, this becomes $n^{q+\varepsilon-o(1)}$ queried occurrences.

The theorem also explains the role of the aggregator. OR has a one-bit certificate for output 1 but needs a full row for output 0; parity has full-row certificate complexity on both sides. The theorem records the two forcing costs separately rather than hiding this asymmetry.

1.4 Equality quotients

Occurrence independence is invalid once a certificate uses sound equality links. If two formulas are known to be equisatisfiable, their labels agree; in a parity row, two occurrences from the same class may cancel without either value being fixed. The correct equality-aware model quotients the generated formula universe by a label-respecting equivalence relation E_n.

For parity, let L_{E_n} be the binary incidence map from quotient labels to source truth tables. Fixing some quotient labels leaves an affine set $\mathcal{A}$ of possible source truth tables. We prove that completion soundness above s_n implies

$$\min_{h\in\mathcal{A}} \min_{b\in\{0,1\}} \mathrm{dist}(h, b\mathbf{1}) = \Omega(s_n/n).$$

Otherwise one affine completion is close to a constant function and can be patched by a small circuit. This is the equality-aware replacement for the private-occurrence completion argument. It does not normalize nonlinear semantic constraints such as SAT self-reduction.

1.5 Application to polylogarithmic SAT parity carriers

The motivating application is a generator of SAT instances of length

$$\ell_n = \lceil (\log n)^d \rceil, \qquad d > 1,$$

with $M_n = n^{q+o(1)}$ outputs and total generation cost $g_n = n^{\rho+o(1)}$. Define

$$H_n^{\mathrm{SAT}}(x) = \bigoplus_{j=1}^{M_n} \mathrm{SAT}(G_{n,j}(x)).$$

If SAT has circuits of size m^c, substituting one such circuit at every generated position gives

$$\mathrm{C}(H_n^{\mathrm{SAT}}) \le g_n + M_n \ell_n^c + O(M_n) = n^{\max\{q,\rho\}+o(1)}.$$

The target exponent c disappears because $\ell_n = n^{o(1)}$. Thus an infinitely-often source lower bound

$$\mathrm{C}(H_n^{\mathrm{SAT}}) > n^{1+\varepsilon-o(1)}, \qquad \max\{q, \rho\} < 1 + \varepsilon,$$

would imply SAT $\notin$ P/poly.

Our certificate theorems do not prove this source lower bound. They show what several local proof interfaces would have to pay in order to certify it. At the threshold above, the costs are summarized in Table 1.

Label granularity	Completion model	Necessary resource at $s_n = n^{1+\varepsilon-o(1)}$
Distinct formulas	Arbitrary labels on unexposed generated strings	$n^{1+\varepsilon-o(1)}$ exposed generated formulas of each polarity
Independent occurrences	Arbitrary completion of each position (x, j)	$\Omega(M_n n^{\varepsilon-o(1)}) = n^{q+\varepsilon-o(1)}$ probes for parity; $n^{\varepsilon-o(1)}$ complete rows of each parity
Equality quotient	Arbitrary completion of unexposed quotient labels	Residual affine source set at distance $n^{\varepsilon-o(1)}$ from both constant truth tables

Table 1: The three completion barriers for a near-threshold SAT parity carrier.

1.6 Status of the theorem package

Table 2 records the role of each claim in the revised paper. This table is part of the mathematical scoping, not a meta-commentary: later sections use the same boundaries. In particular, the formula-footprint theorem is the main technical result, while the row-forcing and quotient-distance theorems are supporting structural lemmas. The SAT carrier discussion is conditional calibration, and the displayed acyclic ER consequence remains a controlled local proof-object theorem rather than an unrestricted lower bound for stronger proof systems.

The admission rule is that every result must play one of five roles in the same arc: flagship theorem, calibration, interface variant, proof-system bridge, or nonclaim boundary. A new lemma is not admitted merely because it is locally correct. It must strengthen the formula-footprint spine, calibrate its sharpness, extend its proof-system reach with controlled overhead, or clarify why a stronger proof-system or SAT-family claim has not yet been proved. This is the paper's cohered-results criterion: the theorem package should not become a fragmented ensemble of results, and technically correct side payoffs are quarantined unless they serve the same before/after understanding shift.

Table 2: Claim status after the referee-objection closure pass.

Claim	Status in this paper	Boundary
Batched dictionary lemma	Patched primitive	Query records and dictionary records are tagged; dictionary records are masked out before aggregation.
Formula-footprint barrier theorem	Main technical result	Gives polarity-refined lower bounds for nondeterministic decision-tree certificates of the induced hardness predicate.
Identity-carrier generic sharpness	Sharpness theorem	Matches the formula-footprint scale up to polylogarithmic factors in the standard truth-table carrier.
Row-forcing theorem	Supporting structural lemma	Applies to completion-sound private occurrence certificates; not a proof-system theorem.
Equality quotient theorem	Supporting structural lemma	Handles sound equality identifications for parity; not nonlinear SAT semantics.
Semantic-relation support certificate	Interface-boundary proposition	Nonlinear relation constraints require ordinary membership certificates inside the declared support; this is not yet a global SAT residual-completion theorem.

Claim	Status in this paper	Boundary
Acyclic OR-tree residual certificate	Bounded semantic-frontier proposition	After eliminating internal OR nodes, root 0 forces the root and all frontier leaves, while root 1 needs the root and one true leaf; shared pure-OR frontiers are handled by the next row.
OR-DAG frontier residual certificate	Bounded shared-frontier proposition	For finite acyclic OR DAGs after sound frontier quotienting, root 0 forces the root and every distinct reachable frontier class, while root 1 needs the root and one true reachable frontier class; exposed non-OR frontiers are handled by the Boolean-frontier row.
Acyclic Boolean frontier residual certificate	Exposed semantic-frontier proposition	For any exposed finite acyclic Boolean frontier circuit, support-locality is exactly the root coordinate plus an ordinary pointwise Boolean certificate for the induced frontier function; visible adaptivity is handled by the next row.
Visible adaptive frontier residual certificate	Exposed adaptive-frontier proposition	For any checker-visible finite adaptive decision tree over frontier coordinates, support-locality is exactly the root coordinate plus an ordinary pointwise Boolean certificate for the induced decision-tree function; the actual transcript path is sufficient support, while shared-state adaptivity is handled by the next row.
Visible adaptive branching-program frontier residual certificate	Exposed shared adaptive-frontier proposition	For any checker-visible finite acyclic branching program over frontier coordinates, support-locality is the root coordinate plus an ordinary pointwise Boolean certificate for the induced branching-program function; the actual computation path is sufficient support, while hidden adaptive control remains outside scope.
Hidden adaptive transcript insufficiency separator	Boundary proposition	A reported adaptive transcript that is not certified complete can be compatible with both a one-variable frontier function and an all-frontier OR function at the same hard tuple; transcript-only support extraction is therefore unsound unless it exposes the controller/dependencies or pays worst-case support.
Controller-complete adaptive audit normalizer	Proof-system bridge proposition	A hidden adaptive controller can be normalized when the proof object exposes an acyclic controller, certifies local equivalence on the claimed completion cylinder, and pays for the replay path support; unaudited transcript summaries remain outside scope.
Trace-complete DPLL decision-log normalizer	Recognizable proof-object bridge	A DPLL-style proof object supplies the controller-complete audit when it logs the full finite decision controller, the replay path, and clause or propagation witnesses certifying local equivalence on the claimed completion cylinder; partial solver transcripts and unaudited learned clauses remain outside scope.
Antecedent-complete CDCL learned-clause normalizer	Recognizable learned-clause bridge	A CDCL-style proof object supplies the same audit when every learned clause used by the replay has an acyclic antecedent DAG down to original local clauses, decisions, propagations, or already audited learned clauses; arbitrary CDCL traces without antecedent provenance remain outside scope.
CDCL antecedent-erasure trichotomy	Boundary for arbitrary CDCL logs	If the learned-clause antecedents and role provenance are erased, a support extractor must either add equivalent provenance or a semantic invariant, pay the conservative role-agnostic support budget, or be unsound on a same-skeleton lift; arbitrary provenance-free CDCL syntax is not a free normalizer.
RUP-complete CDCL support normalizer	Positive non-plain CDCL bridge	A CDCL/DRUP-style log supplies the audit when every used learned clause has a checker-visible RUP propagation certificate over a known live clause database, with label namespace or role parsing and retained support for deleted clauses; raw RUP syntax without support parsing remains outside scope.

Claim	Status in this paper	Boundary
Canonical-namespace RUP-CDCL standardizer	Checkable encoding bridge	A deterministic encoding manifest can replace trusted role advice when it reserves a formula-label variable interval, maps that interval injectively to frontier coordinates, and lets the checker compute every RUP-trace support by intersecting touched live clauses with that interval; unmanifested DIMACS syntax remains outside scope.
Self-auditing RUP-CDCL instantiation	Carrier-independent proof-system bridge	If the trace format itself contains the checked label decoder, live-database reconstruction, RUP propagation witnesses, and retained deleted-clause support, the carrier-independent audit criterion applies with linear audit budget.
Append-only self-auditing RUP-CDCL	Less-annotated RUP proof bridge	If the self-auditing trace is deletion-free, the support audit is recovered from the monotone clause-addition log and RUP witnesses, so retained deleted-clause support annotations are unnecessary.
Prefix-typed append-only RUP-CDCL	Typed proof-system bridge	If formula-label variables are a public syntactic class of an append-only RUP-CDCL system, the audit needs neither a proof-internal label decoder nor retained deleted-clause support records; untyped raw RUP remains outside scope.
SAT parity application	Conditional calibration	Requires an explicit source lower bound not proved here.
Unrestricted standard proof-system lower bound	Not claimed	Frege, Extended Frege, IPS, compressed proofs, and unrestricted ER still require a separate extraction theorem or support budget.
Displayed-atom extraction theorem	Restricted bridge theorem	Applies only when accepted proofs explicitly display faithful completion-sound formula atoms.
Assumption-resolution theorem	Restricted named-system instantiation	Applies to resolution refutations whose measured proof object explicitly lists faithful formula-label assumptions.
Resolution unit-core corollary	Restricted named-system instantiation	Applies to full-assignment resolution refutations by counting the distinct faithful label-unit leaves used by the proof.
Resolution local-clause support theorem	Restricted named-system instantiation	Applies to resolution refutations that use explicit faithful clauses over formula-label variables, by counting their distinct label support.
Support-local auxiliary resolution theorem	Restricted auxiliary-resolution instantiation	Applies to mixed auxiliary encodings whose auxiliary constraints remain satisfiable under every completion preserving a fixed label support.
Support-local extension resolution theorem	Restricted extension-resolution instantiation	Applies to extension definitions computed from a fixed formula-label support.
Dependency-cone extension normalizer	Cone-annotated normalizer theorem	Applies when extension variables carry syntactic formula-label dependency cones whose union controls the final contradiction.
Canonical syntactic-cone normalizer	Automatic extension normalizer	Extracts dependency cones directly from the acyclic extension-definition graph; no separate cone annotations are required.
Final backward-slice lower bound	Proof-object budget corollary	The final extension slice must contain near-threshold many distinct formula-label inputs of each polarity.
Slice-explicit extension-resolution lower bound	Restricted ER-style proof-size corollary	Applies when the proof-size measure charges the distinct formula-label inputs in the final backward slice.
Displayed acyclic ER length lower bound	Restricted displayed-proof corollary	Applies when extension definitions and final clauses are written explicitly; ordinary displayed literal length charges the final slice boundary.
Expansion-bounded compressed ER transfer	Conditional compressed-proof corollary	Applies when a compressed proof expands to a displayed acyclic ER object with a controlled expansion budget.
Macro-compressed ER reuse tradeoff	Conditional macro-compression corollary	Applies when a displayed skeleton and macro library expand to displayed acyclic ER; forces a large skeleton-plus-reuse product.
Support-declared macro ER normalizer	Conditional support-budget corollary	Applies when compressed macro syntax supplies a sound declared formula-label support; forces that support budget to be near-threshold large.
Support-audited LRAT trace normalizer	Checkable support-budget corollary	Applies when an LRAT-style clausal trace includes a sound support audit; forces the audited support budget to be near-threshold large.

Claim	Status in this paper	Boundary
Provenance-typed LRAT support audit	Checkable provenance corollary	Applies when an LRAT-style trace, or DRAT trace with LRAT certificate, carries a trusted variable-role map and block-local clause provenance; the provenance package supplies the support audit.
Namespace-typed LRAT antecedent-closure normalizer	Checkable namespace corollary	Applies when formula-label variables occupy a trusted namespace and the LRAT antecedent closure of the empty clause exposes all used auxiliary clauses; the label variables in those used clauses form the support budget when a hard auxiliary witness exists.
Strict-prefix Qrep replay normalizer	Quotient-compression bridge proposition	Applies when every representative use has an earlier charged same-theory certificate forest; delayed quotient evidence is hidden support.
Support-audit necessity separator	Boundary proposition	Shows that a short checked clausal refutation can coexist with an unsound small support declaration; support locality must be audited separately.
Support-erasure indistinguishability separator	Boundary proposition	Shows that the same untyped LRAT skeleton can have an auxiliary lift with empty formula-label support and a label lift requiring all exposed labels; variable-role provenance is not cosmetic.
Cone-budget extension lower bound	Quantitative proof-object corollary	Forces either many final variables or a large dependency cone in cone-annotated extension proofs.
Bounded-depth extension tradeoff	Quantitative proof-object corollary	Forces a large final interface or logarithmic extension depth for bounded-fanin cone-annotated proofs.
Support-local normalization transfer theorem	Conditional proof-system transfer	Any proof system with a support-local normalizer and controlled support budget inherits the formula-footprint lower bound.
Carrier-independent audit upgrade criterion	Top-paper proof-system target	A system-intrinsic audit extractor, without carrier-specific namespace manifests or hidden role advice, instantiates the transfer theorem and yields a proof-size lower bound with the stated budget exponent.
Randomized bounded-error union footprint	Supporting theorem	Accepting-query union reduces to a deterministic certificate; expected footprint remains open.

1.7 Nonvacuity and sharpness checks

The abstract hardness predicates are not empty. For formula labels, take $M_n = 1$, let G_n be the identity map $x \mapsto x$, and use the identity aggregator. Then the formula-label carrier is the ordinary truth table of a Boolean function. The generic sharpness theorem in Theorem 4.5 shows that, in this standard model, hard functions not only exist but typically have completion-sound certificates at the same scale as the lower bound up to polylogarithmic factors. For occurrence labels, private row labels can realize any source truth table in the same way. For quotient labels under parity, nonvacuity follows whenever the quotient incidence map $L_{E_n} : \mathbb{F}_2^{\mathcal{C}_n} \to \mathbb{F}_2^{2^n}$ has an image containing a truth table outside the size-s circuit ball; full row rank is a sufficient condition.

These examples are intentionally modest. They show that the certificate predicates are meaningful in unrestricted or rank-conditioned models. They do not construct an explicit SAT-labeled carrier satisfying the source lower bound in the unrestricted circuit model. Consequently, the SAT application remains conditional.

The three label granularities are genuinely different. If a parity row contains M_n copies of one generated key u_x, with $x \mapsto u_x$ injective and M_n odd, the formula-label model needs only the shared value $\lambda(u_x)$ for that row, whereas the private occurrence model must query all M_n positions to force the row parity. Conversely, equality quotienting can collapse private occurrence hardness: if each parity row contains two equal occurrences and they are quotient-identified, the row contribution is always zero. These separations justify treating the row-forcing and quotient-distance results as structural tests, while also showing that a full matching-bound theory for explicit hard SAT carriers is still future work.

1.8 Scope and organization

The results are deliberately scoped. We do not prove an unrestricted circuit lower bound for an explicit SAT-labeled carrier. We do not prove a lower bound for Frege, Extended Frege, IPS, or another standard proof system. The occurrence theorem applies only when acceptance remains sound under independent occurrence completions; the quotient theorem applies only to equality constraints; and the formula theorem applies to functional certificates over generated formula labels.

The contribution is a compact theorem package about the nondeterministic query complexity and completion geometry of induced circuit-hardness predicates. Section 2 gives the formal model. Section 3 proves sparse-row patching and batched static lookup. Section 4 proves the formula-footprint theorem. Section 5 proves the row-forcing theorem. Section 6 treats equality quotients. Section 7 gives the SAT magnification application and a brief naturalization observation. Section 8 records the proof dependencies and nonclaim boundaries. Section 9 places the results in the literature and states the main open directions.

2 Preliminaries and hardness predicates

2.1 Circuit complexity

We use nonuniform Boolean circuits over a fixed finite fan-in-two basis. XOR and constants have constant-size implementations and are used with constant-factor overhead. For $f : \{0,1\}^n \to \{0,1\}$, let $\mathrm{C}(f)$ denote its minimum circuit size. For a size function s, $\mathrm{SIZE}(s(n))$ is the class of languages having circuits of size $O(s(n))$ at length n, and

$$\mathrm{P/poly} = \bigcup_{c \geq 1} \mathrm{SIZE}(n^c).$$

All logarithms are base two. Fixed constants may depend on fixed parameters such as q, ρ, d, and ε.

2.2 Inputwise certificates

Let V be a finite set of Boolean variables and let $P : \{0,1\}^V \to \{0,1\}$ be a Boolean property. A partial assignment is a map $\sigma : S \to \{0,1\}$ for some $S \subseteq V$. A total assignment $y \in \{0,1\}^V$ extends σ if $y|_S = \sigma$.

Definition 2.1 (Inputwise 1-certificate)**.** Let $y \in \{0,1\}^V$ satisfy $P(y) = 1$. A partial assignment $\sigma : S \to \{0,1\}$ is a 1-*certificate for* P *at* y if $\sigma = y|_S$ and every total assignment extending σ also satisfies $P = 1$. The inputwise certificate complexity $C_1(P, y)$ is the minimum $|S|$ over such certificates.

This is the standard nondeterministic decision-tree notion of certificate complexity; see, for example, Nisan [14] and the survey of Buhrman and de Wolf [6]. We will prove stronger, structured lower bounds that count separately which polarities or rows a certificate fixes.

2.3 Batched generators and formula-label carriers

Definition 2.2 (Batched generator)**.** For each source length n, a batched generator is a multi-output circuit

$$\mathbf{G}_n : \{0,1\}^n \longrightarrow (\{0,1\}^{\ell_n})^{M_n}, \qquad \mathbf{G}_n(x) = (G_{n,1}(x), \ldots, G_{n,M_n}(x)).$$

Its size g_n is the size of the single circuit producing all $M_n \ell_n$ output bits.

The batched convention is important: the generator is paid once. If n^ρ were instead the cost of producing one query, the total cost could be $n^{q+\rho}$ rather than $n^{\max\{q,\rho\}}$.

Let $A_M : \{0,1\}^M \to \{0,1\}$ be a family of nonconstant aggregators having circuits of size $O(M)$. Define the generated universe

$$U_n := \{G_{n,j}(x) : x \in \{0,1\}^n,\ j \in [M_n]\}.$$

It is a set of distinct strings, so repeated generated occurrences share one formula label.

For a labeling $\lambda : U_n \to \{0,1\}$, define

$$H^\lambda_{n,A}(x) := A_{M_n}\big(\lambda(G_{n,1}(x)), \dots, \lambda(G_{n,M_n}(x))\big). \tag{1}$$

Definition 2.3 (Formula-label hardness predicate)**.** For a threshold s, define

$$\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda) := \mathbf{1}[\mathrm{C}(H^\lambda_{n,A}) > s], \qquad \lambda \in \{0,1\}^{U_n}.$$

A 1-certificate for $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}$ is exactly a partial formula labeling whose every completion induces a source function of circuit complexity greater than s.

2.4 Occurrence-label carriers

Let

$$\Omega_n := \{0,1\}^n \times [M_n].$$

For an occurrence assignment $z : \Omega_n \to \{0,1\}$, define

$$F^z_{n,A}(x) := A_{M_n}\big(z(x,1), \dots, z(x,M_n)\big). \tag{2}$$

Definition 2.4 (Occurrence-label hardness predicate)**.** For a threshold s, define

$$\mathsf{Hard}^{\mathrm{occ}}_{n,s,A}(z) := \mathbf{1}[\mathrm{C}(F^z_{n,A}) > s], \qquad z \in \{0,1\}^{\Omega_n}.$$

Every formula labeling induces an occurrence assignment

$$z_\lambda(x,j) := \lambda(G_{n,j}(x)),$$

but most occurrence assignments are not induced by formula labels. The occurrence model deliberately forgets equality constraints among repeated generated strings.

2.5 Minimum forcing numbers of an aggregator

A partial assignment $\alpha : S \to \{0,1\}$, $S \subseteq [M]$, *forces* $A_M = b$ if every total string extending α has aggregator value b.

Definition 2.5 (Minimum forcing number)**.** For $b \in \{0,1\}$, define

$$\mathrm{force}_b(A_M) := \min\{|S| : \text{some partial assignment on } S \text{ forces } A_M = b\}.$$

We use "minimum forcing number" to distinguish this quantity from the more common worst-input certificate complexity. For the symmetric aggregators used below, the values are immediate:

$$\mathrm{force}_0(\oplus_M) = \mathrm{force}_1(\oplus_M) = M,$$

while

$$\mathrm{force}_0(\mathrm{OR}_M) = M, \qquad \mathrm{force}_1(\mathrm{OR}_M) = 1.$$

3 Two cheap completion lemmas

The main theorems use two elementary upper bounds. The first patches a sparse set of source rows. The second evaluates a common hardwired dictionary on a whole batch of generated strings.

3.1 Sparse-row patching

Lemma 3.1 (Equality patching)**.** *Let $f, g : \{0,1\}^n \to \{0,1\}$, and let*

$$R := \{x \in \{0,1\}^n : f(x) \neq g(x)\}.$$

Then

$$\mathrm{C}(f) \leq \mathrm{C}(g) + O((n+1)|R|). \tag{3}$$

Proof. For each $r \in R$, compute the equality predicate

$$\mathrm{EQ}_r(x) = \bigwedge_{i=1}^{n} \begin{cases} x_i, & r_i = 1, \\ \neg x_i, & r_i = 0. \end{cases}$$

using $O(n)$ gates. Let $R' \subseteq R$ be the rows on which g must be flipped. Since the predicates EQ_r are mutually exclusive,

$$f(x) = g(x) \oplus \bigoplus_{r \in R'} \mathrm{EQ}_r(x).$$

The correction circuit has size $O((n+1)|R|)$. □

Corollary 3.2 (Residual-error threshold)**.** *If $\mathrm{C}(f) > s$ and $\mathrm{C}(g) \leq t < s$, then*

$$\mathrm{dist}(f, g) = \Omega\left(\frac{s-t}{n+1}\right).$$

In particular, a lower bound $\mathrm{C}(f) > n^{1+\varepsilon-o(1)}$ rules out every subthreshold circuit that is wrong on only $n^{\varepsilon-o(1)}$ source rows.

3.2 Batched static dictionary lookup

For $M, \ell \geq 1$ and $K \geq 0$, define

$$\mathrm{Lookup}(M, K, \ell) := (M+K)(\ell + \log(M+K+2)) \log^2(M+K+2). \tag{4}$$

Lemma 3.3 (Batched static dictionary)**.** *Fix a dictionary $D \subseteq \{0,1\}^\ell$ of K distinct hardwired keys, a hardwired bit $a(u)$ for each $u \in D$, and a default bit b. There is a fan-in-two circuit that, on M variable queries $q_1, \ldots, q_M \in \{0,1\}^\ell$, outputs*

$$v_j = \begin{cases} a(q_j), & q_j \in D, \\ b, & q_j \notin D, \end{cases} \qquad j \in [M],$$

in size $O(\mathrm{Lookup}(M, K, \ell))$. If only a symmetric aggregate of the v_j is needed, the same bound holds without restoring the original query order.

Proof. Create M query records and K hardwired dictionary records. A query record has the form

$$(q_i, 1, i, 0),$$

where the second field is a query tag and i is the original query index. A dictionary record for a key $u \in D$ has the form

$$(u, 0, 0, a(u)).$$

Sort the $N = M + K$ records lexicographically by key and then by the tag, placing the unique dictionary record, if present, before all query records with the same key.

Batcher's sorting network has $O(N \log^2 N)$ comparators [5]. A comparator on records of width $O(\ell + \log N)$ has circuit size $O(\ell + \log N)$, including comparison and conditional exchange. After sorting, perform a left-to-right scan inside each equal-key block. The scan records whether a dictionary record has appeared in the current block and propagates its payload value. Each query record therefore obtains $a(q_i)$ when $q_i \in D$, and the default value b otherwise. Dictionary records are masked out at the output stage.

If the final aggregator is symmetric, the circuit may aggregate the recovered values of the query records in sorted order, since only their multiset matters. If the aggregator is not symmetric, sort all records once more by the query tag and then by the stored original index, feed the first M query records to the aggregator, and ignore the dictionary records. The scan, masks, and optional second sort are absorbed by the same $O(\text{Lookup}(M, K, \ell))$ bound. Thus original-order restoration is unnecessary only in the symmetric case; query-record separation is required in all cases. □

Remark 3.4 (A logarithmic improvement)**.** Using the $O(N \log N)$-comparator sorting networks of Ajtai, Komlós, and Szemerédi [1] replaces the final $\log^2(M + K)$ factor in (4) by $\log(M + K)$. We state the transparent Batcher bound because all threshold corollaries below lose only an $n^{o(1)}$ factor.

The lemma is nonuniform in exactly the intended way: the dictionary keys and values are constants in the upper-bound circuit.

For a labeling $\lambda : U_n \to \{0, 1\}$ and a baseline $b \in \{0, 1\}$, define the defect set

$$\text{Def}_b(\lambda) := \{u \in U_n : \lambda(u) \neq b\}.$$

Proposition 3.5 (Sparse formula-defect upper bound)**.** *Let A_M have circuit size $O(M)$. For every formula labeling $\lambda : U_n \to \{0, 1\}$, every baseline b, and*

$$K = |\text{Def}_b(\lambda)|,$$

one has

$$\text{C}(H^{\lambda}_{n,A}) \leq g_n + O\big(\text{Lookup}(M_n, K, \ell_n) + M_n\big). \tag{5}$$

For a symmetric aggregator, including parity, the final reordering network is unnecessary.

Proof. Run the batched generator once. Use the defect set as the hardwired dictionary, label every dictionary key by $1 - b$, and use b as the default. The dictionary circuit recovers all M_n labels, after which the size-$O(M_n)$ aggregator is applied. □

Repeated formulas are automatically handled by one dictionary key. This is the quantitative reason that the formula-label barrier depends nearly linearly on $M_n + K$ rather than on their product.

4 Formula-label certificates and near-threshold footprints

Fix a formula labeling λ with $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda) = 1$, and let $\sigma : S \to \{0,1\}$ be a 1-certificate at λ. Write

$$S_b := \{u \in S : \sigma(u) = b\}, \qquad b \in \{0,1\}.$$

For later algebra define

$$\mathcal{L}_n(s) := (\ell_n + \log(s + M_n + 2)) \log^2(s + M_n + 2). \tag{6}$$

Theorem 4.1 (Two-sided formula-footprint barrier)**.** *There is a constant $c \geq 1$ such that every 1-certificate σ for $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}$ satisfies, for each $b \in \{0,1\}$,*

$$g_n + c\,\mathrm{Lookup}(M_n, |S_b|, \ell_n) + cM_n > s. \tag{7}$$

Consequently,

$$|S_b| \geq \left(\frac{s - g_n - cM_n}{c\mathcal{L}_n(s)} - M_n \right)_+ . \tag{8}$$

Proof. Fix $b \in \{0,1\}$. Complete every unexposed formula label to the opposite baseline $1 - b$:

$$\lambda^{(1-b)}(u) = \begin{cases} \sigma(u), & u \in S, \\ 1 - b, & u \notin S. \end{cases}$$

This total labeling extends the certificate, so

$$\mathrm{C}(H^{\lambda^{(1-b)}}_{n,A}) > s.$$

Relative to the constant baseline $1 - b$, its defect set is exactly S_b. Applying Theorem 3.5 proves (7).

For the explicit inequality, let $K = |S_b|$. If $K > s$, then (8) is immediate after increasing c, since $\mathcal{L}_n(s) \geq 1$. If $K \leq s$, then

$$\mathrm{Lookup}(M_n, K, \ell_n) \leq (M_n + K)\mathcal{L}_n(s).$$

Substitute this into (7) and rearrange. □

The theorem is instancewise: it applies to every hard labeling and every certificate at that labeling. It also counts the two polarities separately. Merely exposing many formulas is not sufficient if nearly all exposed labels have the same value.

Corollary 4.2 (Nondeterministic formula-query decision-tree lower bound)**.** *Let $C_1(\mathsf{Hard}^{\mathrm{form}}_{n,s,A}, \lambda)$ denote the standard inputwise nondeterministic decision-tree certificate complexity of the Boolean property $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}$ at a hard labeling λ. Then*

$$C_1(\mathsf{Hard}^{\mathrm{form}}_{n,s,A}, \lambda) \geq \sum_{b \in \{0,1\}} \left(\frac{s - g_n - cM_n}{c\mathcal{L}_n(s)} - M_n \right)_+ ,$$

where c is the constant from Theorem 4.1. In particular, under the hypotheses of Theorem 4.3,

$$C_1(\mathsf{Hard}^{\mathrm{form}}_{n,s_n,A}, \lambda) \geq 2n^{a - o(1)}.$$

Proof. Every nondeterministic decision-tree 1-certificate is exactly a partial formula-label assignment satisfying Theorem 2.1. Apply the two polarity lower bounds in Theorem 4.1 and sum them. □

Corollary 4.3 (Near-threshold formula cost)**.** *Assume*

$$s_n = n^{a-o(1)}, \qquad M_n = n^{q+o(1)}, \qquad g_n = n^{\rho+o(1)}, \qquad \ell_n = n^{o(1)},$$

where $a > \max\{q, \rho\}$*. Then every* 1*-certificate for* $\mathsf{Hard}^{\mathrm{form}}_{n,s_n,A}$ *satisfies*

$$|S_0|, |S_1| \geq n^{a-o(1)}.$$

In particular, at $a = 1 + \varepsilon$ *each polarity has size* $n^{1+\varepsilon-o(1)}$*.*

Proof. The fixed exponent gap gives $g_n + M_n = o(s_n)$, while $\mathcal{L}_n(s_n) = n^{o(1)}$. Apply Theorem 4.1. □

Corollary 4.4 (Diversity and label-entropy floors)**.** *Under the hypotheses of Theorem 4.3, every hard labeling* λ *itself has*

$$|\lambda^{-1}(0)|, |\lambda^{-1}(1)| \geq n^{a-o(1)}.$$

Consequently,

$$|U_n| \geq n^{a-o(1)}.$$

If $U_n \subseteq \{0,1\}^{\ell_n}$*, then*

$$\ell_n \geq (a - o(1)) \log n.$$

Proof. The full assignment is always a 1-certificate at a hard input. Apply Theorem 4.3; its polarity sets are subsets of $\lambda^{-1}(0)$ and $\lambda^{-1}(1)$. The entropy bound follows from $|U_n| \leq 2^{\ell_n}$. □

4.1 Generic sharpness in the identity carrier

The formula-footprint theorem is close to tight in the standard truth-table model. Let the identity carrier have $M_n = 1$, $G_{n,1}(x) = x$, $A_1(z) = z$, and $U_n = \{0,1\}^n$. Then a formula labeling is exactly a Boolean function $f : \{0,1\}^n \to \{0,1\}$, and $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(f)$ is the predicate $\mathrm{C}(f) > s$.

Theorem 4.5 (Generic sharpness in the identity carrier)**.** *Fix the finite circuit basis. There is a constant* C *such that the following holds for all* $n, s, r \geq 1$*. Let*

$$m \geq Cs \log(s+n) + r \qquad \text{and} \qquad m \leq 2^n.$$

For a uniformly random Boolean function $f : \{0,1\}^n \to \{0,1\}$*, with probability at least*

$$1 - 2^{-r} - 2e^{-m/18},$$

there is a 1*-certificate* $\sigma : T \to \{0,1\}$ *for the predicate* $\mathrm{C}(f) > s$*, at* f*, such that*

$$|T| = m, \qquad |\sigma^{-1}(0)|, |\sigma^{-1}(1)| \leq \frac{2m}{3}.$$

Consequently, in the identity carrier, the lower bound in Theorem 4.1 is sharp up to polylogarithmic factors for generic hard truth tables.

Proof. Let $\mathcal{C}_{n,s}$ be the set of n-input Boolean functions computable by size-s circuits over the fixed basis. A standard circuit count gives

$$|\mathcal{C}_{n,s}| \leq 2^{C_0 s \log(s+n)}$$

for an absolute constant C_0: a circuit of size s is specified by its gate types and by two input wires per gate, each chosen from at most $n + s + O(1)$ possibilities.

Choose f uniformly at random and choose an m-element set $T \subseteq \{0,1\}^n$ uniformly at random, independently of f. For any fixed $h \in \mathcal{C}_{n,s}$,

$$\Pr[h|_T = f|_T] = 2^{-m}.$$

Taking $C \geq C_0 + 1$ and applying the union bound gives

$$\Pr[\exists h \in \mathcal{C}_{n,s} : h|_T = f|_T] \leq 2^{-r}.$$

If no size-s function agrees with f on T, then $f|_T$ is a 1-certificate for $\mathrm{C}(f) > s$: every completion agreeing with f on T is inconsistent with every size-s circuit.

The values $f(x)$ on T are independent fair bits. A Chernoff bound gives

$$\Pr\left[\max_{b\in\{0,1\}} |\{x \in T : f(x) = b\}| > \frac{2m}{3}\right] \leq 2e^{-m/18}.$$

Thus, over the joint choice of f and T, the probability that T fails to be a balanced certificate is at most $2^{-r} + 2e^{-m/18}$. If a fixed f has no such T, then every choice of T fails. Hence the same upper bound holds for the probability, over f, that no balanced certificate T exists.

Taking $m = O(s \log(s+n))$ gives certificates of size $O(s \log(s+n))$, while Theorem 4.1 gives the lower bound $\Omega(s/\operatorname{polylog} s)$ when the generator overhead is negligible. This proves the claimed polylogarithmic sharpness in the identity carrier. □

4.2 Adaptive deterministic verifiers

The certificate formulation immediately covers adaptive oracle verifiers. Let V be deterministic, let π be a proposed proof string, and let $V^\lambda(\pi)$ denote its execution with formula-label oracle λ.

Proposition 4.6 (Accepted query transcripts are certificates)**.** *Suppose*

$$V^\lambda(\pi) = 1 \quad \Longrightarrow \quad \mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda) = 1$$

for every formula labeling λ. Fix an accepted execution and let σ be its query-answer transcript. Then σ is a 1*-certificate for $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}$, so Theorem 4.1 applies to its* 0*- and* 1*-answers separately.*

Proof. Every labeling extending the transcript answers all adaptive queries exactly as in the accepted execution, follows the same path, and is accepted. Functional soundness then makes every such completion hard. □

Thus the formula theorem lower-bounds nondeterministic formula-query complexity without any largeness assumption on the accepted set and without requiring truth-table constructivity.

4.3 Bounded-error accepting-union certificates

The deterministic transcript argument has a bounded-error analogue for the union of labels seen on accepting executions. This does not lower-bound the expected number of queries in one randomized run, but it gives a clean randomized robustness statement for the resource most directly comparable to a proof's displayed local information.

Theorem 4.7 (Accepting-query union is a certificate)**.** *Let $P : \{0,1\}^V \to \{0,1\}$ be any Boolean property and let V_R be a randomized adaptive oracle verifier with randomness $r \in R$. Fix a total assignment y with $P(y) = 1$. Let*

$$A_y := \{r \in R : V_R^y(r) = 1\}$$

be the accepting random strings on y, and let $U_y \subseteq V$ be the union of all variables queried during the executions $V_R^y(r)$ with $r \in A_y$. Suppose there is a soundness threshold β such that every y' with $P(y') = 0$ satisfies

$$\Pr_r[V_R^{y'}(r) = 1] \le \beta,$$

while

$$\Pr_r[V_R^y(r) = 1] > \beta.$$

Then the partial assignment $y|_{U_y}$ is a 1-certificate for P at y.

Proof. Assume toward a contradiction that $y|_{U_y}$ is not a 1-certificate. Then some y' with $P(y') = 0$ agrees with y on every variable in U_y. For each $r \in A_y$, the adaptive execution under y' follows the same query path as under y: every query made on the accepting execution under y lies in U_y and receives the same answer under y'. Hence $V_R^{y'}(r) = 1$ for every $r \in A_y$. Therefore

$$\Pr_r[V_R^{y'}(r) = 1] \ge \Pr_r[V_R^y(r) = 1] > \beta,$$

contradicting soundness. □

Corollary 4.8 (Bounded-error formula union footprint)**.** *Let V_R be a randomized adaptive formula-label verifier for $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}$. Suppose it accepts a hard labeling λ with probability $> \beta$, and accepts every labeling λ' with $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda') = 0$ with probability at most β. Let U_λ be the union of generated formula labels queried on accepting executions at λ, and write*

$$U_{\lambda,b} := \{u \in U_\lambda : \lambda(u) = b\}.$$

Then, for each $b \in \{0,1\}$,

$$g_n + c\,\mathrm{Lookup}(M_n, |U_{\lambda,b}|, \ell_n) + cM_n > s,$$

with the same constant c as in Theorem 4.1. In the near-threshold regime of Theorem 4.3, every bounded-error verifier has accepting union footprint

$$|U_{\lambda,0}|, |U_{\lambda,1}| \ge n^{a-o(1)}.$$

Proof. Apply Theorem 4.7 to the property $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}$. The accepting union assignment $\lambda|_{U_\lambda}$ is a deterministic 1-certificate, so Theorem 4.1 and Theorem 4.3 apply to its two polarity classes. □

The union resource is deliberately strong: a randomized verifier may have small expected query complexity while spreading accepting executions over a large union. Proving expected-footprint, information, or transcript-support lower bounds requires additional distributional arguments and remains open.

4.4 Displayed-atom proof interfaces

The preceding verifier propositions can be stated as an extraction theorem for a restricted proof interface. This is the formal bridge used by the paper; stronger proof-complexity consequences require normalizing a concrete proof system into this interface.

Definition 4.9 (Displayed formula-atom interface)**.** Fix n, s, A. A displayed formula-atom interface for $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}$ is an acceptance relation together with an extractor that maps every accepted proof π to a partial assignment

$$\sigma_\pi : S_\pi \to \{0,1\}, \qquad S_\pi \subseteq U_n.$$

For a labeling λ, the extracted assignment is *faithful at* λ if $\sigma_\pi(u) = \lambda(u)$ for every $u \in S_\pi$. It is *completion-sound* if every formula labeling extending σ_π satisfies $\mathsf{Hard}^{\mathrm{form}}_{n,s,A} = 1$. The interface is *literal-listing* if each extracted atom is displayed in the proof by writing its generated key $u \in \{0,1\}^{\ell_n}$ and its bit value.

Theorem 4.10 (Displayed-atom extraction)**.** *Let λ satisfy $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda) = 1$, and let π be an accepted proof in a displayed formula-atom interface whose extracted assignment $\sigma_\pi : S_\pi \to \{0,1\}$ is faithful at λ and completion-sound. Write*

$$S_{\pi,b} := \{u \in S_\pi : \sigma_\pi(u) = b\}.$$

Then, for each $b \in \{0,1\}$,

$$g_n + c\,\mathrm{Lookup}(M_n, |S_{\pi,b}|, \ell_n) + cM_n > s,$$

with the same constant c as in Theorem 4.1. In the near-threshold regime of Theorem 4.3,

$$|S_{\pi,0}|, |S_{\pi,1}| \geq n^{a-o(1)}.$$

If the interface is literal-listing, every such proof displays at least

$$(|S_{\pi,0}| + |S_{\pi,1}|)(\ell_n + 1)$$

atom bits before any additional proof overhead.

Proof. Faithfulness says that σ_π is a partial assignment at the actual hard labeling. Completion soundness says that every extension of σ_π remains in the hardness predicate. Hence σ_π is a 1-certificate for $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}$ at λ. The footprint inequalities follow from Theorem 4.1, and the near-threshold lower bound follows from Theorem 4.3. In a literal-listing interface, each extracted atom explicitly contains an ℓ_n-bit key and a one-bit value, giving the displayed atom-bit lower bound by counting the listed atoms. □

Remark 4.11 (What this extraction theorem does and does not say)**.** Theorem 4.10 is intentionally restricted. It covers proof interfaces whose local semantic content is an explicit faithful list of formula labels and whose soundness is stable under all completions of unlisted labels. It does not by itself lower-bound Frege, Extended Frege, IPS, implicit proofs, or algebraic proof systems, which may encode semantic information nonlocally. Its role is to make the required normalization target precise: to use the formula-footprint theorem in a standard proof system, one must extract a displayed completion-sound atom set with controlled overhead.

4.5 A restricted resolution instantiation

The displayed-atom theorem gives a concrete proof-system consequence for a standard but deliberately restricted resolution format: resolution refutations with explicit formula-label assumptions. This is not a lower bound for unrestricted resolution encodings of circuit lower bounds. It is a normalization theorem for the case where the proof derives contradiction from an explicitly listed set of local label facts.

Definition 4.12 (Easy-completion CNF)**.** An *easy-completion CNF* for $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}$ is a CNF

$$\Phi_{n,s,A}(Y, Z)$$

over formula-label variables $Y = \{Y_u : u \in U_n\}$ and auxiliary variables Z such that, for every total formula labeling $\mu : U_n \to \{0,1\}$,

$$\exists z\ \Phi_{n,s,A}(\mu, z) = 1 \quad \Longleftrightarrow \quad \mathsf{Easy}^{\mathrm{form}}_{n,s,A}(\mu),$$

where $\mathsf{Easy}^{\mathrm{form}}_{n,s,A}$ is the complement of $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}$.

Theorem 4.13 (Resolution with explicit label assumptions)**.** *Let λ satisfy $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda) = 1$. Let $\Phi_{n,s,A}$ be an easy-completion CNF, and suppose there is a resolution refutation of*

$$\Phi_{n,s,A}(Y, Z) \wedge \Gamma$$

where Γ is a set of unit clauses over formula-label variables of the form $Y_u = \lambda(u)$. Let

$$S_\Gamma := \{u \in U_n : Y_u = \lambda(u) \text{ appears in } \Gamma\}, \qquad S_{\Gamma,b} := \{u \in S_\Gamma : \lambda(u) = b\}.$$

Then, for each $b \in \{0,1\}$,

$$g_n + c\,\mathrm{Lookup}(M_n, |S_{\Gamma,b}|, \ell_n) + cM_n > s,$$

with the same constant c as in Theorem 4.1. In the near-threshold regime of Theorem 4.3,

$$|S_{\Gamma,0}|, |S_{\Gamma,1}| \geq n^{a-o(1)}.$$

Consequently, any resolution proof format that counts these label assumptions as displayed unit clauses has at least

$$|S_{\Gamma,0}| + |S_{\Gamma,1}|$$

displayed label assumptions, and at least

$$(|S_{\Gamma,0}| + |S_{\Gamma,1}|)(\ell_n + 1)$$

displayed label-atom bits, before counting the rest of the resolution derivation.

Proof. Resolution soundness implies that $\Phi_{n,s,A} \wedge \Gamma$ has no satisfying assignment. By the defining completeness of the easy-completion CNF, there is therefore no total formula labeling μ extending the unit assumptions in Γ with $\mathsf{Easy}^{\mathrm{form}}_{n,s,A}(\mu)$. Equivalently, every formula labeling extending Γ lies in $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}$. Since the assumptions are faithful to the actual hard labeling λ, the partial assignment specified by Γ is a 1-certificate for $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}$ at λ. The two polarity inequalities and the near-threshold bounds are exactly Theorems 4.1 and 4.3. If the proof format explicitly lists label assumptions as unit clauses, each distinct assumption displays one generated key and one bit, giving the stated displayed-assumption and displayed-bit lower bounds. □

Remark 4.14 (Scope of the resolution instantiation)**.** Theorem 4.13 is a genuine resolution statement, but only for assumption-based encodings where local label facts are part of the measured proof object. It does not rule out short resolution, Frege, Extended Frege, IPS, or implicit proofs that encode the needed semantic information without explicitly listing faithful label assumptions. Its value is as a first controlled normalization theorem: the proof-system theorem becomes strong exactly when an external argument shows that a natural proof format must expose such assumptions.

Corollary 4.15 (Resolution unit-core lower bound)**.** *Let λ satisfy $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda) = 1$, and let Λ_λ be the full set of unit clauses $Y_u = \lambda(u)$ for $u \in U_n$. For any resolution refutation Π of*

$$\Phi_{n,s,A}(Y, Z) \wedge \Lambda_\lambda,$$

let $T_\Pi \subseteq U_n$ be the set of formula-label variables whose unit clauses from Λ_λ occur as initial clauses used in Π, and set

$$T_{\Pi,b} := \{u \in T_\Pi : \lambda(u) = b\}.$$

Then, for each $b \in \{0, 1\}$,

$$g_n + c\,\mathrm{Lookup}(M_n, |T_{\Pi,b}|, \ell_n) + cM_n > s.$$

In the near-threshold regime of Theorem 4.3,

$$|T_{\Pi,0}|, |T_{\Pi,1}| \geq n^{a-o(1)}.$$

Thus the lower bound applies to the used unit core *of any full-assignment resolution refutation, not only to a preselected assumption set.*

Proof. Discard every initial unit clause from Λ_λ that is not used as a leaf of the refutation Π. The remaining proof is still a resolution refutation of $\Phi_{n,s,A} \wedge \Gamma_\Pi$, where

$$\Gamma_\Pi := \{Y_u = \lambda(u) : u \in T_\Pi\}.$$

Apply Theorem 4.13 to Γ_Π. □

Theorem 4.16 (Resolution local-clause support)**.** *Let λ satisfy $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda) = 1$, and let $\Phi_{n,s,A}$ be an easy-completion CNF. Let Γ be any CNF over formula-label variables $Y = \{Y_u : u \in U_n\}$, with no auxiliary variables, such that $\lambda \models \Gamma$. Suppose there is a resolution refutation of*

$$\Phi_{n,s,A}(Y, Z) \wedge \Gamma.$$

Let $S_\Gamma \subseteq U_n$ be the set of formula labels whose variables occur in Γ, and set

$$S_{\Gamma,b} := \{u \in S_\Gamma : \lambda(u) = b\}.$$

Then, for each $b \in \{0, 1\}$,

$$g_n + c\,\mathrm{Lookup}(M_n, |S_{\Gamma,b}|, \ell_n) + cM_n > s.$$

In the near-threshold regime of Theorem 4.3,

$$|S_{\Gamma,0}|, |S_{\Gamma,1}| \geq n^{a-o(1)}.$$

Thus the lower bound applies to the two-polarity label support of any explicitly displayed local label clauses, even when those clauses are not unit clauses.

Proof. Resolution soundness implies that $\Phi_{n,s,A} \wedge \Gamma$ is unsatisfiable. Let $\mu : U_n \to \{0,1\}$ be any formula labeling extending $\lambda|_{S_\Gamma}$. Since Γ contains only variables from S_Γ and $\lambda \models \Gamma$, the labeling μ also satisfies Γ. If μ were easy, then by Theorem 4.12 there would be an auxiliary assignment z satisfying $\Phi_{n,s,A}(\mu, z)$, and hence (μ, z) would satisfy $\Phi_{n,s,A} \wedge \Gamma$, a contradiction. Therefore every extension of $\lambda|_{S_\Gamma}$ lies in $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}$. The partial assignment $\lambda|_{S_\Gamma}$ is a 1-certificate at λ, and Theorems 4.1 and 4.3 give the claimed polarity bounds. □

Remark 4.17 (Local clauses versus nonlocal encodings)**.** Theorem 4.16 counts the explicit label support of clauses over formula-label variables. It does not apply to encodings that mix label variables with auxiliary variables in a way that lets the proof hide semantic information outside a fixed label support. Such encodings are precisely where a stronger normalization theorem would be needed.

Theorem 4.18 (Support-local auxiliary resolution)**.** *Let λ satisfy $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda) = 1$, and let $\Phi_{n,s,A}$ be an easy-completion CNF. Let $\Gamma(Y, W)$ be a CNF over formula-label variables $Y = \{Y_u : u \in U_n\}$ and auxiliary variables W. Suppose there is a set $S \subseteq U_n$ such that Γ is* support-local at λ over *S: for every formula labeling $\mu : U_n \to \{0,1\}$ with $\mu|_S = \lambda|_S$, there exists an assignment w to W satisfying $\Gamma(\mu, w)$. If there is a resolution refutation of*

$$\Phi_{n,s,A}(Y, Z) \wedge \Gamma(Y, W),$$

then, writing

$$S_b := \{u \in S : \lambda(u) = b\},$$

we have, for each $b \in \{0,1\}$,

$$g_n + c\,\mathrm{Lookup}(M_n, |S_b|, \ell_n) + cM_n > s.$$

In the near-threshold regime of Theorem 4.3,

$$|S_0|, |S_1| \geq n^{a - o(1)}.$$

Proof. Resolution soundness implies that $\Phi_{n,s,A} \wedge \Gamma$ is unsatisfiable. Let μ be any formula labeling extending $\lambda|_S$. By support-locality, choose w satisfying $\Gamma(\mu, w)$. If μ were easy, then by Theorem 4.12 there would be an auxiliary assignment z satisfying $\Phi_{n,s,A}(\mu, z)$. Then (μ, z, w) would satisfy $\Phi_{n,s,A} \wedge \Gamma$, contradicting the resolution refutation. Thus every completion of $\lambda|_S$ lies in $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}$, so $\lambda|_S$ is a 1-certificate at λ. The bounds follow from Theorems 4.1 and 4.3. □

Remark 4.19 (Where auxiliary encodings still escape)**.** Theorem 4.18 allows auxiliary variables and mixed clauses, but only under a semantic locality invariant: once the labels in S are fixed, the auxiliary constraints remain satisfiable for every completion of the other labels. A proof system that uses auxiliary variables to encode global information about the unexposed labels need not satisfy this invariant. Proving that such a system admits a support-local normalization is exactly the stronger proof-system problem left open here.

Theorem 4.20 (Support-local extension resolution)**.** *Let λ satisfy $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda) = 1$, and let $\Phi_{n,s,A}$ be an easy-completion CNF. Fix $S \subseteq U_n$. Let C be an acyclic Boolean extension circuit whose input variables are the formula-label variables $Y_S = \{Y_u : u \in S\}$, and let W be its extension gates. Let $E_C(Y_S, W)$ be the standard CNF encoding of the extension definitions, so every assignment to Y_S has a satisfying extension $w = C(Y_S)$. Let $\Theta(W)$ be a CNF over extension variables only, and suppose $\Theta(C(\lambda|_S)) = 1$. If there is a resolution refutation of*

$$\Phi_{n,s,A}(Y, Z) \wedge E_C(Y_S, W) \wedge \Theta(W),$$

then, writing

$$S_b := \{u \in S : \lambda(u) = b\},$$

we have, for each $b \in \{0,1\}$,

$$g_n + c\,\mathrm{Lookup}(M_n, |S_b|, \ell_n) + cM_n > s.$$

In the near-threshold regime of Theorem 4.3,

$$|S_0|, |S_1| \geq n^{a-o(1)}.$$

Proof. Let $\mu : U_n \to \{0,1\}$ be any formula labeling with $\mu|_S = \lambda|_S$. Because C reads only Y_S, the extension value $w = C(\mu|_S)$ equals $C(\lambda|_S)$, and hence satisfies both $E_C(\mu|_S, w)$ and $\Theta(w)$. Therefore $E_C \wedge \Theta$ is support-local at λ over S. Applying Theorem 4.18 to $\Gamma(Y, W) = E_C(Y_S, W) \wedge \Theta(W)$ shows that $\lambda|_S$ is a completion-sound formula-label certificate. The claimed inequalities are exactly the formula-footprint and near-threshold bounds. □

Remark 4.21 (Scope of the extension-resolution statement)**.** Theorem 4.20 is an extended-resolution-style instantiation of the support-local bridge, not a lower bound for unrestricted Extended Frege, unrestricted extended resolution, or algebraic proof systems. Its extension definitions may compute derived predicates, but only from the fixed formula-label support S, and its final extension clauses must be stable once $\lambda|_S$ is fixed. A proof system whose extension variables depend on nonlocal semantic information still requires a separate normalizer.

Definition 4.22 (Dependency-cone extension package)**.** Let C be an acyclic Boolean extension circuit with formula-label inputs $Y = \{Y_u : u \in U_n\}$ and extension variables W. A *dependency-cone annotation* assigns to every label variable Y_u the cone $\Delta(Y_u) = \{u\}$, and to every extension variable $w \in W$ a set $\Delta(w) \subseteq U_n$, such that if w is defined from earlier variables $v_1, \ldots, v_r$, then

$$\Delta(w) \supseteq \Delta(v_1) \cup \cdots \cup \Delta(v_r).$$

For a CNF Θ over label and extension variables, its *declared final support* is

$$S_\Theta := \bigcup_{X \text{ occurs in } \Theta} \Delta(X).$$

Theorem 4.23 (Dependency-cone extension normalizer)**.** *Let* λ *satisfy* $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda) = 1$*, and let* $\Phi_{n,s,A}$ *be an easy-completion CNF. Let* C *be an acyclic Boolean extension circuit with a dependency-cone annotation* Δ*, let* $E_C(Y, W)$ *be the standard CNF encoding of its extension definitions, and let* $\Theta(Y, W)$ *be a CNF over label and extension variables. Suppose the extension assignment obtained by evaluating* C *on* λ *satisfies* Θ*. If there is a resolution refutation of*

$$\Phi_{n,s,A}(Y, Z) \wedge E_C(Y, W) \wedge \Theta(Y, W),$$

then, writing

$$S_{\Theta,b} := \{u \in S_\Theta : \lambda(u) = b\},$$

we have, for each $b \in \{0,1\}$,

$$g_n + c\,\mathrm{Lookup}(M_n, |S_{\Theta,b}|, \ell_n) + cM_n > s.$$

In the near-threshold regime of Theorem 4.3,

$$|S_{\Theta,0}|, |S_{\Theta,1}| \geq n^{a-o(1)}.$$

In particular, any proof system whose accepted extension proof π *supplies such a cone annotation with* $|S_\Theta| \leq B(|\pi|)$ *has a support-local normalization with budget* B*.*

Proof. Let $\mu : U_n \to \{0,1\}$ be any formula labeling with $\mu|_{S_\Theta} = \lambda|_{S_\Theta}$. We first prove by induction over the acyclic extension order that every variable X with $\Delta(X) \subseteq S_\Theta$ has the same value under evaluation on μ as under evaluation on λ. This is immediate for label variables because their cones are singletons. For an extension variable w, every input variable to the definition of w has its cone contained in $\Delta(w)$. If $\Delta(w) \subseteq S_\Theta$, the induction hypothesis gives equality of all input values, so the value of w is also the same under μ and λ.

Thus the evaluation assignment $w = C(\mu)$ satisfies $\Theta(\mu, w)$, because $C(\lambda)$ satisfies $\Theta(\lambda, C(\lambda))$. The extension clauses $E_C(\mu, w)$ are satisfied by construction. Hence $E_C \wedge \Theta$ is support-local at λ over S_Θ. Applying Theorem 4.18 to $\Gamma(Y, W) = E_C(Y, W) \wedge \Theta(Y, W)$ yields that $\lambda|_{S_\Theta}$ is a completion-sound formula-label certificate, and Theorems 4.1 and 4.3 give the polarity lower bounds. The final sentence is exactly Theorem 4.54 with the extracted support S_Θ. □

Remark 4.24 (What cone annotations buy)**.** Theorem 4.23 is still not an unrestricted extended-resolution lower bound. It says that a proof artifact becomes usable by the formula-footprint theorem once every extension variable exposes a syntactic upper bound on the formula labels it can semantically depend on. This is a normalizer one step closer to ordinary proof objects: the support is not chosen in advance, but extracted from the declared dependency cones of the final contradiction.

Corollary 4.25 (Canonical syntactic-cone normalizer)**.** *Let λ, $\Phi_{n,s,A}$, C, E_C, and Θ be as in Theorem 4.23, but assume no external cone annotation is supplied. Define the canonical syntactic cone $\mathrm{Syn}_C(X) \subseteq U_n$ by*

$$\mathrm{Syn}_C(Y_u) = \{u\}, \qquad \mathrm{Syn}_C(w) = \mathrm{Syn}_C(v_1) \cup \cdots \cup \mathrm{Syn}_C(v_r)$$

whenever the extension variable w is defined from earlier variables $v_1, \ldots, v_r$. Let

$$S^{\mathrm{syn}}_\Theta := \bigcup_{X \text{ occurs in } \Theta} \mathrm{Syn}_C(X).$$

If the extension assignment obtained by evaluating C on λ satisfies Θ, and there is a resolution refutation of

$$\Phi_{n,s,A}(Y, Z) \wedge E_C(Y, W) \wedge \Theta(Y, W),$$

then, writing

$$S^{\mathrm{syn}}_{\Theta,b} := \{u \in S^{\mathrm{syn}}_\Theta : \lambda(u) = b\},$$

we have, for each $b \in \{0,1\}$,

$$g_n + c\,\mathrm{Lookup}(M_n, |S^{\mathrm{syn}}_{\Theta,b}|, \ell_n) + cM_n > s.$$

In the near-threshold regime of Theorem 4.3,

$$|S^{\mathrm{syn}}_{\Theta,0}|, |S^{\mathrm{syn}}_{\Theta,1}| \geq n^{a-o(1)}.$$

In particular, any explicit acyclic extension proof whose final syntactic-cone support satisfies $|S^{\mathrm{syn}}_\Theta| \leq B(|\pi|)$ has a support-local normalization with budget B.

Proof. The recursively defined sets $\mathrm{Syn}_C(X)$ form a valid dependency-cone annotation: label variables have singleton cones, and every extension variable's cone contains the cones of its defining inputs with equality. Applying Theorem 4.23 to this canonical annotation gives the stated support S^{syn}_Θ, the two polarity inequalities, and the near-threshold bounds. The final sentence is the support-local normalization clause of Theorem 4.23 with the canonical support. □

Remark 4.26 (Scope of the canonical cone normalizer)**.** Theorem 4.25 removes the need for separate cone annotations: the cone is read from the extension-definition graph itself. It still does not prove a lower bound for unrestricted extended resolution, Extended Frege, Frege, IPS, or implicit proofs. If the final contradiction mentions variables with huge syntactic cones, or if a stronger proof system compresses nonlocal semantic information in a way not represented by acyclic label-dependency syntax, the bound may be vacuous. The theorem's payoff is therefore precise: ordinary acyclic extension syntax is enough for normalization, but a selective proof-system consequence still needs a nontrivial syntactic-cone support bound.

Corollary 4.27 (Final backward-slice lower bound)**.** *In the setting of Theorem 4.25, let* $\mathrm{Slice}_C(\Theta)$ *be the backward slice of the acyclic extension-definition graph containing every label or extension variable that reaches a variable occurring in* Θ*. Let*

$$\partial_Y \mathrm{Slice}_C(\Theta) := \{u \in U_n : Y_u \text{ occurs as a label input in } \mathrm{Slice}_C(\Theta)\}.$$

If the extension assignment induced by λ *satisfies* Θ*, and*

$$\Phi_{n,s,A}(Y,Z) \wedge E_C(Y,W) \wedge \Theta(Y,W)$$

has a resolution refutation, then, for each $b \in \{0,1\}$*,*

$$g_n + c\,\mathrm{Lookup}(M_n, |\partial_Y \mathrm{Slice}_C(\Theta) \cap \lambda^{-1}(b)|, \ell_n) + cM_n > s.$$

In the near-threshold regime of Theorem 4.3,

$$|\partial_Y \mathrm{Slice}_C(\Theta) \cap \lambda^{-1}(0)|,\ |\partial_Y \mathrm{Slice}_C(\Theta) \cap \lambda^{-1}(1)| \geq n^{a-o(1)}.$$

Consequently, any explicit acyclic extension proof representation whose length counts the distinct formula-label inputs appearing in the final backward slice has length at least $2n^{a-o(1)}$ *in this regime.*

Proof. By construction, $\partial_Y \mathrm{Slice}_C(\Theta) = S^{\mathrm{syn}}_\Theta$: a label u lies in the final backward slice exactly when there is a directed path in the extension-definition graph from Y_u to some variable occurring in Θ, which is exactly the recursive definition of the canonical syntactic cone union. The two polarity inequalities and near-threshold lower bounds are therefore Theorem 4.25. The final statement follows because the proof representation is assumed to count each distinct formula-label input in the backward slice at least once, and the two polarity sets are disjoint. □

Remark 4.28 (What the slice lower bound does and does not say)**.** Theorem 4.27 is the proof-object budget form of the canonical syntactic-cone normalizer. It applies to explicit acyclic extension syntax and says that the final contradiction cannot be supported by a small local backward slice. It is still not an unrestricted extended-resolution, Extended Frege, Frege, or IPS lower bound: a stronger proof system can escape the statement if the relevant semantic information is compressed outside such an explicit acyclic label-dependency slice, or if the final backward slice is itself large.

Definition 4.29 (Slice-explicit extension refutation)**.** A *slice-explicit extension refutation* of the easy-completion CNF consists of:

1. an explicit acyclic extension circuit C over formula-label inputs $Y = \{Y_u : u \in U_n\}$;
2. its CNF definition $E_C(Y,W)$;
3. a final CNF $\Theta(Y,W)$;

4. a resolution refutation of $\Phi_{n,s,A}(Y,Z) \wedge E_C(Y,W) \wedge \Theta(Y,W)$.

Its slice size $\mathrm{size}_{\mathrm{slice}}(\pi)$ is any proof-size measure that is at least the number of distinct formula-label inputs in $\partial_Y \mathrm{Slice}_C(\Theta)$.

Corollary 4.30 (Slice-explicit extension-resolution lower bound)**.** *Let λ satisfy $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda) = 1$, and let π be a slice-explicit extension refutation whose final CNF Θ is satisfied by evaluating its extension circuit C on λ. Then, for each $b \in \{0,1\}$,*

$$g_n + c\,\mathrm{Lookup}(M_n, |\partial_Y \mathrm{Slice}_C(\Theta) \cap \lambda^{-1}(b)|, \ell_n) + cM_n > s.$$

In the near-threshold regime of Theorem 4.3,

$$\mathrm{size}_{\mathrm{slice}}(\pi) \geq 2n^{a-o(1)}.$$

Proof. Theorem 4.27 gives the two polarity lower bounds for $\partial_Y \mathrm{Slice}_C(\Theta)$. In the near-threshold regime, both polarity subsets have size at least $n^{a-o(1)}$, and they are disjoint. By the definition of slice-explicit size, $\mathrm{size}_{\mathrm{slice}}(\pi)$ is at least the number of distinct formula-label inputs in their union, so it is at least $2n^{a-o(1)}$. □

Remark 4.31 (Restricted ER-style payoff)**.** Theorem 4.30 is a restricted extended-resolution-style lower bound for proof representations that explicitly charge the formula-label boundary of the final extension slice. It is more recognizable than an abstract support-local normalizer, because the measured object is read from an ordinary acyclic extension-definition graph. It is still not an unrestricted extended-resolution, Extended Frege, Frege, or IPS lower bound: the theorem assumes the final slice is explicitly represented and charged.

Corollary 4.32 (Displayed acyclic ER length lower bound)**.** *Let λ satisfy $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda) = 1$, and let π be a slice-explicit extension refutation as in Theorem 4.29. Assume that every extension definition is displayed by CNF clauses in E_C with its operand variables written explicitly, and define the displayed extension-object length $\mathrm{len}_{\mathrm{disp}}(\pi)$ to be the number of literal occurrences in E_C and Θ, plus the literal occurrences in the resolution derivation. If the final CNF Θ is satisfied by evaluating C on λ, then, in the near-threshold regime of Theorem 4.3,*

$$\mathrm{len}_{\mathrm{disp}}(\pi) \geq 2n^{a-o(1)}.$$

Proof. Theorem 4.27 gives

$$|\partial_Y \mathrm{Slice}_C(\Theta) \cap \lambda^{-1}(0)|,\ |\partial_Y \mathrm{Slice}_C(\Theta) \cap \lambda^{-1}(1)| \geq n^{a-o(1)}.$$

It remains to compare this boundary with ordinary displayed length. If $u \in \partial_Y \mathrm{Slice}_C(\Theta)$, then Y_u lies on a directed path to a variable occurring in Θ. If the path has length 0, then Y_u itself occurs in Θ. Otherwise, let X be the first extension variable on such a path. The definition of X has Y_u as a displayed operand, so the CNF clauses defining X in E_C contain a literal over Y_u. Thus every distinct formula-label input in the final backward slice contributes at least one displayed literal occurrence in $E_C \cup \Theta$. The two polarity subsets are disjoint, so the displayed extension-object length is at least their sum, which is at least $2n^{a-o(1)}$. □

Remark 4.33 (Displayed length versus unrestricted ER)**.** Theorem 4.32 removes the separate slice-size charging convention for explicitly written acyclic extension objects: ordinary displayed literal length already pays for every formula-label input in the final backward slice. The boundary is now the standard one for compressed proof systems. If a proof format hides extension definitions

behind macros, encodes label identities implicitly, or supplies nonlocal semantic information without a displayed acyclic dependency graph, this corollary does not apply. In particular, it is not a lower bound for Frege, Extended Frege, IPS, implicit proofs, or unrestricted ER encodings whose effective dependency support is not visible in the displayed clauses.

Corollary 4.34 (Expansion-bounded compressed ER transfer)**.** *Let λ satisfy $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda) = 1$. Let ρ be a compressed acyclic extension refutation that expands, by a sound decompression map* Exp*, to a displayed acyclic extension refutation $\pi = \mathrm{Exp}(\rho)$ satisfying the hypotheses of Theorem 4.32. Suppose the displayed expansion length is bounded by a monotone function B of compressed length:*

$$\mathrm{len}_{\mathrm{disp}}(\mathrm{Exp}(\rho)) \le B(|\rho|).$$

Then, in the near-threshold regime of Theorem 4.3,

$$B(|\rho|) \ge 2n^{a-o(1)}.$$

In particular, if $B(L) \le CL^k$ for constants C, k, then every such compressed refutation has

$$|\rho| \ge C^{-1/k} 2^{1/k} n^{a/k-o(1)}.$$

Proof. The expanded object $\pi = \mathrm{Exp}(\rho)$ is a displayed acyclic extension refutation satisfying the hypotheses of Theorem 4.32. Therefore

$$\mathrm{len}_{\mathrm{disp}}(\mathrm{Exp}(\rho)) \ge 2n^{a-o(1)}.$$

Combining this with the assumed expansion upper bound gives $B(|\rho|) \ge 2n^{a-o(1)}$. If $B(L) \le CL^k$, rearranging $C|\rho|^k \ge 2n^{a-o(1)}$ yields the stated compressed-length lower bound. □

Remark 4.35 (What compression must prove)**.** Theorem 4.34 is not a lower bound for arbitrary compressed proofs. It says that compression helps only by making the displayed expansion large, by hiding the relevant dependency support, or by leaving the displayed acyclic ER model. Thus a future proof-system theorem must either prove a controlled expansion/support budget for the compressed representation, or explain why its proof object is genuinely outside the local completion models used here.

Corollary 4.36 (Macro-compressed ER reuse tradeoff)**.** *Let λ satisfy $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda) = 1$. Suppose a macro-compressed acyclic extension refutation ρ consists of:*

1. *a displayed skeleton of literal length S;*
2. *a displayed macro library of total literal length D;*
3. *r macro invocations, each expanding to a library body;*
4. *a sound expansion* Exp(ρ) *to a displayed acyclic extension refutation satisfying the hypotheses of Theorem 4.32.*

Then, in the near-threshold regime of Theorem 4.3,

$$S + rD \ge 2n^{a-o(1)}.$$

Consequently, if the encoded length L of ρ satisfies $S, D, r \le L$, then

$$L^2 + L \ge 2n^{a-o(1)}, \qquad \text{and hence} \qquad L \ge n^{a/2-o(1)}.$$

Proof. The displayed expansion consists of the skeleton plus, for each of the r invocations, one displayed macro body from a library whose total body length is D. Hence

$$\mathrm{len}_{\mathrm{disp}}(\mathrm{Exp}(\rho)) \le S + rD.$$

By Theorem 4.32, the same displayed expansion has length at least $2n^{a-o(1)}$. Thus $S+rD \ge 2n^{a-o(1)}$. If $S, D, r \le L$, then $S + rD \le L + L^2$, giving $L^2 + L \ge 2n^{a-o(1)}$. The final lower bound follows by taking square roots and absorbing constants and the lower-order exponent loss. □

Remark 4.37 (Macro reuse is the remaining escape)**.** Theorem 4.36 is the concrete macro-library version of Theorem 4.34. It does not rule out proof systems whose macros have nonlocal semantics rather than displayed acyclic expansions, nor systems where a small proof denotes an exponentially large object without a controlled expansion accounting. It says that ordinary macro sharing can compress the displayed proof only by paying a visible reuse product rD.

Corollary 4.38 (Support-declared macro ER normalizer)**.** *Let λ satisfy $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda) = 1$, and let $\Phi_{n,s,A}$ be an easy-completion CNF. Suppose a macro-compressed extension refutation ρ supplies:*

1. *a compressed macro or extension constraint package $\Gamma_\rho(Y, W)$, representing its skeleton, macro library, macro invocations, and final clauses;*
2. *a resolution refutation of $\Phi_{n,s,A}(Y, Z) \wedge \Gamma_\rho(Y, W)$;*
3. *a declared formula-label support $S_\rho \subseteq U_n$;*
4. *the support-soundness invariant that, for every formula labeling μ with $\mu|_{S_\rho} = \lambda|_{S_\rho}$, there is an auxiliary assignment w_μ satisfying $\Gamma_\rho(\mu, w_\mu)$.*

Write $S_{\rho,b} := \{u \in S_\rho : \lambda(u) = b\}$. Then, for each $b \in \{0, 1\}$,

$$g_n + c\,\mathrm{Lookup}(M_n, |S_{\rho,b}|, \ell_n) + cM_n > s.$$

In the near-threshold regime of Theorem 4.3,

$$|S_{\rho,0}|, |S_{\rho,1}| \ge n^{a-o(1)}.$$

Consequently, if the support declaration has budget $|S_\rho| \le B(|\rho|)$, then

$$B(|\rho|) \ge 2n^{a-o(1)}.$$

Proof. The support-soundness invariant says exactly that Γ_ρ is support-local at λ over S_ρ. Since $\Phi_{n,s,A} \wedge \Gamma_\rho$ has a resolution refutation, Theorem 4.18 applies with $S = S_\rho$. Hence $\lambda|_{S_\rho}$ is a completion-sound formula-label 1-certificate, and Theorems 4.1 and 4.3 give the two polarity inequalities and near-threshold lower bounds. The budget statement follows from

$$B(|\rho|) \ge |S_\rho| \ge |S_{\rho,0}| + |S_{\rho,1}| \ge 2n^{a-o(1)}.$$

□

Remark 4.39 (Support declarations are an obligation)**.** Theorem 4.38 is not a lower bound for arbitrary compressed proof systems. It gives the support-budget counterpart to Theorem 4.36: a compressed macro proof may avoid displaying its whole expansion, but if it claims a local support summary, that summary must itself be large near threshold. The theorem does not apply to macros with unsound declarations, hidden nonlocal semantics, or proof systems whose compressed witnesses affect unexposed labels without a separate support-local normalizer.

Corollary 4.40 (Support-audited LRAT trace normalizer)**.** *Let λ satisfy $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda) = 1$, and let $\Phi_{n,s,A}$ be an easy-completion CNF. Suppose an LRAT-style checked clausal trace τ certifies unsatisfiability of*

$$\Phi_{n,s,A}(Y, Z) \wedge \Gamma_\tau(Y, W),$$

where Γ_τ is the collection of checker-visible auxiliary, macro, or support clauses supplied with the trace. Assume that the trace also carries a support audit: a declared set $S_\tau \subseteq U_n$ and a verified invariant saying that, for every formula labeling μ with $\mu|_{S_\tau} = \lambda|_{S_\tau}$, there is an auxiliary assignment w_μ satisfying $\Gamma_\tau(\mu, w_\mu)$. Write $S_{\tau,b} := \{u \in S_\tau : \lambda(u) = b\}$. Then, for each $b \in \{0, 1\}$,

$$g_n + c\,\mathrm{Lookup}(M_n, |S_{\tau,b}|, \ell_n) + cM_n > s.$$

In the near-threshold regime of Theorem 4.3,

$$|S_{\tau,0}|, |S_{\tau,1}| \geq n^{a-o(1)}.$$

Consequently, if the trace or its checker-visible annotation has support budget $|S_\tau| \leq B(|\tau|)$, then

$$B(|\tau|) \geq 2n^{a-o(1)}.$$

Proof. LRAT soundness gives an ordinary clausal refutation of $\Phi_{n,s,A} \wedge \Gamma_\tau$. The verified support audit says exactly that Γ_τ is support-local at λ over S_τ. Applying Theorem 4.18 with $S = S_\tau$ shows that $\lambda|_{S_\tau}$ is a completion-sound formula-label certificate. The polarity inequalities and near-threshold bounds follow from Theorems 4.1 and 4.3; the final budget inequality follows from $|S_\tau| \geq |S_{\tau,0}| + |S_{\tau,1}|$. □

Remark 4.41 (LRAT without a support audit is outside the theorem)**.** Theorem 4.40 is a checker-facing version of the support-declared macro normalizer. It does not prove a lower bound for arbitrary LRAT or DRAT traces. The new object is the support audit: without a sound, independently checkable reason why all completions preserving S_τ satisfy the auxiliary trace constraints, the formula-footprint theorem has no local object to count.

Corollary 4.42 (Provenance-typed LRAT support audit)**.** *Let λ satisfy $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda) = 1$, and let $\Phi_{n,s,A}$ be an easy-completion CNF. Suppose an LRAT-style checked trace, or a DRAT trace accompanied by an LRAT certificate, refutes*

$$\Phi_{n,s,A}(Y, Z) \wedge \bigwedge_{j \in J} \Gamma_j(Y, W_j).$$

Assume that the trace carries a trusted provenance package with the following data:

1. *a variable-role map identifying the formula-label variables Y_u and the auxiliary variables;*
2. *pairwise disjoint auxiliary blocks W_j;*
3. *a declared formula-label support $P_j \subseteq U_n$ for each auxiliary block Γ_j;*
4. *a checked local witness invariant saying that, for every formula labeling μ with $\mu|_{P_j} = \lambda|_{P_j}$, there is an assignment $w_j(\mu)$ satisfying $\Gamma_j(\mu, w_j(\mu))$;*
5. *a provenance rule that every derived clause is assigned the union of the provenance sets of its antecedents, with final declared support $S_\tau := \bigcup_{j \in J} P_j$.*

Write $S_{\tau,b} := \{u \in S_\tau : \lambda(u) = b\}$. *Then, for each* $b \in \{0,1\}$,

$$g_n + c\,\mathrm{Lookup}(M_n, |S_{\tau,b}|, \ell_n) + cM_n > s.$$

In the near-threshold regime of Theorem 4.3,

$$|S_{\tau,0}|, |S_{\tau,1}| \geq n^{a-o(1)}.$$

Consequently, if the trace or checker-visible provenance annotation has support budget $|S_\tau| \leq B(|\tau|)$, *then*

$$B(|\tau|) \geq 2n^{a-o(1)}.$$

Proof. It is enough to show that the provenance package supplies the support audit required by Theorem 4.40. Let μ be any formula labeling with $\mu|_{S_\tau} = \lambda|_{S_\tau}$. Since $P_j \subseteq S_\tau$ for every j, the same μ agrees with λ on every P_j. The local witness invariant gives an assignment $w_j(\mu)$ satisfying $\Gamma_j(\mu, w_j(\mu))$. Because the auxiliary blocks W_j are pairwise disjoint, these assignments combine into one assignment w_μ satisfying all clauses in $\bigwedge_j \Gamma_j(Y, W_j)$. Thus the whole auxiliary package is support-local at λ over S_τ.

LRAT soundness, or the supplied LRAT certificate for the DRAT trace, gives the required clausal refutation. Applying Theorem 4.40 yields the polarity inequalities, the near-threshold support lower bounds, and the final budget inequality. □

Remark 4.43 (Typed provenance versus raw traces)**.** Theorem 4.42 closes the trusted-provenance route, not the raw-trace route. It does not extract formula-label support from untyped LRAT or DRAT syntax. If auxiliary blocks share variables, a separate compatibility proof or global witness constructor must replace the disjoint-block assumption. Without such typed provenance or compatibility data, Theorem 4.48 still applies.

Corollary 4.44 (Namespace-typed LRAT normalizer)**.** *Let* λ *satisfy* $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda) = 1$, *and let* $\Phi_{n,s,A}$ *be an easy-completion CNF. Suppose an LRAT-style checked trace refutes*

$$\Phi_{n,s,A}(Y, Z) \wedge \Gamma(Y, W),$$

where the input uses a trusted syntactic namespace distinguishing formula-label variables Y_u *from auxiliary variables* W. *Let* $\mathcal{A}_\tau$ *be the LRAT antecedent closure of the final empty clause, and let* Γ_τ *be the original auxiliary clauses from* Γ *that occur in* $\mathcal{A}_\tau$. *Define*

$$S_\tau := \{u \in U_n : Y_u \text{ occurs in some clause of } \Gamma_\tau\}.$$

Assume there is an auxiliary assignment w_λ *satisfying* $\Gamma_\tau(\lambda, w_\lambda)$. *Write* $S_{\tau,b} := \{u \in S_\tau : \lambda(u) = b\}$. *Then, for each* $b \in \{0,1\}$,

$$g_n + c\,\mathrm{Lookup}(M_n, |S_{\tau,b}|, \ell_n) + cM_n > s.$$

In the near-threshold regime of Theorem 4.3,

$$|S_{\tau,0}|, |S_{\tau,1}| \geq n^{a-o(1)}.$$

Consequently, if the trace length or checker-visible namespace accounting bounds $|S_\tau| \leq B(|\tau|)$, *then*

$$B(|\tau|) \geq 2n^{a-o(1)}.$$

Proof. Clauses outside the LRAT antecedent closure of the empty clause are unused by the checked derivation, so the same trace restricted to $\mathcal{A}_\tau$ gives an LRAT refutation of $\Phi_{n,s,A} \wedge \Gamma_\tau$. It remains to show that S_τ is a sound support audit for Γ_τ.

Let μ be any formula labeling with $\mu|_{S_\tau} = \lambda|_{S_\tau}$. Every formula-label variable that occurs in Γ_τ is indexed by an element of S_τ. Therefore every label literal in $\Gamma_\tau(\mu, w_\lambda)$ has the same truth value as in $\Gamma_\tau(\lambda, w_\lambda)$, while the auxiliary assignment is unchanged. Since w_λ satisfies $\Gamma_\tau(\lambda, w_\lambda)$, the same w_λ satisfies $\Gamma_\tau(\mu, w_\lambda)$. Thus Γ_τ is support-local at λ over S_τ.

The support-audited LRAT theorem *Theorem* 4.40 now applies to the restricted trace and support S_τ, giving the polarity inequalities, near-threshold support bounds, and final budget lower bound. □

Remark 4.45 (Namespace typing is weaker than full provenance but stronger than raw syntax)**.** Theorem 4.44 removes the need for a separate derived-clause provenance annotation in the special case where the proof log has a trusted formula-label namespace and the used auxiliary clauses have a hard witness. It still does not make plain untyped LRAT or DRAT a support normalizer: erasing the namespace returns to the ambiguity in Theorem 4.48, and clauses whose auxiliary part has no hard witness still require a separate support-locality proof.

Proposition 4.46 (Support-audit necessity separator)**.** *Let $T \subseteq U_n$ be finite and let $\alpha : T \to \{0,1\}$. Introduce formula-label variables $\{Y_u : u \in T\}$. Let Γ_T be the unit CNF fixing each Y_u to $\alpha(u)$, and let Φ_T be the single blocking clause falsified exactly by that joint assignment:*

$$\Phi_T := \bigvee_{u \in T : \alpha(u)=1} \neg Y_u \ \vee \bigvee_{u \in T : \alpha(u)=0} Y_u.$$

Then $\Phi_T \wedge \Gamma_T$ has an LRAT-style clausal refutation of length $O(|T|)$. However, if a declared support $S \subseteq T$ is sound in the support-local sense that every labeling μ with $\mu|_S = \alpha|_S$ satisfies $\Gamma_T(\mu)$, then $T \subseteq S$.

Proof. Resolve the blocking clause successively against the unit clause for each $u \in T$. Each resolution step removes the corresponding literal, so after $|T|$ steps one derives the empty clause. This derivation is a standard linear clausal refutation and can be written with LRAT-style antecedent hints.

For the support claim, suppose $u \in T \setminus S$. Define μ to agree with α on S, to flip $\alpha(u)$ at u, and arbitrarily on $T \setminus (S \cup \{u\})$. Then $\mu|_S = \alpha|_S$, but the unit clause in Γ_T fixing Y_u to $\alpha(u)$ is false under μ. Since Γ_T has no auxiliary variables, no witness can repair this failure. Hence a sound support declaration must contain u. As u was arbitrary, $T \subseteq S$. □

Remark 4.47 (What the separator says)**.** Theorem 4.46 does not prove a lower bound for unaudited LRAT or DRAT. It proves the narrower boundary needed by Theorem 4.40: clausal refutation checking and completion-local support checking are different obligations. A short trace may certify unsatisfiability while every sound support audit has to expose the whole set of labels fixed by the auxiliary constraints.

Proposition 4.48 (Support-erasure indistinguishability separator)**.** *Let T be finite and let $\alpha : T \to \{0,1\}$. Let C_T be the clausal skeleton over variables $\{P_u : u \in T\}$ consisting of the unit clauses fixing each P_u to $\alpha(u)$ and the single blocking clause falsified exactly by that joint assignment. The same LRAT-style linear resolution trace refutes C_T under both of the following semantic lifts.*

In the auxiliary lift, the variables P_u are auxiliary variables unrelated to formula-label coordinates. The constraint package has empty formula-label support: for every formula labeling μ, setting $P_u = \alpha(u)$ satisfies the unit constraints.

In the label lift, the variables P_u are the formula-label variables Y_u for $u \in T$. Then every sound support declaration for the unit constraints must contain T. Consequently the untyped clausal skeleton and its LRAT antecedent hints do not determine the completion-local formula-label support.

Proof. The syntactic refutation is the same in both lifts. Resolve the blocking clause successively against the unit clauses fixing P_u to $\alpha(u)$; after all variables in T have been resolved away, the empty clause is derived. This gives one LRAT-style trace with the same clauses and antecedent hints.

For the auxiliary lift, the variables P_u are witnesses. No formula-label coordinate is constrained: for every labeling of the formula variables, choose the auxiliary assignment $P_u = \alpha(u)$. Thus the unit part of the trace is support-local over the empty formula-label support.

For the label lift, identify P_u with Y_u. If a declared support $S \subseteq T$ omits some u, choose a completion μ agreeing with α on S and flipping u. The unit clause fixing Y_u to $\alpha(u)$ is then false, with no auxiliary witness able to repair it. Hence every sound support declaration contains all of T.

The raw clausal refutation is identical in the two cases; only the erased role map saying which variables are formula labels and which are auxiliaries differs. Therefore support locality is not certified by untyped LRAT syntax alone. □

Remark 4.49 (Support provenance)**.** Theorem 4.48 is not an impossibility theorem for every unannotated clausal proof. It says that a trace checker which sees only untyped clauses and antecedent hints has not checked the semantic support object needed by Theorem 4.40. A trusted variable-role map, clause provenance, or support-locality checker supplies extra information; that extra information is precisely the support audit being charged.

Proposition 4.50 (Plain LRAT extraction trichotomy)**.** *Call an extractor* plain *if its input is only the untyped clausal trace: clause strings together with LRAT antecedent hints, but no variable-role map, clause provenance, support declaration, or support-locality checker. Suppose such an extractor returns a set $S(\tau)$ of variables and claims that, whenever τ is accepted, $S(\tau)$ is a sound completion-local formula-label support for the semantic instance represented by the trace.*

For the unit-plus-blocking skeleton C_T from Theorem 4.48, exactly one of the following must occur.

1. *The extractor uses extra semantic information, such as a variable-role map, clause provenance, or support-locality checker. Then it is not plain; it has supplied the additional audit object charged in Theorems 4.40 and 4.42.*

2. *The extractor is sound for every semantic lift of the raw trace. Then its output on C_T contains all variables in T.*

3. *The extractor outputs a strict subset of T from the raw trace alone. Then the label lift in Theorem 4.48 falsifies its claimed support soundness.*

Consequently, plain unaudited LRAT or DRAT syntax alone does not give a nontrivial support-extraction normalizer for the present formula-footprint theorem. A positive proof-system application must either charge provenance or prove a separate system-specific invariant that rules out the bad semantic lift.

Proof. The auxiliary and label lifts in Theorem 4.48 have the same untyped clauses and the same LRAT antecedent hints. A plain extractor therefore receives the same input in both lifts and must return the same set $S(\tau)$.

In the label lift, the variables of the skeleton are the formula-label variables Y_u. By the support claim in Theorem 4.46, every sound completion-local support for the unit constraints fixing those labels must contain T. Hence, if $S(\tau)$ is sound for every semantic lift, $T \subseteq S(\tau)$.

If instead $S(\tau)$ omits some $u \in T$, choose the label lift and flip Y_u while preserving all coordinates in $S(\tau)$, as in the proof of Theorem 4.46. The corresponding unit clause is falsified and no auxiliary witness repairs it. Thus the claimed support is unsound for that semantic lift.

The only way for the extractor to return different supports on the auxiliary and label lifts is to use information not present in the raw trace, namely a role map, provenance, support declaration, or an equivalent semantic invariant. That is the first case and is precisely the additional object charged by the audited/provenance-typed LRAT theorems. □

Remark 4.51 (What remains after the trichotomy)**.** Theorem 4.50 is a no-free-normalizer statement, not a lower bound for every LRAT or DRAT refutation. It closes only the route that tries to read nontrivial formula-label support from raw untyped trace syntax. A proof format may still enter the theorem package by supplying trusted provenance, by making support-locality checker-visible, or by proving a system-specific invariant that prevents the ambiguous semantic lift.

Corollary 4.52 (Cone-budget extension lower bound)**.** *In the setting of Theorem 4.23, suppose the final CNF Θ mentions at most q distinct label or extension variables, and every mentioned variable X has declared cone size $|\Delta(X)| \le d$. Then*

$$qd \ge |S_\Theta| \ge |S_{\Theta,0}| + |S_{\Theta,1}|.$$

In the near-threshold regime of Theorem 4.3,

$$qd \ge 2n^{a-o(1)}.$$

Consequently a cone-annotated extension proof with $q \le L$ final variables and cone bound $d \le D(L)$ must satisfy

$$L\,D(L) \ge 2n^{a-o(1)}.$$

Proof. The support S_Θ is the union of at most q cones, each of size at most d, so $|S_\Theta| \le qd$. The sets $S_{\Theta,0}$ and $S_{\Theta,1}$ are disjoint subsets of S_Θ, and Theorem 4.23 gives the near-threshold lower bound $|S_{\Theta,0}|, |S_{\Theta,1}| \ge n^{a-o(1)}$. The final inequality follows by substituting $q \le L$ and $d \le D(L)$. □

Corollary 4.53 (Bounded-depth extension tradeoff)**.** *In the setting of Theorem 4.52, suppose every extension definition has fan-in at most r, and every label or extension variable mentioned in the final CNF Θ has extension depth at most h, where label variables have depth 0. If Θ mentions at most q distinct label or extension variables, then in the near-threshold regime of Theorem 4.3,*

$$q\,r^h \ge 2n^{a-o(1)}.$$

Consequently, if $q \le L$, then

$$h \ge \log_r\left(\frac{2n^{a-o(1)}}{L}\right)$$

whenever the right-hand side is positive.

Proof. A label variable has cone size $1 = r^0$. If an extension variable at depth t is computed from at most r earlier variables, and every input variable has cone size at most r^{t-1}, then its cone has size at most $r \cdot r^{t-1} = r^t$. Induction on the extension depth therefore gives $|\Delta(X)| \le r^h$ for every variable X mentioned in Θ. Apply Theorem 4.52 with $d = r^h$. The logarithmic form is just the rearranged inequality when the quotient is larger than 1. □

Definition 4.54 (Support-local normalization with support budget)**.** Fix an easy-completion CNF $\Phi_{n,s,A}$. A proof or certificate system $\mathcal{P}$ has a *support-local normalization with support budget* B for a hard labeling λ if every accepted $\mathcal{P}$-proof π of the corresponding lower-bound statement yields:

1. a support set $S_\pi \subseteq U_n$;
2. a CNF $\Gamma_\pi(Y, W)$ that is support-local at λ over S_π;
3. a sound refutation, in the underlying refutation calculus, of
$$\Phi_{n,s,A}(Y, Z) \wedge \Gamma_\pi(Y, W);$$
4. the size control $|S_\pi| \le B(|\pi|)$, where $|\pi|$ is the measured proof length.

Theorem 4.55 (Support-local normalization transfer)**.** *Let λ satisfy $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}(\lambda) = 1$. Suppose a proof or certificate system $\mathcal{P}$ has a support-local normalization with support budget B for λ. For every accepted proof π, write*
$$S_{\pi,b} := \{u \in S_\pi : \lambda(u) = b\}.$$
Then, for each $b \in \{0, 1\}$,
$$g_n + c\,\mathrm{Lookup}(M_n, |S_{\pi,b}|, \ell_n) + cM_n > s.$$
In the near-threshold regime of Theorem 4.3,
$$|S_{\pi,0}|, |S_{\pi,1}| \ge n^{a-o(1)}, \qquad B(|\pi|) \ge 2n^{a-o(1)}.$$
In particular, if $B(L) = O(L)$, then every such accepted proof has length $n^{a-o(1)}$.

Proof. Let Γ_π and S_π be supplied by the normalizer. The sound refutation of $\Phi_{n,s,A} \wedge \Gamma_\pi$ implies that this CNF is unsatisfiable. The proof of Theorem 4.18 used only this unsatisfiability, the easy-completion semantics of $\Phi_{n,s,A}$, and support-locality of Γ_π. Therefore $\lambda|_{S_\pi}$ is a 1-certificate for $\mathsf{Hard}^{\mathrm{form}}_{n,s,A}$, and Theorems 4.1 and 4.3 give the two polarity inequalities and the two near-threshold lower bounds. Since $S_{\pi,0}$ and $S_{\pi,1}$ are disjoint subsets of S_π, the support budget satisfies
$$B(|\pi|) \ge |S_\pi| \ge |S_{\pi,0}| + |S_{\pi,1}| \ge 2n^{a-o(1)}.$$
The final statement is immediate when $B(L) = O(L)$. □

Corollary 4.56 (Carrier-independent audit upgrade criterion)**.** *Let $\mathcal{P}$ be a recognizable proof or certificate system for the easy-completion lower-bound statements $\Phi_{n,s,A}$. Suppose there is a polynomial-time, proof-system-intrinsic audit extractor*
$$\mathsf{Aud}_{\mathcal{P}}(\pi, n) = (S_\pi, \Gamma_\pi, \mathcal{W}_\pi)$$
with the following properties for every accepted $\mathcal{P}$-proof π.

(i) $S_\pi \subseteq U_n$ is a set of formula-label coordinates computed from the proof object and the public proof-system syntax, not from a carrier-specific namespace manifest, hidden role advice, or semantic oracle.

(ii) $\Gamma_\pi(Y, W)$ is support-local at the hard labeling λ over S_π, with $\mathcal{W}_\pi$ certifying this support-locality by exposed controller completeness, local equivalence, dependency cones, provenance, or another system-intrinsic invariant.

(iii) The underlying checker verifies a sound refutation of

$$\Phi_{n,s,A}(Y,Z) \wedge \Gamma_\pi(Y,W).$$

(iv) A budget $B(|\pi|)$ upper-bounds the extracted support and audit data:

$$|S_\pi| + |\mathcal{W}_\pi| \le B(|\pi|).$$

Then $\mathcal{P}$ satisfies the support-local normalization hypothesis of Theorem 4.55. In particular, in the near-threshold regime of Theorem 4.3,

$$B(|\pi|) \ge 2n^{a-o(1)}.$$

If $B(L) = O(L^k)$, then every accepted proof has length

$$|\pi| \ge n^{a/k-o(1)}.$$

Proof. The data output by $\mathsf{Aud}_{\mathcal{P}}$ are exactly the four pieces in the definition of support-local normalization: a support set, a support-local CNF, a sound refutation after adjoining that CNF to the easy-completion formula, and a size budget. The additional requirement that the audit is proof-system-intrinsic records what is needed for a top-paper proof-system upgrade: the support object must be forced by the proof format itself, rather than imported through a carrier-specific manifest or hidden role annotation. Thus *Theorem 4.55* applies with the same budget B, giving $B(|\pi|) \ge 2n^{a-o(1)}$. If $B(L) = O(L^k)$, then $O(|\pi|^k) \ge 2n^{a-o(1)}$, which rearranges to the stated proof-length lower bound.

This is an upgrade criterion, not an instantiation for Frege, Extended Frege, IPS, arbitrary compressed proofs, or raw proof traces. To use it for such a system one must still prove the system-intrinsic audit extractor and its support-locality certificate. □

Corollary 4.57 (Self-auditing RUP-CDCL instantiation)**.** *Consider an enriched RUP-CDCL proof format for the easy-completion CNF $\Phi_{n,s,A}$ in which the proof object contains:*

(i) a checked, proof-internal label decoder mapping formula-label coordinates $u \in U_n$ to the Boolean variables representing them in the clausal encoding;

(ii) a live-database reconstruction log for clause additions and deletions;

(iii) for every learned or deleted-clause consequence used in the final replay, a checker-visible RUP propagation certificate over the reconstructed live database;

(iv) retained support annotations obtained by intersecting every touched clause in those RUP certificates with the decoded formula-label variables.

Assume the checker verifies that all auxiliary clauses touched by the replay are satisfiable under every completion preserving the retained support, and that the final reconstructed database refutes $\Phi_{n,s,A} \wedge \Gamma_\pi$. Then this self-auditing RUP-CDCL format instantiates the carrier-independent audit upgrade criterion with linear audit budget. Hence, in the near-threshold regime of Theorem 4.3, every accepted proof in this format has length $n^{a-o(1)}$.

Proof. The extractor is the generic checker for the enriched proof syntax. From the proof-internal decoder and live-database log it computes the formula-label variables touched by the RUP certificates used in the final replay. Their coordinate preimages form S_π. The clauses whose touched support is

contained in this set form Γ_π, while the RUP propagation certificates, live-database reconstruction, deleted-clause support records, and support-local auxiliary witnesses form $\mathcal{W}_\pi$. All of this data is read from the proof artifact itself, with no external carrier-specific namespace manifest, hidden role advice, or semantic oracle.

The soundness and support-locality assumptions are precisely the hypotheses needed by the RUP-complete and canonical-namespace corollaries, Theorems 6.18 and 6.19, except that the namespace parser is internalized into the proof format rather than supplied as a separate manifest. Therefore the extractor returns the data required by Theorem 4.56. Because every support element, propagation step, live-database event, and retained deleted-clause support record is explicitly present in the proof object, the audit budget is $B(L) = O(L)$. Applying *Theorem* 4.56 gives the near-threshold support lower bound and hence proof length $n^{a-o(1)}$.

This is not a lower bound for raw DIMACS, raw RUP/DRUP, arbitrary CDCL, Frege, Extended Frege, IPS, or an unannotated compressed proof system. Its role is to show that the carrier-independent criterion is an honest proof-format theorem once the proof system itself carries the label decoder and support audit that the separators prove cannot be erased for free. □

Corollary 4.58 (Append-only self-auditing RUP-CDCL)**.** *Consider the self-auditing RUP-CDCL format of Theorem 4.57 under the additional syntactic restriction that the reconstructed clause database is append-only: once a clause is added, it remains available until the final empty-clause replay. Then the proof object need not carry separate retained deleted-clause support records. It is enough that it contains the checked proof-internal label decoder, the append-only clause-addition log, and checker-visible RUP propagation certificates for the used learned consequences. If the checker verifies support-locality for the touched auxiliary clauses relative to the decoded formula-label support, then this deletion-free format instantiates the carrier-independent audit upgrade criterion with linear audit budget. Hence, in the near-threshold regime of Theorem 4.3, every accepted proof in this append-only self-auditing RUP-CDCL format has length $n^{a-o(1)}$.*

Proof. Run the extractor from Theorem 4.57, but compute support by a single scan over the monotone clause database and the RUP propagation certificates used in the final replay. Since no clause is deleted, every clause that can participate in a later propagation remains present in the append-only log. The support of a used clause is therefore just the set of decoded formula-label variables occurring in that clause, and no retained deleted-clause support record is needed to remember information that vanished from the live database.

The support set S_π, support-local CNF Γ_π, RUP witnesses, and support-local auxiliary witnesses are all read from the ordinary append-only proof artifact plus its internal decoder. The size of this audit data is linear in the proof length. Thus the hypotheses of Theorem 4.56 hold with $B(L) = O(L)$, giving the near-threshold support lower bound and hence proof length $n^{a-o(1)}$.

This removes one annotation layer from Theorem 4.57; it does not remove the need for a checked label decoder, RUP propagation witnesses, or support-locality checks. In particular, it is still not a lower bound for raw DIMACS, arbitrary deletion traces, provenance-free CDCL, Frege, Extended Frege, IPS, or unannotated compressed proofs. □

Corollary 4.59 (Prefix-typed append-only RUP-CDCL)**.** *Fix a typed append-only RUP-CDCL proof system for the easy-completion CNF $\Phi_{n,s,A}$ in which the formula-label variables form a public syntactic class: from a variable name alone the checker can decide whether it is a formula-label variable and, if so, recover its coordinate $u \in U_n$. Suppose the proof object contains an append-only clause-addition log, checker-visible RUP propagation certificates for every used learned consequence, and support-local auxiliary witnesses for every touched auxiliary clause. Then no proof-internal label decoder and no retained deleted-clause support records are needed. This typed append-only*

RUP-CDCL system instantiates the carrier-independent audit upgrade criterion with linear audit budget. Hence, in the near-threshold regime of Theorem 4.3, every accepted proof in this format has length $n^{a-o(1)}$.

Proof. The public typed namespace is itself the decoder. The checker scans the append-only clause-addition log and the RUP propagation certificates used in the final replay, collects exactly the variables belonging to the public formula-label syntactic class, and maps them to their coordinates in U_n. This gives the support set S_π without any proof-specific decoder annotation.

Because the clause database is append-only, no later replay step can depend on a deleted clause whose formula-label support must be remembered separately. Thus the monotone scan also recovers the support-local CNF Γ_π, the RUP witnesses, and the touched auxiliary witnesses with total size linear in the proof length. The assumed support-local auxiliary witnesses certify that the touched auxiliary clauses are satisfiable under every completion preserving S_π, and the RUP replay certifies the final clausal contradiction for $\Phi_{n,s,A} \wedge \Gamma_\pi$. Therefore the hypotheses of Theorem 4.56 hold with $B(L) = O(L)$, giving the claimed lower bound.

The theorem removes the proof-internal label decoder only because the typed proof system makes the formula-label namespace public and checkable. It does not prove a lower bound for untyped DIMACS, raw deletion-bearing RUP/DRUP, arbitrary CDCL, provenance-free learned clauses, Frege, Extended Frege, IPS, or arbitrary compressed proofs, and it does not remove the support-locality witness obligation. □

Proposition 4.60 (Support-local witnesses are necessary for prefix-typed append-only RUP-CDCL)**.** *In Theorem 4.59, the support-local auxiliary witness hypothesis cannot be replaced by append-only RUP checking plus public formula-label typing alone. More precisely, there is no sound extractor that, for every typed append-only RUP-CDCL trace, scans only the public formula-label variables occurring in the append-only log and RUP certificates and then certifies completion-local support for touched auxiliary clauses without an additional support-locality witness or an equivalent proof-system invariant.*

Proof. This is the same-skeleton obstruction from Theorems 4.46 and 4.48 with the variable-role ambiguity removed. Let W be the formula-label support obtained by the public typed scan of the append-only log and the RUP certificates. Add a touched auxiliary clause, or a bounded auxiliary gadget, whose displayed clausal use and RUP replay are unchanged but whose semantic soundness depends on a hidden formula-label coordinate $h \notin W$. There are two completions that agree on every coordinate in W and differ only at h. The typed append-only trace, the public support scan, and the RUP certificates are identical under the two completions, but the auxiliary clause is completion-sound in one completion and not in the other.

Thus W is not a completion cylinder support for the touched auxiliary object. Public typing identifies formula-label variables that occur in the trace, append-only logging prevents deleted-clause support from being lost, and RUP certificates verify displayed clausal implication over the replayed database. None of these syntactic checks proves that every touched auxiliary constraint is satisfiable under all completions preserving W. That last fact is exactly what the support-local auxiliary witness, or an equivalent intrinsic invariant, supplies. The positive conclusion of Theorem 4.59 is therefore unchanged, but its remaining witness assumption is a real hypothesis rather than removable bookkeeping. □

Proposition 4.61 (Strict-prefix Qrep replay normalizer)**.** *Let π be a finite proof transcript with an ordinary displayed support set $K_0(\pi)$. Suppose the only compressed support steps in π are*

quotient-representative dictionary entries

$$e = (a_e, r_e, F_e, t_e, u_e, \mathrm{rank}(e)),$$

where a_e is the row or local object replaced by representative r_e, F_e is a finite same-theory certificate forest proving that a_e and r_e have the same label or semantic value in the declared local theory, t_e is the line where F_e is displayed, and u_e is the first line using r_e in place of a_e. Assume every used entry is strict-prefix and well-founded:

1. *$t_e < u_e$;*
2. *every dictionary dependency in F_e has strictly smaller rank and has already been accepted in the prefix before t_e;*
3. *dictionary entries are append-only, and no representative is rebound after first use;*
4. *the entries, ranks, dependency pointers, certificate forests, and first-use lines are determined by N and the transcript π, with no SAT-truth or row-selection advice.*

Then the quotient uses in π have a canonical replay expansion: each representative use can be replaced, in line order and increasing rank, by the representative row together with the already charged forest F_e. The expanded transcript has local support contained in

$$K_{\mathrm{Qrep}}(\pi) := K_0(\pi) \cup \bigcup_{e \in \mathrm{Used}(\pi)} \big(\mathrm{supp}(F_e) \cup \{a_e, r_e\}\big),$$

and its displayed support budget is enlarged by at most a constant factor times

$$D_Q(\pi) := \sum_{e \in \mathrm{Used}(\pi)} \big(|F_e| + |\,\mathrm{supp}(F_e)| + 1\big).$$

Consequently, if a proof-system normalizer is support-local after expanding all non-Qrep steps and its only remaining compression is strict-prefix Qrep, then Theorem 4.55 applies with support budget enlarged by $O(D_Q(\pi))$.

If any first representative use is not strict-prefix, rank-decreasing, dependency-closed, append-only, or advice-free, no transparent-support conclusion follows from this proposition. In the abstract label model there is a private-row falsifier at the first unsafe use: two completions can agree on the visible prefix while differing on the hidden row, hidden certificate bit, or hidden representative choice that controls the quotient merge.

Proof. Process dictionary entries in proof-prefix order, breaking ties by increasing rank. The strict inequality $t_e < u_e$ says that when a representative is first used, its certificate forest is already displayed. The rank condition says that every quotient dependency needed by that forest has already been replayed or belongs to the ordinary displayed support. Thus the recursive replay terminates and is independent of later proof lines. Append-only dictionaries and the no-rebinding condition prevent a later line from changing the meaning of an earlier representative.

For each use of r_e in place of a_e, splice in the certificate forest F_e and the displayed representative row. Since F_e is a same-theory certificate, the splice preserves the local semantic value used by the original line. Since all leaves of F_e, together with a_e and r_e, are included in $K_{\mathrm{Qrep}}(\pi)$, the replay introduces no uncharged local object. Summing the certificate forest sizes and newly exposed supports over used entries gives the stated $O(D_Q(\pi))$ budget increase.

After replay, the transcript is an ordinary displayed-support transcript for purposes of any support-local normalizer. Applying Theorem 4.55 to the replayed proof gives the advertised inherited footprint bound.

For the boundary statement, choose the earliest unsafe representative use. All earlier lines are fixed. If the missing condition is late certification, cyclic or nondecreasing dependency, rebinding, or advice-chosen representative selection, introduce a private row or certificate bit not determined by the visible prefix and set two completions to agree on every displayed object before that use while disagreeing on the hidden object controlling the merge. The two completions make the same visible prefix acceptable but disagree on whether the representative use is sound. Hence the unexpanded quotient step is not transparent support. □

Example 4.62 (Two-line prefix scan for Qrep)**.** The finite checker used in the accompanying artifact has two one-entry transcripts. In the valid transcript, $a = \texttt{row_B}$ is replaced by $r = \texttt{row_A}$, the same-theory certificate forest $F = \{\texttt{cert_A_equiv_B}\}$ is displayed at line 3, and the representative is first used at line 4. The prefix scan accepts. Its support consists of the final lower-bound line, the FormulaEval parent, and the two rows $\texttt{row_A}, \texttt{row_B}$, so

$$|K_{\mathrm{Qrep}}| = 4, \qquad D_Q = |F| + 2 + 1 = 4.$$

In the delayed transcript the same representative is used at line 4, but the certificate is displayed only at line 5. The checker rejects with the reason `certificate_not_strict_prefix`. This is exactly the temporal private-row falsifier in Theorem 4.61: before the first use, the visible prefix has not paid for the same-theory evidence.

5 Occurrence-label certificates and full rows

Fix z with $\mathsf{Hard}^{\mathrm{occ}}_{n,s,A}(z) = 1$, and let $\sigma : Q \to \{0,1\}$ be a 1-certificate at z, where $Q \subseteq \Omega_n$. For a row $x \in \{0,1\}^n$, define

$$Q_x := \{j \in [M_n] : (x, j) \in Q\}$$

and let σ_x be the induced partial assignment on that row.

Definition 5.1 (Forced rows)**.** For $b \in \{0,1\}$, let

$$R_b(\sigma) := \{x \in \{0,1\}^n : \sigma_x \text{ forces } A_{M_n} = b\}.$$

The sets $R_0(\sigma)$ and $R_1(\sigma)$ are disjoint.

Theorem 5.2 (Row-forcing theorem)**.** *There is an absolute constant $c \geq 1$ such that every 1-certificate σ for $\mathsf{Hard}^{\mathrm{occ}}_{n,s,A}$ satisfies, for each $b \in \{0,1\}$,*

$$|R_b(\sigma)| > \frac{s - c}{c(n+1)}. \tag{9}$$

Consequently,

$$|Q| \geq \mathrm{force}_0(A_{M_n})|R_0(\sigma)| + \mathrm{force}_1(A_{M_n})|R_1(\sigma)|, \tag{10}$$

and therefore, for $s = \Omega(n)$,

$$|Q| = \Omega\left((\mathrm{force}_0(A_{M_n}) + \mathrm{force}_1(A_{M_n}))\frac{s}{n}\right). \tag{11}$$

Proof. Fix $b \in \{0,1\}$. For every row $x \notin R_b(\sigma)$, the partial assignment σ_x does not force output b. Hence it has a completion whose aggregator value is $1 - b$. Choose such a completion independently

for every row outside $R_b(\sigma)$, and complete the rows in $R_b(\sigma)$ arbitrarily. The resulting total occurrence labeling $z^{(b)}$ extends the certificate and satisfies

$$F_{n,A}^{z^{(b)}}(x) = 1 - b \qquad \text{for every } x \notin R_b(\sigma).$$

On a row in $R_b(\sigma)$, every completion has output b. Thus $F_{n,A}^{z^{(b)}}$ differs from the constant function $1 - b$ exactly on $R_b(\sigma)$.

By Theorem 3.1,

$$\mathrm{C}(F_{n,A}^{z^{(b)}}) \le c + c(n+1)|R_b(\sigma)|$$

for an absolute constant c. Since $z^{(b)}$ extends a 1-certificate, its induced function has circuit complexity greater than s. Rearranging proves (9).

Every row in $R_b(\sigma)$ contains at least $\mathrm{force}_b(A_{M_n})$ queried coordinates. The forced-row sets are disjoint, so summing their rowwise query counts proves (10). □

5.1 Examples of aggregator costs

For the threshold aggregator $T_{M,t}(u) = 1$ iff $\sum_i u_i \ge t$,

$$\mathrm{force}_1(T_{M,t}) = t, \qquad \mathrm{force}_0(T_{M,t}) = M - t + 1.$$

This includes OR and AND. The resulting bounds are shown in Table 3.

Aggregator A_M	$\mathrm{force}_0(A_M)$	$\mathrm{force}_1(A_M)$	Query lower bound
OR_M	M	1	$\Omega((M+1)s/n)$
AND_M	1	M	$\Omega((M+1)s/n)$
Majority, odd M	$(M+1)/2$	$(M+1)/2$	$\Omega(Ms/n)$
Parity $\oplus_M$	M	M	$\Omega(Ms/n)$

Table 3: Minimum forcing numbers and the consequence of Theorem 5.2. Parity is maximally expensive on both output values.

The total lower bound for OR is still $\Omega(Ms/n)$ because many rows must be forced to zero, but its positive rows can be certified by one occurrence each. Parity has no such one-sided shortcut.

Corollary 5.3 (Full-row barrier for parity)**.** *For parity, every* 1*-certificate for* $\mathsf{Hard}^{\mathrm{occ}}_{n,s,\oplus}$ *satisfies*

$$|Q| = \Omega(M_n s/n).$$

Moreover, every row in $R_0(\sigma) \cup R_1(\sigma)$ *is complete: all* M_n *coordinates of that row belong to* Q*. If*

$$M_n = n^{q+o(1)}, \qquad s_n = n^{1+\varepsilon-o(1)},$$

then

$$|Q| \ge n^{q+\varepsilon-o(1)}.$$

Proof. Any proper partial assignment to parity leaves a free coordinate whose flip changes the output. Hence both forcing numbers equal M_n. Apply Theorem 5.2. □

5.2 Adaptive occurrence-query verifiers

Proposition 5.4 (Occurrence-query lower bound)**.** *Let V be deterministic and suppose*

$$V^z(\pi) = 1 \quad \Longrightarrow \quad \mathsf{Hard}^{\mathrm{occ}}_{n,s,A}(z) = 1$$

for every occurrence labeling z. Then the query-answer transcript of every accepted execution satisfies (10). For parity it makes $\Omega(M_n s/n)$ oracle queries.

The conclusion is a lower bound on semantic occurrence access. Turning it into a proof-length lower bound requires a separate statement relating proof length to the number of concrete occurrence labels exposed by the proof system.

6 Equality quotients and residual affine distance

The occurrence model treats all positions as independent. Equality links can invalidate that relaxation.

Example 6.1 (Parity cancellation under an equality link)**.** Suppose a parity row contains two formula strings u and u' and the admissible labelings satisfy $\lambda(u) = \lambda(u')$. Their contribution is always

$$\lambda(u) \oplus \lambda(u') = 0,$$

although neither label is fixed. Independent occurrence completion would incorrectly allow the pair to contribute 1.

We therefore quotient labels before taking row parity.

6.1 The quotient incidence map

Let E_n be an equivalence relation on U_n, and write

$$\mathcal{C}_n := U_n/E_n$$

for its set of classes. For $C \in \mathcal{C}_n$, define the parity incidence

$$B_{E_n}(x, C) := \bigoplus_{j : G_{n,j}(x) \in C} 1.$$

For a quotient labeling $z \in \{0,1\}^{\mathcal{C}_n}$, define

$$H^z_{n,E_n}(x) := \bigoplus_{C \in \mathcal{C}_n} B_{E_n}(x, C) z_C. \tag{12}$$

Equivalently, $H^z_{n,E_n} = L_{E_n} z$, where

$$L_{E_n} : \mathbb{F}_2^{\mathcal{C}_n} \longrightarrow \mathbb{F}_2^{\{0,1\}^n}$$

is the binary incidence map.

Define

$$\mathsf{Hard}^{\mathrm{quot}}_{n,s,E_n}(z) := \mathbf{1}[\mathrm{C}(H^z_{n,E_n}) > s].$$

Let $\sigma : W \to \{0,1\}$ be a 1-certificate at some hard quotient labeling. Its set of source completions is the affine set

$$\mathcal{A}(\sigma) := \{L_{E_n} z : z|_W = \sigma\} \subseteq \mathbb{F}_2^{\{0,1\}^n}. \tag{13}$$

Definition 6.2 (Residual constant distance)**.** Define

$$\delta_{\pm}(\sigma) := \min_{h \in \mathcal{A}(\sigma)} \min_{b \in \{0,1\}} \operatorname{dist}(h, b\mathbf{1}), \tag{14}$$

where $\mathbf{1}$ denotes the all-one source truth table.

Theorem 6.3 (Quotient residual-distance barrier)**.** *There is an absolute constant $c \geq 1$ such that every 1-certificate σ for $\mathsf{Hard}^{\mathrm{quot}}_{n,s,E_n}$ satisfies*

$$\delta_{\pm}(\sigma) > \frac{s - c}{c(n+1)}. \tag{15}$$

In particular, if $s_n = n^{1+\varepsilon-o(1)}$, then

$$\delta_{\pm}(\sigma) \geq n^{\varepsilon - o(1)}.$$

Proof. Choose $h = L_{E_n} z \in \mathcal{A}(\sigma)$ and $b \in \{0,1\}$ attaining (14). The quotient labeling z extends the certificate, so $\mathrm{C}(h) > s$. On the other hand, h differs from the constant function b on exactly $\operatorname{dist}(h, b\mathbf{1})$ source rows. Equality patching gives

$$\mathrm{C}(h) \leq c + c(n+1) \operatorname{dist}(h, b\mathbf{1}).$$

Rearranging proves the claim. □

Remark 6.4 (Coding interpretation)**.** Before labels are fixed, $\operatorname{im}(L_{E_n})$ is a binary linear code of source truth tables. A certificate restricts the label vector to an affine subcube, whose image is an affine subset of this code. Theorem 6.3 says that completion soundness forces the entire residual affine set outside Hamming balls of radius $\Theta(s/n)$ around both constant functions.

Remark 6.5 (Scope of quotienting)**.** The theorem handles constraints saying that certain labels are equal. It does not by itself handle nonlinear semantic laws. For example, SAT self-reduction imposes

$$Y(u) = Y(u|_{x_i=0}) \vee Y(u|_{x_i=1}),$$

which is not captured by an equality quotient. The next proposition gives the local support-accounting analogue for one relation instance. A global residual-completion theorem for a whole SAT self-reduction closure remains an open direction discussed in Section 9.

Proposition 6.6 (Semantic-relation support certificate)**.** *Let J be a finite index set, let $R \subseteq \{0,1\}^J$ be a Boolean relation, and let $a \in R$. A set $S \subseteq J$ is support-local for R at a if every $b \in \{0,1\}^J$ with $b|_S = a|_S$ also lies in R. Then S is exactly a standard 1-certificate for membership of a in R.*

For the SAT self-reduction relation

$$R_{\vee} := \{(y, y_0, y_1) : y = y_0 \vee y_1\},$$

the tuple $(0,0,0)$ requires all three coordinates in any support-local certificate. The satisfying tuples $(1,1,0)$, $(1,0,1)$, and $(1,1,1)$ have support-local certificates of size 2, namely $y = 1$ together with one child equal to 1.

Proof. The first statement is a restatement of the two definitions: S is support-local exactly when fixing $a|_S$ forces every completion to remain in R, which is precisely the usual certificate condition for the relation membership predicate.

For $R_{\vee}$, if $a = (0,0,0)$ and any one of the three coordinates is omitted, flipping that coordinate while preserving the others violates $y = y_0 \vee y_1$. Thus all three coordinates are necessary and sufficient. If $a = (1,1,0)$ or $(1,1,1)$, fixing $y = 1$ and $y_0 = 1$ forces the relation regardless of y_1. If $a = (1,0,1)$, fixing $y = 1$ and $y_1 = 1$ does the same. □

Proposition 6.7 (Acyclic OR-tree residual certificate)**.** *Let T be a finite rooted tree whose internal nodes compute OR of their children. Let $L(T)$ be the leaf set and let r be the root. After eliminating internal nodes, the frontier relation on $\{r\} \cup L(T)$ is*

$$R_T := \left\{ (y_r, (y_\ell)_{\ell \in L(T)}) : y_r = \bigvee_{\ell \in L(T)} y_\ell \right\}.$$

Let $a \in R_T$, and let $S \subseteq \{r\} \cup L(T)$ be support-local for R_T at a. If $a_r = 0$, then $r \in S$ and $L(T) \subseteq S$. If $a_r = 1$, then $r \in S$ and S contains at least one leaf ℓ with $a_\ell = 1$. Conversely, these conditions are sufficient.

Proof. The frontier relation is the ordinary OR relation on the leaves. If $a_r = 0$, then every leaf has value 0. If $r \notin S$, changing only y_r to 1 violates R_T. If some leaf $\ell \notin S$, changing only y_ℓ to 1 violates R_T. Thus r and all leaves are necessary, and fixing them to 0 is plainly sufficient.

If $a_r = 1$, then at least one true leaf exists. If $r \notin S$, changing y_r to 0 while leaving the true leaf unchanged violates R_T. If S contains no true leaf, complete every leaf not in S to 0; the root remains fixed to 1, so the relation is violated. Hence S must contain r and a true leaf. Conversely, fixing $r = 1$ and any leaf ℓ with $a_\ell = 1$ forces the OR relation under all completions. □

Proposition 6.8 (OR-DAG frontier residual certificate)**.** *Let D be a finite acyclic directed graph with root r. Each non-frontier node computes the OR of its children. Let Q_r be the set of distinct quotient classes of frontier coordinates reachable from r, where the quotient identifies only soundly equal frontier formulas. After eliminating internal OR nodes, the residual frontier relation is*

$$R_D := \left\{ (y_r, (z_q)_{q \in Q_r}) : y_r = \bigvee_{q \in Q_r} z_q \right\}.$$

Let $a \in R_D$, and let $S \subseteq \{r\} \cup Q_r$ be support-local for R_D at a. If $a_r = 0$, then $r \in S$ and $Q_r \subseteq S$. If $a_r = 1$, then $r \in S$ and S contains at least one reachable frontier class $q \in Q_r$ with $a_q = 1$. Conversely, these conditions are sufficient.

Proof. Eliminate the internal nodes in reverse topological order. Each internal node computes the OR of the frontier coordinates reachable below it. Shared subgraphs only repeat the same OR subexpression, and repeated frontier occurrences collapse under idempotence of OR. Sound frontier quotienting merely replaces repeated equal formulas by one coordinate. Hence the root relation is exactly the displayed OR over the distinct reachable frontier classes Q_r.

The support-local certificates for this relation are the ordinary OR certificates. If $a_r = 0$, every $q \in Q_r$ has $a_q = 0$. Omitting r lets us flip y_r to 1, and omitting any reachable frontier class q lets us flip z_q to 1; either completion violates R_D. Thus r and all classes in Q_r are necessary and plainly sufficient. If $a_r = 1$, omitting r lets us flip the root to 0 while keeping a true reachable class fixed. If no true reachable class is included in S, complete all unexposed reachable classes to 0, leaving the fixed root equal to 1; this violates R_D. Fixing $r = 1$ and one true reachable class forces the relation under all completions. □

Remark 6.9 (Boundary of the acyclic tree theorem)**.** Theorems 6.7 and 6.8 are residual-completion theorems only for eliminated pure-OR frontier relations. The DAG theorem removes the tree-only and repeated-frontier caveats for this pure OR fragment after sound quotienting. It does not say that a proof may ignore local relation audits on displayed internal nodes. Exposed acyclic non-OR frontiers are handled next by Theorem 6.10, checker-visible adaptive decision trees by

Theorem 6.11, and checker-visible adaptive branching programs by Theorem 6.12; learned clauses, cyclic dependencies, and arbitrary hidden real-SAT semantic influence still require a separate normalizer or realizer.

Proposition 6.10 (Acyclic Boolean frontier residual certificate)**.** *Let C be a finite acyclic Boolean circuit with frontier coordinates Q, root coordinate r, and induced function $F_C : \{0,1\}^Q \to \{0,1\}$. After eliminating internal gates, consider the residual frontier relation*

$$R_C := \{(y_r, z) \in \{0,1\}^{\{r\}\cup Q} : y_r = F_C(z)\}.$$

Let $a \in R_C$, and let $S \subseteq \{r\} \cup Q$. Then S is support-local for R_C at a iff $r \in S$ and $S \cap Q$ is a standard pointwise certificate forcing $F_C(z) = a_r$ at the frontier assignment $a|_Q$. In particular, the minimum support size is

$$1 + C_{a_r}(F_C, a|_Q),$$

where $C_b(F_C, a|_Q)$ denotes ordinary Boolean certificate complexity at $a|_Q$ for forcing value b.

Proof. If $r \notin S$, keep the frontier assignment equal to $a|_Q$ and flip only y_r. The completion preserves S but violates $y_r = F_C(z)$. Hence $r \in S$.

Assume $r \in S$. If $S \cap Q$ is not a certificate forcing $F_C(z) = a_r$, then there is a frontier assignment z' agreeing with a on $S \cap Q$ but satisfying $F_C(z') \neq a_r$. Keeping $y_r = a_r$ and using z' gives a completion preserving S that violates R_C. Thus support-locality implies that $S \cap Q$ is a certificate.

Conversely, if $r \in S$ and $S \cap Q$ certifies $F_C(z) = a_r$, then every completion preserving S keeps $y_r = a_r$ and keeps $F_C(z) = a_r$. Therefore $y_r = F_C(z)$ holds for every such completion, which is exactly support-locality. The minimum-size identity follows by minimizing the frontier certificate size and adding the necessary root coordinate. □

Proposition 6.11 (Visible adaptive frontier residual certificate)**.** *Let T be a finite binary decision tree whose internal nodes query frontier coordinates in Q and whose leaves are labelled by bits. Let r be a root/output coordinate and let $F_T : \{0,1\}^Q \to \{0,1\}$ be the Boolean function computed by following the queried branch outcomes to a leaf. Consider the residual frontier relation*

$$R_T := \{(y_r, z) \in \{0,1\}^{\{r\}\cup Q} : y_r = F_T(z)\}.$$

Let $a \in R_T$, and let $S \subseteq \{r\} \cup Q$. Then S is support-local for R_T at a iff $r \in S$ and $S \cap Q$ is a standard pointwise certificate forcing $F_T(z) = a_r$ at the frontier assignment $a|_Q$. In particular, the minimum support size is

$$1 + C_{a_r}(F_T, a|_Q),$$

and the frontier coordinates queried along the actual $a|_Q$-transcript path form one sufficient, though not always minimum, frontier certificate.

Proof. The first three steps are the same root-plus-certificate argument as in Theorem 6.10. If $r \notin S$, keep the frontier assignment equal to $a|_Q$ and flip y_r, violating $y_r = F_T(z)$. If $r \in S$ but $S \cap Q$ is not a certificate for $F_T(z) = a_r$, choose a frontier assignment z' that agrees with a on $S \cap Q$ but has $F_T(z') \neq a_r$; with $y_r = a_r$, this preserves S and violates R_T. Conversely, if $r \in S$ and $S \cap Q$ certifies $F_T(z) = a_r$, every completion preserving S keeps both sides of $y_r = F_T(z)$ equal to a_r.

For the transcript-path claim, fix the values of all coordinates queried along the path followed by $a|_Q$. Every completion preserving those values follows the same path to the same leaf and hence has the same value $F_T(z) = a_r$. Thus the path variables certify the decision-tree value. A smaller certificate may exist if different subtrees have the same leaf value, so the path certificate is sufficient rather than necessarily minimum. □

Proposition 6.12 (Visible adaptive branching-program frontier residual certificate)**.** *Let B be a finite acyclic deterministic branching program with start node s, internal nodes labelled by queries to frontier coordinates in Q, two outgoing edges labelled 0 and 1 from each internal node, and sink nodes labelled by bits. Let r be a root/output coordinate and let $F_B : \{0,1\}^Q \to \{0,1\}$ be the Boolean function computed by following the queried edge labels from s to a sink. Consider the residual frontier relation*

$$R_B := \{(y_r, z) \in \{0,1\}^{\{r\}\cup Q} : y_r = F_B(z)\}.$$

Let $a \in R_B$, and let $S \subseteq \{r\} \cup Q$. Then S is support-local for R_B at a iff $r \in S$ and $S \cap Q$ is a standard pointwise certificate forcing $F_B(z) = a_r$ at the frontier assignment $a|_Q$. In particular, the minimum support size is

$$1 + C_{a_r}(F_B, a|_Q).$$

The frontier coordinates queried along the actual $a|_Q$-computation path form one sufficient, though not always minimum, frontier certificate.

Proof. The root-coordinate and certificate-complexity directions are identical to Theorem 6.11, with F_B in place of F_T. If $r \notin S$, keep the frontier assignment equal to $a|_Q$ and flip y_r. This preserves S but violates $y_r = F_B(z)$. If $r \in S$ and $S \cap Q$ is not a certificate for $F_B(z) = a_r$, choose a frontier completion z' agreeing with a on $S \cap Q$ but satisfying $F_B(z') \neq a_r$. With $y_r = a_r$, this completion preserves S and violates R_B. Conversely, root plus a frontier certificate for $F_B(z) = a_r$ forces both sides of the residual equation for every completion preserving S.

For the path-support claim, fix all coordinate values queried along the computation path followed by $a|_Q$. Any completion preserving those queried values traverses the same visible branching-program edges to the same sink, hence has the same output $F_B(z) = a_r$. Thus the path variables are a sufficient support certificate. Sharing in the branching program may make the represented function smaller than its unfolded decision tree, and other certificates may be smaller than the actual path, so the path bound is not asserted to be minimum. □

Proposition 6.13 (Hidden adaptive transcript insufficiency)**.** *Fix $m \geq 2$, frontier coordinates $Q = \{q_1, \ldots, q_m\}$, root coordinate r, and hard tuple $a_r = 0$, $a_{q_i} = 0$ for all i. Consider the reported adaptive transcript summary*

$$\tau = (q_1 = 0,\ \text{output} = 0),$$

with no certified claim that the transcript is a complete computation path or that the controller has no further hidden dependencies. Then τ is compatible at a with both residual relations

$$R_1 = \{(y_r, z) : y_r = z_{q_1}\} \qquad \text{and} \qquad R_\vee = \{(y_r, z) : y_r = \bigvee_{i=1}^{m} z_{q_i}\}.$$

For R_1 at a, the support $\{r, q_1\}$ is support-local. For $R_\vee$ at a, every support-local set contains r and all of Q. Consequently any support extractor that sees only τ and outputs a proper subset of $\{r\} \cup Q$ is unsound for some hidden adaptive controller compatible with the same reported transcript. A sound transcript-based normalizer must therefore either certify controller/path completeness, expose enough dependency support, or pay worst-case frontier support.

Proof. Both displayed relations agree with the reported summary at the all-zero tuple: the first queried value is $q_1 = 0$ and the output/root value is 0. For R_1, fixing $r = 0$ and $q_1 = 0$ forces $y_r = z_{q_1}$ under every completion preserving those two coordinates, so $\{r, q_1\}$ is support-local.

For $R_\vee$, omitting r lets us flip the root while keeping the frontier fixed. Omitting any frontier coordinate q_j lets us preserve all coordinates in the alleged support, set $z_{q_j} = 1$, keep all other unforced

frontier coordinates arbitrary, and keep $y_r = 0$. The resulting completion violates $y_r = \bigvee_i z_{q_i}$. Thus r and every coordinate in Q are necessary, and they are plainly sufficient.

The same reported transcript summary τ cannot distinguish these two cases. Hence a transcript-only extractor returning less than $\{r\} \cup Q$ is correct for the first relation but unsound for the second. The conclusion is exactly the hidden-controller boundary: the positive visible-adaptive propositions rely on an audited controller and a certified complete path, not merely on a small reported transcript summary. □

Proposition 6.14 (Controller-complete adaptive audit normalizer)**.** *Let $H : \{0,1\}^Q \to \{0,1\}$ be a hidden adaptive controller and let*

$$R_H = \{(y_r, z) \in \{0,1\}^{\{r\}\cup Q} : y_r = H(z)\}.$$

Fix $a \in R_H$. A controller-complete adaptive audit for a and a support $S \subseteq \{r\} \cup Q$ consists of a finite acyclic deterministic branching program B over Q, a replay path π of B on $a|_Q$, and a certificate of local controller equivalence such that:

1. *$r \in S$, and every frontier coordinate queried along π belongs to $S \cap Q$;*
2. *the sink reached by π has label a_r;*
3. *for every $z' \in \{0,1\}^Q$ agreeing with a on $S \cap Q$, the hidden controller agrees with the exposed one: $H(z') = F_B(z')$.*

Then S is support-local for R_H at a. If the audit certifies the global identity $H = F_B$ on all of $\{0,1\}^Q$, then the exact minimum support is again

$$1 + C_{a_r}(F_B, a|_Q),$$

as in Theorem 6.12.

Proof. Let (y_r, z') be any completion preserving S. Since $r \in S$, the root coordinate remains $y_r = a_r$. Since every query along π lies in $S \cap Q$, the branching program B sees the same answers along that path on z' as it sees on $a|_Q$. Therefore the exposed controller follows π and outputs $F_B(z') = a_r$.

The local controller-equivalence certificate transfers this value from the exposed controller to the hidden controller on the entire completion cylinder preserving $S \cap Q$: $H(z') = F_B(z') = a_r$. Hence $y_r = H(z')$, so the residual relation holds for every completion preserving S. This is exactly support-locality.

If the audit certifies $H = F_B$ globally, then $R_H = R_B$. The exact minimum support statement is therefore precisely the visible branching-program certificate theorem *Theorem* 6.12. The point of the audit is not that a transcript alone determines B; it is that a proof object paying for controller completeness and local equivalence restores the visible-adaptive normalizer that *Theorem* 6.13 says is unavailable from the transcript summary alone. □

Corollary 6.15 (Trace-complete DPLL log normalizer)**.** *Consider a DPLL-style decision-log object for a residual relation*

$$R_H = \{(y_r, z) \in \{0,1\}^{\{r\}\cup Q} : y_r = H(z)\},$$

where the hidden computation may make adaptive queries to frontier coordinates in Q. Suppose the log is trace-complete at a hard tuple $a \in R_H$: it contains a finite acyclic DPLL decision controller B_Π over Q, the replay path π followed by $a|_Q$, terminal output a_r, and clause, propagation, or

pruning witnesses certifying that $H(z') = F_{B_\Pi}(z')$ for every completion z' agreeing with a on the root and all frontier coordinates queried along π. Let

$$S_\Pi := \{r\} \cup \{q \in Q : q \text{ is queried along } \pi\} \cup W_\Pi,$$

where W_Π is the set of formula-label frontier coordinates appearing in those local witness certificates. Then S_Π is support-local for R_H at a. Consequently any proof-size measure or checker budget $B(|\Pi|)$ that upper-bounds $|S_\Pi|$ is subject to the same near-threshold formula-footprint lower bound when the DPLL log is used as the proof object for $\Phi_{n,s,A}$:

$$B(|\Pi|) \geq 2n^{a-o(1)}$$

for $s = n^{a-o(1)}$ in the polarity-balanced regime of Theorem 4.3.

Proof. The trace-complete DPLL log is exactly a controller-complete adaptive audit after adding the witness coordinates W_Π to the support: B_Π is the exposed acyclic controller, π is the replay path, and the logged clause, propagation, or pruning witnesses are the local equivalence certificate on the completion cylinder determined by S_Π. Theorem 6.14 therefore makes S_Π support-local for R_H at a. When the residual relation is the easy-completion proof object for $\Phi_{n,s,A}$, the support-local normalization transfer and Theorem 4.3 give the displayed support-budget lower bound. □

Corollary 6.16 (Antecedent-complete CDCL normalizer)**.** *Consider a CDCL-style learned-clause log for the same residual relation R_H. Suppose the log is antecedent-complete at the hard tuple a: it contains a finite acyclic decision-and-restart controller B_Π, the replay path π, terminal output a_r, and, for every learned clause, propagation, or pruning step used along π, an acyclic antecedent DAG whose leaves are original support-local clauses, explicit decisions, unit-propagation witnesses, or earlier learned clauses with smaller antecedent rank. Let W_Π be the set of formula-label frontier coordinates appearing in the antecedent closure of all learned clauses and propagation witnesses used along π, and set*

$$S_\Pi := \{r\} \cup \{q \in Q : q \text{ is queried along } \pi\} \cup W_\Pi.$$

Assume the antecedent DAGs certify local equivalence $H(z') = F_{B_\Pi}(z')$ for every completion z' agreeing with a on $S_\Pi \cap Q$. Then S_Π is support-local for R_H at a. Hence any proof-size measure or checker budget $B(|\Pi|)$ upper-bounding this replay path plus antecedent-closure support obeys

$$B(|\Pi|) \geq 2n^{a-o(1)}$$

near the polarity-balanced threshold of Theorem 4.3 when the CDCL log is used as the proof object for $\Phi_{n,s,A}$.

Proof. Expand the learned-clause uses along their antecedent DAGs. Acyclicity and antecedent ranks ensure that the expansion terminates in original local clauses, explicit decisions, unit-propagation witnesses, or already expanded lower-rank learned clauses. By definition, all formula-label frontier coordinates appearing in this expansion are included in W_Π, hence in S_Π. The antecedent-completeness certificate therefore supplies the local controller-equivalence field required by Theorem 6.14: on every completion preserving S_Π, the exposed controller and all learned-clause justifications replay with the same values and yield the same terminal output a_r. Thus S_Π is support-local. Applying the support-local normalization transfer and the threshold form of the formula-footprint theorem gives the budget lower bound. □

Proposition 6.17 (CDCL antecedent-erasure trichotomy)**.** *Let a CDCL-erased extractor be a procedure that sees only a finite decision/restart skeleton, the replay path, the sequence of learned-clause names and uses on that path, and the untyped displayed clauses, but does not see learned-clause antecedent DAGs, variable-role provenance, clause-deletion provenance, or an independent support-locality checker. Suppose the extractor outputs a formula-label support intended to certify completion-local soundness for R_H. For the unit-plus-blocking skeleton C_T of Theorem 4.48, packaged as a single learned-clause use on the replay path, exactly one of the following holds.*

(i) The extractor obtains additional antecedent provenance, a trusted variable-role map, a namespace certificate, or a system-specific semantic invariant. Then it is no longer an antecedent-erased CDCL extractor; it has supplied the support audit or learned-clause provenance charged by the positive theorems above.

(ii) The extractor is sound for every same-skeleton semantic lift of the erased log. Then, on C_T, its output must contain all variables in T, hence it pays the role-agnostic support budget for the possible label lift.

(iii) The extractor outputs a strict subset of T while using only the erased CDCL skeleton. Then the label lift from Theorem 4.48, with the same erased learned-clause skeleton, falsifies its claimed support soundness.

Consequently, arbitrary CDCL logs with erased learned-clause antecedents do not instantiate Theorem 4.55 by syntax alone. A positive CDCL theorem must either expose antecedent-complete learned-clause DAGs, add equivalent provenance or namespace data, charge a conservative support budget, or prove a genuinely non-plain semantic invariant that rules out the bad lift.

Proof. Use the auxiliary and label lifts from Theorem 4.48. In the auxiliary lift, the displayed learned clause can be realized through auxiliary variables, so formula-label support is empty after roles are known. In the label lift, the same untyped clause skeleton is a blocking constraint over the variables in T, and Theorem 4.46 forces every sound completion-local support to contain all of T. Erasing learned-clause antecedents, variable roles, and deletion provenance makes these two lifts identical to the CDCL-erased extractor: it sees the same controller skeleton, replay path, learned-clause name, and untyped displayed clause. Therefore it must output the same set in both lifts. If it outputs all of T, it is conservative for this skeleton and pays the role-agnostic budget. If it outputs a strict subset of T, the label lift witnesses unsoundness. If it avoids the ambiguity by distinguishing the two lifts, then it used antecedent provenance, role information, namespace information, a support-locality checker, or another semantic invariant beyond the erased CDCL syntax. These are precisely the three cases. □

Corollary 6.18 (RUP-complete CDCL support normalizer)**.** *Consider a CDCL/DRUP-style learned-clause log for the residual relation R_H. Suppose the log is RUP-complete at the hard tuple a: it contains a finite acyclic decision-and-restart controller B_Π, the replay path π, terminal output a_r, and, for every learned clause, propagation, pruning step, or clause-deletion-dependent use along π, a checker-visible RUP propagation certificate over the known live clause database at that moment. Each RUP certificate records the unit-propagation trace, the clauses touched, the propagated literals, the conflict clause, and the formula-label frontier coordinates touched by those clauses. The proof object also carries either a trusted formula-label namespace, a trusted variable-role map, or an equivalent local-support parser separating label variables from auxiliary variables; if a deleted clause is used only through a retained learned clause, the retained RUP certificate and its support are still charged.*

Let W_Π be the union of all formula-label frontier coordinates appearing in the replayed RUP traces, in the local-support parser output for clauses used by those traces, and in retained certificates for deleted clauses whose consequences are still used. Set

$$S_\Pi := \{r\} \cup \{q \in Q : q \text{ is queried along } \pi\} \cup W_\Pi.$$

Assume the RUP traces certify local equivalence $H(z') = F_{B_\Pi}(z')$ for every completion z' agreeing with a on $S_\Pi \cap Q$. Then S_Π is support-local for R_H at a. Consequently any proof-size measure or checker budget $B(|\Pi|)$ upper-bounding the replay path together with the parsed RUP-trace support obeys

$$B(|\Pi|) \geq 2n^{a-o(1)}$$

near the polarity-balanced threshold of Theorem 4.3 when the RUP-complete CDCL log is used as the proof object for $\Phi_{n,s,A}$.

Proof. A RUP certificate is an executable local witness: starting from the known live clause database, the recorded unit-propagation trace derives the displayed conflict after adding the negation of the learned clause. Because the certificate lists the clauses touched, propagated literals, conflict clause, and parsed formula-label coordinates, preserving S_Π preserves the truth values of every label coordinate on which that local check depends. The trusted namespace, role map, or equivalent parser is what makes this a formula-label support statement rather than an untyped syntactic statement. Retained certificates for deleted clauses ensure that deletion does not erase already used support.

Induct over the accepted learned clauses and propagation steps in the order in which the replay uses them. The base clauses are original local clauses, explicit decisions, or previously retained audited certificates whose parsed formula-label support lies in W_Π. The induction step is the RUP trace itself: every clause and literal read by the unit-propagation check has support inside W_Π, so the same local implication is valid on every completion preserving S_Π. Thus the replayed controller, all learned clauses used by it, and the terminal output remain locally equivalent to the hidden residual computation on the completion cylinder determined by S_Π.

This is exactly the local controller-equivalence field required by *Theorem* 6.14. Therefore S_Π is support-local for R_H at a. Applying the support-local normalization transfer and the formula-footprint threshold lower bound gives the displayed budget bound. The statement is deliberately non-plain: without the live-database record, parsed formula-label namespace or role map, and retained support for deleted-clause consequences, the erasure ambiguity of *Theorem* 6.17 remains. □

Corollary 6.19 (Canonical-namespace RUP-CDCL standardizer)**.** *In the setting of Theorem 6.18, suppose the CDCL/DRUP-style trace is accompanied by a deterministic encoding manifest $\mathcal{M}$ with the following checker-visible fields.*

(i) A reserved formula-label variable interval I_{lab} and an injective map $\Lambda : I_{\mathrm{lab}} \to Q$ from those variables to frontier coordinates.

(ii) A disjoint auxiliary-variable region I_{aux}, with every displayed variable belonging to exactly one of I_{lab} or I_{aux}.

(iii) A rule saying that the formula-label support of any displayed clause C is exactly $\Lambda(C \cap I_{\mathrm{lab}})$, ignoring literal signs and auxiliary variables.

(iv) A live-database log sufficient for the RUP checker to reconstruct, at each learned-clause check, the currently live original clauses, accepted learned clauses, and deletions.

(v) *A retention rule saying that when a deleted clause contributes through a retained learned clause, the already checked RUP certificate for that retained learned clause remains available and its parsed label support remains charged.*

Assume every RUP propagation certificate in the trace is checker-visible over the live database reconstructed from $\mathcal{M}$. Then the manifest induces an equivalent local-support parser, so the trace satisfies the parsing hypothesis of Theorem 6.18 without an additional trusted role map. Hence the canonical support

$$S_{\Pi}^{\mathcal{M}} := \{r\} \cup \{q \in Q : q \text{ is queried along } \pi\} \cup \bigcup_{C \text{ touched by replayed RUP traces}} \Lambda(C \cap I_{\mathrm{lab}})$$

together with retained deleted-clause certificate supports is support-local for R_H at a. Any checker budget upper-bounding this canonical parsed support obeys the same near-threshold formula-footprint lower bound.

Proof. The manifest makes variable roles syntactic rather than trusted semantic advice. The checker verifies disjointness of I_{lab} and I_{aux}, verifies that every displayed variable lies in exactly one region, and computes the label support of each touched clause by the declared interval intersection $C \cap I_{\mathrm{lab}}$ followed by Λ. This computation is deterministic and local to the displayed clause and manifest; no hidden role judgment is used.

The live-database log reconstructs the clause database used by each RUP check, while the retention rule prevents deletion from erasing support already charged to retained consequences. Therefore every replayed RUP certificate has exactly the data required by *Theorem* 6.18: touched clauses, propagated literals, conflict clause, live database, and a parsed formula-label support for all touched clauses. Applying *Theorem* 6.18 gives support-locality for $S_{\Pi}^{\mathcal{M}}$ and then the formula-footprint budget bound.

This corollary standardizes the namespace route; it does not make unmanifested DIMACS, raw RUP, or raw DRUP syntax into a support normalizer. If the interval map, live database, or retention rule is erased, the extractor is back in the ambiguity regime of *Theorem* 6.17. □

Remark 6.20 (Boundary of exposed Boolean frontiers)**.** Theorems 6.7 and 6.8 are the pure-OR special cases of Theorem 6.10. The Boolean-frontier theorem allows exposed finite acyclic AND/OR/NOT frontier circuits, Theorem 6.11 records the checker-visible adaptive decision-tree case, and Theorem 6.12 records the shared-state acyclic branching-program case. Theorem 6.13 shows why merely reporting a small adaptive transcript is not a substitute for that visibility, while Theorem 6.14 records the positive audit that is sufficient once controller completeness and local equivalence are paid for. Theorem 6.15 packages the first recognizable proof-object instantiation of that audit: a DPLL decision log is admissible only when it is trace-complete and carries the local witnesses that make the replayed controller sound on the claimed completion cylinder. Theorem 6.16 adds the learned-clause version: CDCL-style compression is admissible when the used learned clauses carry acyclic antecedent DAGs whose leaves expose the local support they depend on. Theorem 6.17 records the matching boundary for erased learned-clause logs: add provenance, pay conservative support, or become unsound on a same-skeleton lift. Theorem 6.18 gives the positive non-plain response: a CDCL/DRUP-style log can supply the same audit through checker-visible RUP propagation certificates, a known live clause database, trusted label namespace or role parsing, and retained deleted-clause support. Theorem 6.19 standardizes one such parsing route: a deterministic encoding manifest with a reserved label interval lets the checker compute support directly from touched clauses. The positive results still assume that the semantic influence has been exposed as an auditable finite object. They do not handle raw RUP/DRUP traces without a manifest or support

parsing, unaudited learned clauses, hidden clause minimization, cyclic restart/search graphs, hidden proof-system influence, or an arbitrary SAT self-reduction proof whose frontier circuit, adaptive control, and quotienting have not been made checker-visible. Those cases still require a separate normalizer or realizer.

7 Application to polylogarithmic SAT parity carriers

We now specialize the labels to satisfiability. Fix a standard total encoding of Boolean formulas: malformed strings are assigned SAT value 0. Let

$$\ell_n = \lceil (\log n)^d \rceil, \qquad d > 1,$$

and let the batched generator have

$$M_n = n^{q+o(1)}, \qquad g_n = n^{\rho+o(1)}.$$

Define

$$H_n^{\mathrm{SAT}}(x) := \bigoplus_{j=1}^{M_n} \mathrm{SAT}_{\ell_n}(G_{n,j}(x)). \tag{16}$$

The condition $d > 1$ is not needed for substitution. It merely prevents the complete ℓ_n-bit SAT table, of size 2^{ℓ_n}, from being polynomial in n.

7.1 Exponent debt and the all-exponent regime

The following accounting proposition separates the general issue from this carrier.

Proposition 7.1 (Exponent-window lemma)**.** *Suppose that for every fixed $c \geq 1$,*

$$\mathrm{SAT} \in \mathrm{SIZE}(m^c) \quad \Longrightarrow \quad \mathrm{C}(K_n) \leq n^{q+\beta c+o(1)}.$$

If $\mathrm{C}(K_n) > n^{a-o(1)}$ infinitely often, then

$$\mathrm{SAT} \notin \mathrm{SIZE}(m^c)$$

for every fixed c satisfying $q + \beta c < a$. Hence one fixed source exponent excludes all polynomial target exponents only when $\beta = 0$ and $a > q$.

Proof. For a fixed c with $q + \beta c < a$, choose a positive constant gap between the two exponents. The conditional upper bound then contradicts the infinitely-often lower bound on all sufficiently large hard lengths. □

Polylogarithmic query length has $\beta = 0$.

Lemma 7.2 (Batched substitution)**.** *If a label function $y : \{0,1\}^{\ell_n} \to \{0,1\}$ has circuit size t_n, then for every size-$O(M_n)$ aggregator A,*

$$\mathrm{C}(H_{n,A}^y) \leq g_n + M_n t_n + O(M_n).$$

Proof. Compute the whole query batch once, feed every query into a separate copy of the size-t_n label circuit, and apply the aggregator. □

Theorem 7.3 (Polylogarithmic-query transfer)**.** *For every fixed $c \geq 1$,*

$$\mathrm{SAT} \in \mathrm{SIZE}(m^c) \quad \Longrightarrow \quad \mathrm{C}(H_n^{\mathrm{SAT}}) \leq n^{\max\{q,\rho\}+o(1)}. \tag{17}$$

Consequently, if for some $a > \max\{q, \rho\}$,

$$\mathrm{C}(H_n^{\mathrm{SAT}}) > n^{a-o(1)}$$

holds infinitely often, then $\mathrm{SAT} \notin \mathrm{P/poly}$.

Proof. Under the target assumption, SAT_{ℓ_n} has circuits of size ℓ_n^c. Since $\ell_n^c = (\log n)^{dc+o(1)} = n^{o(1)}$, Theorem 7.2 gives

$$\mathrm{C}(H_n^{\mathrm{SAT}}) \leq n^{\rho+o(1)} + n^{q+o(1)} = n^{\max\{q,\rho\}+o(1)}.$$

The final claim follows from Theorem 7.1, or directly by fixing the polynomial exponent of a hypothetical SAT circuit family. □

7.2 Reverse calibration

The same substitution inequality can be read backward.

Proposition 7.4 (Reverse substitution burden)**.** *If $\mathrm{C}(H_n^{\mathrm{SAT}}) > s_n$, then*

$$\mathrm{C}(\mathrm{SAT}_{\ell_n}) > \frac{s_n - g_n - O(M_n)}{M_n}. \tag{18}$$

The conclusion is vacuous when the numerator is nonpositive.

Proof. Apply Theorem 7.2 with an optimal circuit for SAT_{ℓ_n} and rearrange. □

At $s_n = n^{1+\varepsilon-o(1)}$ with $q, \rho < 1 + \varepsilon$, (18) yields

$$\mathrm{C}(\mathrm{SAT}_{\ell_n}) > n^{1+\varepsilon-q-o(1)} = 2^{\Omega(\ell_n^{1/d})}$$

on the corresponding hard lengths. Thus the transfer removes dependence on the unknown polynomial exponent, but a direct source lower bound already contains a stretched-exponential lower bound for the matched short SAT slices.

7.3 Certificate consequences

Let

$$\lambda_n^{\mathrm{SAT}} := \mathrm{SAT}_{\ell_n}|_{U_n}$$

be the actual formula labeling, and let

$$z_n^{\mathrm{SAT}}(x, j) := \mathrm{SAT}_{\ell_n}(G_{n,j}(x))$$

be its induced occurrence labeling.

Corollary 7.5 (Local certificate costs for a hard SAT carrier)**.** *Assume*

$$\mathrm{C}(H_n^{\mathrm{SAT}}) > s_n.$$

Then the following hold.

1. *Every* 1*-certificate for* $\mathsf{Hard}^{\mathrm{form}}_{n,s_n,\oplus}$ *at* λ_n^{SAT} *satisfies the two inequalities in Theorem 4.1; in the near-threshold regime it exposes* $n^{1+\varepsilon-o(1)}$ *distinct satisfiable generated formulas and the same number of unsatisfiable generated formulas.*

2. *Every* 1*-certificate for* $\mathsf{Hard}^{\mathrm{occ}}_{n,s_n,\oplus}$ *at* z_n^{SAT} *queries* $\Omega(M_n s_n/n)$ *occurrences and contains* $\Omega(s_n/n)$ *complete rows of each parity.*

3. *Let* E_n *be any equivalence relation on* U_n *on whose classes SAT is constant. Every* 1*-certificate for the resulting quotient hardness predicate has residual constant distance* $\Omega(s_n/n)$*.*

The three statements have different hypotheses. A formula-local proof need not be sound under independent occurrence completions, and a proof using SAT self-reduction need not be sound over all equality-quotient completions. The corollary should therefore be read as a menu of barriers for explicitly specified certificate interfaces, not as an exhaustive theorem about arbitrary SAT-semantic proofs.

7.4 A brief global naturalization observation

The local certificate results do not require largeness. At the opposite extreme, a dense truth-table property that is functionally sound for the carrier enters the natural-proofs regime.

Proposition 7.6 (Functional naturalization)**.** *Let* $\mathcal{P}_n$ *be a property of full truth tables* $y : \{0,1\}^{\ell_n} \to \{0,1\}$ *such that*

$$y \in \mathcal{P}_n \quad \Longrightarrow \quad \mathrm{C}(H_{n,\oplus}^{y|_{U_n}}) > n^{a-o(1)}$$

for some $a > \max\{q, \rho\}$*. Then, for every fixed* c *and all sufficiently large relevant* n*,*

$$y \in \mathcal{P}_n \quad \Longrightarrow \quad \mathrm{C}(y) > \ell_n^c.$$

If $\mathcal{P}_n$ *is decidable in time* $2^{O(\ell_n)}$ *from the truth table and has density at least* $2^{-O(\ell_n)}$*, it is a constructive large property useful against polynomial-size circuits on the matched lengths.*

Proof. If y had a circuit of size ℓ_n^c, Theorem 7.2 would give

$$\mathrm{C}(H_{n,\oplus}^{y|_{U_n}}) \le n^{\max\{q,\rho\}+o(1)},$$

contradicting functional soundness. □

Under the usual pseudorandom-function assumptions, the final combination of constructivity, largeness, and usefulness is subject to the Razborov–Rudich barrier [17]. The point is not that every certificate naturalizes: sparse, SAT-specific, or merely local certificates can fall outside these hypotheses.

7.5 What the application does not establish

No explicit generator is presently shown here to make H_n^{SAT} require $n^{1+\varepsilon}$ unrestricted gates. Such a theorem would already imply SAT $\notin$ P/poly by Theorem 7.3 and would contain the short-length lower bound in Theorem 7.4. Likewise, the certificate theorems do not automatically yield lower bounds for Frege, Extended Frege, IPS, implicit proofs, or another standard proof system. Theorem 4.10 supplies this bridge only for restricted displayed-atom interfaces. A standard proof-system application would still require a normalization theorem showing that short proofs induce one of the completion-sound certificate models analyzed above with controlled overhead.

The randomized theorem in Theorem 4.7 gives a bounded-error lower bound for the union of labels queried on accepting executions. It should not be confused with an expected-query lower bound. A verifier could, in principle, query few labels on each random string while spreading its accepting paths over a large union. Controlling expected footprint, transcript support, or information revealed about the labeling would require a minimax, distributional, or information-theoretic argument beyond the deterministic certificate reduction.

8 Proof dependency and audit map

Table 4 records the logical dependency structure of the theorem package. Its purpose is to make clear which claims use only elementary completion arguments, which claims import circuit-counting or sorting-network facts, and which advertised motivations remain outside the proved dependency graph.

Table 4: Proof dependency and audit map. Every theorem-level claim is traced to its local dependencies and its explicit nonclaim boundary.

Claim	Dependencies	Audit boundary
Equality patching, Theorem 3.1	Equality tests for explicit rows and XOR patching.	Nonuniform upper bound; no lower-bound assumption.
Residual-error threshold, Theorem 3.2	Theorem 3.1.	Converts source hardness into distance from any subthreshold circuit.
Batched static dictionary, Theorem 3.3	Sorting networks with tagged query and dictionary records.	Nonuniform dictionary; dictionary records are masked out before aggregation.
Sparse formula-defect upper bound, Theorem 3.5	Batched generator plus Theorem 3.3.	Applies to formula-label completions, with repeated generated strings sharing one key.
Formula-footprint barrier, Theorem 4.1	Certificate completion to a constant baseline plus Theorem 3.5.	Standard nondeterministic certificate lower bound; not a proof-system lower bound by itself.
Decision-tree lower bound and near-threshold corollaries, Theorems 4.2 to 4.4	Theorem 4.1 and exponent accounting.	Gives standard certificate-complexity payoff; no SAT lower bound.
Identity-carrier generic sharpness, Theorem 4.5	Circuit counting, union bound, and Chernoff bound.	Generic truth-table sharpness; not an explicit SAT carrier.
Deterministic and bounded-error formula verifiers, Theorems 4.6 to 4.8	Transcript/certificate equivalence plus Theorem 4.1.	Bounded-error result is for accepting-query union, not expected query complexity.
Displayed-atom extraction, Theorem 4.10	Definition of faithful completion-sound displayed atoms plus Theorem 4.1.	Restricted proof interface only; no Frege, Extended Frege, IPS, or implicit-proof lower bound.
Resolution support-local bridge, Theorems 4.13, 4.15, 4.16, 4.18 and 4.20	Resolution soundness, easy-completion CNF semantics, unused-leaf deletion, local-clause support, support-local auxiliary witnesses, support-local extension definitions, and Theorem 4.1.	Named resolution statements only when faithful label information is explicit, auxiliary constraints are support-local, or extension variables are computed from a fixed label support.

Claim	Dependencies	Audit boundary
Dependency-cone and canonical syntactic-cone extension normalizers, final backward-slice, slice-explicit ER, displayed acyclic ER, expansion-bounded compressed ER, macro-compressed ER, support-declared macro ER, support-audited LRAT, provenance-typed LRAT, namespace-typed LRAT, strict-prefix Qrep replay, support-audit necessity, support-erasure indistinguishability, plain LRAT extraction trichotomy, cone-budget, and bounded-depth lower bounds, Theorems 4.22, 4.23, 4.25, 4.27, 4.29, 4.30, 4.32, 4.34, 4.36, 4.38, 4.40, 4.42, 4.44, 4.46, 4.48, 4.50, 4.52, 4.53 and 4.61	Acyclic extension evaluation, syntactic cone monotonicity, canonical graph reachability, backward-slice identity, displayed-literal charging, expansion accounting, macro-reuse accounting, support-declaration soundness, LRAT soundness, typed variable-role provenance, namespace parsing, LRAT antecedent closure, block-local witness composition, strict-prefix quotient replay, support-locality, the unit-clause separator for unaudited support, provenance-erasure ambiguity, the raw-trace extractor trichotomy, cone counting, depth induction, and Theorem 4.1.	Extension and clausal traces only when final cone or slice support is visible in the displayed acyclic extension object, when the compressed representation expands to such an object with controlled overhead or visible macro reuse, when a sound declared/audited support budget is supplied, when typed block provenance or a trusted label-variable namespace plus antecedent closure composes into such an audit, or when quotient representatives replay from already charged same-theory forests; unrestricted extension variables, delayed quotient certificates, hidden semantic pruning, and raw unaudited or untyped LRAT/DRAT traces may still have huge, compressed, nonlocal, or semantically ambiguous dependency support, and checked clausal unsatisfiability plus raw trace syntax alone does not certify completion-local support.
Support-local normalization transfer, Theorems 4.54 and 4.55	Sound refutation, support-locality, easy-completion semantics, and Theorem 4.1.	Conditional transfer theorem; a concrete proof system still needs a normalizer.
Row-forcing theorem and occurrence consequences, Theorems 5.2 to 5.4	Row completion to a constant and Theorem 3.1; aggregator forcing numbers.	Private occurrence model; equality or nonlinear SAT constraints require separate treatment.
Quotient residual-distance barrier, Theorem 6.3	Affine quotient completion plus Theorem 3.1.	Equality constraints only; nonlinear semantic constraints need separate support accounting.
Semantic-relation support certificate, Theorem 6.6	Definition of support-locality for a finite relation; enumeration of $y = y_0 \vee y_1$.	Local relation-instance audit only; the OR-frontier theorems below are bounded eliminated-frontier cases, not full SAT self-reduction closure theorems.
Acyclic OR-tree residual certificate, Theorem 6.7	Elimination of internal OR nodes to the frontier relation $y_r = \bigvee_{\ell \in L(T)} y_\ell$, plus ordinary OR certificate complexity.	Bounded eliminated-frontier theorem only; shared DAGs and frontier quotienting are handled separately by Theorem 6.8.
OR-DAG frontier residual certificate, Theorem 6.8	Topological elimination of finite acyclic OR DAGs, idempotence of OR, and sound quotienting of repeated reachable frontier formulas.	Pure-OR shared-frontier theorem; the exposed acyclic Boolean generalization is handled by Theorem 6.10.
Acyclic Boolean frontier residual certificate, Theorem 6.10	Exposed finite acyclic Boolean frontier circuit and ordinary pointwise certificate complexity of the induced frontier function.	Exposed-frontier theorem only; visible adaptive control is handled by Theorem 6.11, while learned clauses, cyclic semantic systems, and hidden semantic proof influence remain outside scope.
Visible adaptive frontier residual certificate, Theorem 6.11	Exposed finite adaptive decision tree over frontier coordinates and ordinary pointwise certificate complexity of the induced decision-tree function.	Visible-adaptive theorem only; the whole control tree must be checker-visible, and hidden adaptive pruning or learned semantic influence remains outside scope.
Visible adaptive branching-program frontier residual certificate, Theorem 6.12	Exposed finite acyclic deterministic branching program over frontier coordinates and ordinary pointwise certificate complexity of the induced branching-program function.	Visible shared-adaptive theorem only; the whole branching program must be checker-visible, and hidden adaptive controllers or learned semantic influence remain outside scope.
Hidden adaptive transcript insufficiency, Theorem 6.13	Same reported transcript summary is compatible with a one-variable function and an all-frontier OR function at the all-zero tuple.	Boundary theorem only; it proves transcript-only extraction is unsound without controller/path completeness, exposed dependencies, or worst-case support.
Controller-complete adaptive audit normalizer, Theorem 6.14	Exposed acyclic controller, replay path, and local equivalence certificate on the claimed completion cylinder, plus Theorem 6.12 for the global case.	Positive audited-hidden-adaptive theorem only; it assumes the controller-completeness audit and does not show that arbitrary hidden SAT self-reduction semantics or raw proof traces supply one.

Claim	Dependencies	Audit boundary
Trace-complete DPLL decision-log normalizer, Theorem 6.15	Complete finite DPLL decision controller, replay path, terminal output, and clause or propagation witnesses certifying local equivalence on the claimed completion cylinder, plus Theorem 6.14.	Recognizable DPLL-style bridge only for trace-complete logs; partial solver transcripts, unaudited learned clauses, restarts, cyclic search graphs, and arbitrary CDCL traces still need separate provenance or controller-completeness certificates.
Antecedent-complete CDCL learned-clause normalizer, Theorem 6.16	Complete decision/restart controller, replay path, learned-clause antecedent DAGs, rank-decreasing antecedent closure, and local equivalence certificates, plus Theorems 6.14 and 6.15.	Recognizable CDCL-style bridge only when every used learned clause has acyclic antecedent provenance down to original local witnesses; arbitrary CDCL logs, hidden clause minimization, deleted-clause provenance gaps, and cyclic restart/search graphs remain outside scope.
CDCL antecedent-erasure trichotomy, Theorem 6.17	Support-erasure indistinguishability, support-audit necessity, and the same-skeleton learned-clause lift of the unit-plus-blocking separator.	Boundary result for arbitrary provenance-free CDCL logs: add learned-clause antecedent provenance or an equivalent invariant, pay role-agnostic support, or lose support soundness on an indistinguishable semantic lift.
RUP-complete CDCL support normalizer, Theorem 6.18	Checker-visible RUP propagation certificates over a known live clause database, trusted label namespace or role parsing, retained deleted-clause support, and Theorem 6.14.	Positive non-plain CDCL bridge only for logs whose RUP traces expose parsed formula-label support; raw RUP/DRUP syntax, erased role information, and uncharged deleted-clause influence remain outside scope.
Canonical-namespace RUP-CDCL standardizer, Theorem 6.19	Deterministic encoding manifest, reserved label-variable interval, injective coordinate map, live-database reconstruction, retained deleted-clause certificate support, and Theorem 6.18.	Checkable namespace route only; unmanifested DIMACS/RUP/DRUP syntax, erased interval maps, and hidden variable roles remain outside scope.
SAT transfer and reverse calibration, Theorems 7.1 to 7.6	Batched substitution and the certificate theorems above.	Conditional on a source lower bound; does not prove SAT $\notin$ P/poly.

9 Related work and open directions

9.1 Hardness magnification and locality

Hardness magnification was developed systematically by Oliveira and Santhanam [16], with further near-frontier results by Oliveira, Pich, and Santhanam [15] and compression-based magnification by McKay, Murray, and Williams [13]. Chen, Hirahara, Oliveira, Pich, Rajgopal, and Santhanam studied the interaction between magnification, natural proofs, and locality [8]. Recent work of Atserias and Müller develops general magnification through sparse distinguisher matrices and obtains sharp thresholds in several settings [3].

The present paper does not offer a new general magnification theorem. Its SAT substitution calculation is an application of the elementary fact that polylogarithmic target length removes the target exponent from the source exponent. The new focus is the local certificate complexity of the resulting source-hardness predicate. Formula footprints concern the number of distinct generated labels that must be fixed; row forcing concerns the number and type of aggregate rows that must be made immutable; quotient distance concerns the geometry left after sound equality identifications.

9.2 Certificate and decision-tree complexity

Certificate complexity is the nondeterministic analogue of deterministic query complexity and is a standard measure of Boolean functions [14, 6]. Our hardness predicates have extremely large and highly structured variable sets: distinct generator outputs, generated occurrences, or quotient classes. The results are inputwise lower bounds for 1-certificates of these properties.

The row-forcing theorem is not merely the observation that parity has certificate complexity M. Sparse-row patching first forces $\Omega(s/n)$ rows of each output value, after which the minimum forcing number of the aggregator converts rows into label probes. The formula theorem is different: repeated

occurrences share variables, and a batched dictionary upper bound converts a small two-sided formula footprint into a small source circuit.

9.3 Circuit test suites, anti-checkers, and exact specification

There is a close conceptual relation between our certificates and finite sets of examples that refute small circuits. Chapman and Williams study the circuit-input game and the data complexity of testing circuits with labeled examples, showing a duality between small test suites and circuit lower bounds [7]. That line of work asks for labeled source inputs that distinguish a fixed target function from all small circuits. Our formula-label theorem instead fixes labels on generated objects and asks whether every completion of the ambient label table still induces a hard source function. Thus the roles of examples are different: test suites constrain candidate circuits for a fixed function, whereas formula footprints constrain completions of a generator-induced labeling.

Theorem 4.5 deliberately uses the standard counting side of this picture as a sharpness check. In the identity carrier, generated labels are source-input labels, so the formula-footprint certificate becomes an ordinary finite test set against all size-s circuits. The theorem does not replace the main lower bound; it shows that the lower bound has the right scale in the baseline model.

The same distinction separates the present work from teaching dimension and exact specification. Exact specification by examples studies how many labeled examples uniquely specify a target concept inside a hypothesis class [4]; teaching-dimension work studies the information complexity of selected examples for learning or teaching concept classes [10]. Our certificates need not identify the whole labeling or teach a unique concept. They only force membership in the hardness predicate: every completion must induce a function outside the size-s circuit ball. This is why the lower bounds count polarity footprints, forced rows, and quotient residual distance rather than uniqueness of the target labeling.

9.4 Partial MCSP and partial functions

The formula-label predicate is also near partial-function variants of MCSP. MCSP asks whether a complete truth table has a circuit of size at most s, while partial MCSP variants ask whether a partial function has a small circuit completion. Hirahara proves NP-hardness for a partial-function variant of MCSP under randomized reductions and places it in the meta-complexity landscape [11]. The present predicate is the complementary, certificate-complexity viewpoint for a generator-induced partial labeling: a certificate for hardness must rule out every small-circuit completion after the unexposed generated labels are filled in. The batched dictionary lemma is the point at which the generator structure matters; it turns a small set of formula-label exceptions into a small source circuit, yielding the near-threshold two-polarity footprint lower bound.

9.5 Novelty accounting and narrative cohesion

The comparison above is also a cohesion test. Each theorem-level contribution must identify the nearest established object, the shared mathematical feature, the novelty delta, and its role in the formula-footprint arc. This prevents the paper from becoming a list of locally correct but separately motivated results: an admitted result must either be the flagship obstruction, calibrate its sharpness, expose the same obstruction in a new interface, bridge it to proof objects, or mark a boundary where the bridge is unsound.

Thus the changed understanding is a single statement rather than five separate payoffs: completion-sound evidence that a generated labeling is hard must either expose a large local footprint, or supply an auditable proof-system mechanism that computes the same footprint from

Nearby line	Shared feature	Distinction in this paper
Partial MCSP and partial functions	Completion of a partial object must avoid or admit small circuits.	The partial object is induced by a generator; the lower bound is an inputwise certificate lower bound for forcing hardness under all completions.
Circuit test suites and anti-checkers	Labeled examples can refute all small circuits for a target.	Formula footprints fix generated labels, not source inputs, and the theorem exploits shared batched dictionary lookup.
Teaching and exact specification	Examples force a concept inside a hypothesis class.	The certificate need not identify a unique labeling; it only forces membership in the hardness predicate.
Decision-tree certificate complexity	A partial assignment forces a Boolean property.	The variables have formula, occurrence, or quotient geometry, yielding polarity, row, and affine-distance refinements.
Hardness magnification	A source lower bound could imply a stronger target lower bound.	The SAT section is conditional calibration; the paper does not prove the source lower bound.
Proof complexity	Proofs may expose local semantic information.	Displayed acyclic ER objects inherit a length lower bound here; expansion-bounded compressed ER objects inherit the corresponding decompression tradeoff; macro-compressed ER objects inherit a reuse-product tradeoff; support-declared macro objects, support-audited LRAT traces, provenance-typed LRAT/DRAT-with-LRAT-certificate traces, and namespace-typed LRAT antecedent-closure traces inherit support-budget lower bounds; strict-prefix Qrep objects inherit a replay condition for quotient representatives; trace-complete DPLL decision logs inherit the controller-complete adaptive audit; antecedent-complete CDCL logs inherit the same audit after expanding learned clauses through acyclic antecedent DAGs; self-auditing RUP-CDCL traces instantiate the carrier-independent audit criterion when the decoder and support audit are internal to the proof object, append-only self-auditing traces remove retained deleted-clause support records by forbidding deletions, and prefix-typed append-only traces remove the proof-internal decoder by making formula-label variables public syntax. The support-audit necessity, support-erasure, hidden-transcript, and plain LRAT trichotomy results show why raw unaudited, untyped, partial-transcript, or antecedent-erased support is not enough. Frege, Extended Frege, IPS, arbitrary compressed proofs, arbitrary CDCL without antecedent provenance or typed audit fields, plain unaudited LRAT/DRAT without extra semantic invariants, delayed quotient certificates, and unrestricted nonlocal ER still require a separate extraction theorem.

Table 5: Nearest-work comparison and novelty boundary.

Object	Closest literature object	Novelty delta	Role in the flagship arc
Formula-footprint barrier	Certificate complexity, Partial MCSP, circuit test suites	Completion-sound certificates for generator-induced labels must expose many distinct generated formulas of both polarities; the batched dictionary lemma gives the near-threshold count.	Flagship: the paper's main before/after shift is from examples-as-source-inputs to formula labels as a structured completion footprint.
Identity-carrier sharpness	Counting lower bounds against small circuits	In the standard truth-table carrier, generic certificates match the lower bound up to polylogarithmic factors.	Calibration: the flagship lower bound is not merely an artifact of the proof.
Row and quotient interfaces	Decision-tree certificates and equality-constrained partial functions	Source rows, output polarities, and quotient residual distance are separated as different interfaces of the same completion obstruction.	Interface variant: these results explain what changes when the local object is not a generated formula label.
Semantic frontiers and adaptive certificates	Boolean certificate complexity, decision trees, branching programs, SAT self-reduction frontiers	Exposed finite frontiers reduce to ordinary pointwise certificate complexity, while hidden transcript summaries are separated from sound support audits.	Boundary: semantic constraints help only when the support object is checker-visible.
Proof-object bridges	Resolution, ER-style syntax, clausal traces, and implicit proofs	Support-local normalizers, provenance or namespace conditions, strict-prefix replay, RUP-complete audits, the carrier-independent audit criterion, the self-auditing RUP-CDCL instantiation, its append-only deletion-free variant, and the prefix-typed append-only refinement say exactly when proof size inherits the footprint lower bound.	Proof-system bridge: proof-complexity applications require an auditable extractor, not a rhetorical analogy.

Table 6: Novelty-delta accounting for the cohered result package.

the proof artifact. The surrounding nonclaims are part of this thesis. The paper is not a new general hardness magnification theorem, not a Partial MCSP theorem, and not a Frege, Extended Frege, or IPS lower bound. Its contribution is the local obstruction and the exact audit target needed before such proof-system consequences can be claimed.

9.6 Sorting networks and static lookup

Sorting networks originate with Batcher [5]; Ajtai, Komlós, and Szemerédi gave asymptotically optimal $O(N \log N)$ comparator networks [1]. We use sorting only as a static nonuniform dictionary primitive. The construction is elementary, but it is quantitatively decisive: replacing pairwise comparison cost $MK\ell$ by nearly linear batch-plus-dictionary cost upgrades the formula lower bound from roughly $s/(M\ell)$ to $s/\operatorname{polylog}(s)$ in the relevant regime.

9.7 Natural proofs and constructive lower bounds

Razborov and Rudich formalized the natural-proofs barrier [17]. Carmosino, Impagliazzo, Kabanets, and Kolokolova showed that natural lower bounds yield learning algorithms for the same circuit class [9]; later work developed broader connections between constructive lower bounds, algorithms, and hardness magnification. Our functional naturalization proposition is a direct substitution observation. It is included only to distinguish the global dense regime from the sparse certificate regimes that drive the main results.

9.8 Proof complexity motivation

A long-term motivation is to understand whether lower-bound proofs in strong systems must expose local semantic information. Krajíček's implicit proofs [12] and recent interactive-protocol connections between proof and circuit complexity [2] illustrate that strong proof systems may use compressed or nonlocal witnesses. Therefore no conclusion about such systems follows merely from an occurrence-query lower bound. Theorem 4.10 gives the restricted positive case: if a proof literally displays faithful formula atoms and is completion-sound, the formula-footprint theorem lower-bounds the displayed local content. Theorems 4.13, 4.15, 4.16, 4.18 and 4.20 instantiate this bridge for resolution refutations of an easy-completion CNF when the proof explicitly lists faithful label assumptions, when a full-assignment refutation uses faithful label-unit leaves, when the proof uses faithful clauses over formula-label variables, when auxiliary constraints are support-local under completions, or when extension definitions read only a fixed support of formula-label variables. Theorems 4.23, 4.25, 4.27, 4.30, 4.32, 4.34, 4.36, 4.38, 4.40, 4.42, 4.44, 4.46, 4.48, 4.50, 4.52, 4.53, 4.56 to 4.59, 4.61 and 6.15 to 6.19 move the extension, compressed-replay, clausal-trace, DPLL-log, learned-clause, and carrier-independent audit routes one step closer to an auditable proof artifact: nested extension variables are allowed, the needed dependency cones can be declared or read canonically from the acyclic extension-definition graph, the final backward slice must expose near-threshold label mass, slice-explicit ER-style representations must pay for that boundary, ordinary displayed acyclic extension length pays for it without a separate slice-size convention, expansion-bounded compressed representations inherit a decompression tradeoff, macro libraries inherit a skeleton-plus-reuse product tradeoff, sound support declarations, support-audited LRAT traces, provenance-typed LRAT/DRAT-with-LRAT-certificate traces, and namespace-typed LRAT antecedent-closure traces inherit support-budget lower bounds, strict-prefix quotient representatives replay from already charged same-theory forests, trace-complete DPLL logs inherit the controller-complete adaptive audit when the complete controller and local witnesses are present, antecedent-complete CDCL logs inherit it when every used learned clause has acyclic antecedent provenance, the CDCL antecedent-erasure trichotomy shows that

provenance-free learned-clause syntax must add an equivalent invariant, pay conservative support, or be unsound, RUP-complete CDCL logs give the positive non-plain route when every used learned clause has a checker-visible RUP propagation certificate over the known live database and parsed formula-label support, the canonical-namespace RUP-CDCL standardizer replaces trusted parsing by a checkable label-interval manifest and live-database reconstruction, the self-auditing RUP-CDCL corollary internalizes that parser and the retained support into the proof object itself, the append-only self-auditing corollary removes the retained deletion-support records when the trace forbids deletions, and the prefix-typed append-only corollary replaces the proof-internal decoder by a public syntactic namespace for formula-label variables. The support-audit and support-erasure separators explain why unaudited clausal traces do not automatically supply that support, the plain LRAT trichotomy says a raw-trace extractor must add provenance, pay worst-case support, or be unsound on a same-skeleton semantic lift, the final-interface size, cone width, and bounded-fanin extension depth obey explicit tradeoffs, and a system-intrinsic audit extractor would be enough to remove the carrier-specific manifest from the proof-system bridge. Theorem 4.55 packages the general implication: once a proof system has a support-local normalizer with a controlled support budget, proof-size lower bounds follow. A stronger application to Frege, Extended Frege, IPS, arbitrary compressed proofs, unrestricted nonlocal extended resolution, raw untyped LRAT/DRAT, raw RUP/DRUP without typed support fields, deletion-bearing traces without retained support, delayed quotient certificates, CDCL traces with hidden learned-clause provenance and no RUP-complete support audit, or another proof system would still need to supply that normalizer.

9.9 The normalization target for proof-system applications

The required proof-system bridge can be stated independently of any particular system. A *controlled local normalization theorem* for a proof system $\mathcal{P}$ would consist of:

1. a family of lower-bound statements L_n asserting that the induced carrier lies in one of the hardness predicates studied here;
2. an extractor mapping every accepted $\mathcal{P}$-proof of L_n to one of the following local objects:

   formula labels, private occurrence rows, equality-quotient labels;

3. a faithfulness invariant saying that extracted labels agree with the intended carrier labeling;
4. a completion-soundness invariant saying that every admissible completion of the unextracted labels still satisfies the relevant hardness predicate;
5. an overhead bound relating proof size to extracted local size.

If the extractor returns formula labels, Theorems 4.10, 4.13, 4.15, 4.16, 4.18, 4.20, 4.23, 4.25, 4.27, 4.30, 4.32, 4.34, 4.36, 4.38, 4.40, 4.42, 4.44, 4.52, 4.53, 4.55, 4.57 to 4.59, 4.61, 6.15, 6.16, 6.18 and 6.19 and Theorem 4.1 give the footprint lower bound. If it returns private occurrence rows, Theorem 5.2 applies. If it returns equality-quotient labels, Theorem 6.3 applies. Theorem 4.46 records a boundary case: checked clausal unsatisfiability does not by itself return a support-local formula-label object. Theorem 4.48 records the sharper provenance boundary: untyped LRAT syntax does not determine whether a variable is auxiliary or label support. Theorem 4.50 packages these two separators into the operational trichotomy for plain trace extractors: add provenance, pay role-agnostic worst-case support, or fail on a same-skeleton semantic lift. Theorem 4.42 records the positive trusted-provenance route: typed roles, block-local supports, and compatible witnesses do supply the support audit. Theorem 4.44 records the lighter namespace route: when formula-label

variables are syntactically reserved, the LRAT antecedent closure of the empty clause identifies the used label support of witnessed auxiliary clauses. Theorem 4.61 records the corresponding boundary for quotient compression: a representative use is replayable only when its same-theory certificate is already charged before use. Theorem 6.15 records the recognizable DPLL route: a decision log supplies the adaptive audit only when it logs the complete finite controller, replay path, and local clause or propagation witnesses. Theorem 6.16 records the learned-clause route: a CDCL log supplies the audit only when every used learned clause has acyclic antecedent provenance down to local support. Theorem 6.17 records the provenance-free CDCL boundary: erasing learned-clause antecedents forces added provenance or a non-plain invariant, conservative support, or unsoundness on a same-skeleton lift. Theorem 6.18 records the positive non-plain CDCL route: RUP propagation certificates over a known live database supply the audit when their formula-label support is parsed and retained. Theorem 6.19 records the checkable manifest route: a reserved label interval and live-database log make that parser deterministic. Theorem 4.57 records the carrier-independent positive route: the label decoder, live-database reconstruction, RUP certificates, and retained support are all part of the proof artifact. Theorem 4.58 records the deletion-free refinement: if the live database is append-only, the same audit needs no retained deleted-clause support records. Theorem 4.59 records the typed proof-system refinement: if formula-label variables are public syntax, the same audit needs no proof-internal label decoder. Theorem 4.60 records the matching boundary: public label syntax and append-only RUP checking still do not certify completion-local auxiliary support without a witness or equivalent intrinsic invariant. This is the exact point at which stronger proof-system theorems would enter the paper; beyond the explicit-local, support-local auxiliary, support-local extension-resolution, dependency-cone extension-normalizer, canonical syntactic-cone normalizer, final backward-slice lower bound, slice-explicit ER lower bound, displayed acyclic ER length lower bound, expansion-bounded compressed ER transfer, macro-compressed ER reuse tradeoff, support-declared macro ER normalizer, support-audited LRAT trace normalizer, provenance-typed LRAT support audit, namespace-typed LRAT antecedent-closure normalizer, strict-prefix Qrep replay normalizer, trace-complete DPLL decision-log normalizer, antecedent-complete CDCL learned-clause normalizer, CDCL antecedent-erasure trichotomy, RUP-complete CDCL support normalizer, canonical-namespace RUP-CDCL standardizer, self-auditing RUP-CDCL instantiation, append-only self-auditing RUP-CDCL instantiation, prefix-typed append-only RUP-CDCL instantiation, prefix-typed RUP support-witness necessity, support-audit necessity separator, support-erasure indistinguishability separator, plain LRAT extraction trichotomy, cone-budget, and bounded-depth cases, none is assumed here.

Table 7: Normalization targets for proof-system applications. The explicit-local, support-local auxiliary, support-local extension-resolution, dependency-cone extension, canonical syntactic-cone, final backward-slice, slice-explicit ER, displayed acyclic ER, expansion-bounded compressed ER, macro-compressed ER, support-declared macro ER, support-audited LRAT, provenance-typed LRAT, namespace-typed LRAT, strict-prefix Qrep replay, trace-complete DPLL decision-log, antecedent-complete CDCL learned-clause, CDCL antecedent-erasure trichotomy, RUP-complete CDCL support normalizer, canonical-namespace RUP-CDCL standardizer, self-auditing RUP-CDCL instantiation, append-only self-auditing RUP-CDCL instantiation, prefix-typed append-only RUP-CDCL instantiation, prefix-typed RUP support-witness necessity, support-audit necessity, support-erasure indistinguishability, plain LRAT extraction trichotomy, semantic-relation, acyclic OR-frontier, quotiented OR-DAG frontier, acyclic Boolean frontier, visible adaptive decision-tree frontier, visible adaptive branching-program frontier, hidden adaptive transcript boundary, cone-budget, and bounded-depth routes are proved here, and the support-local normalizer row gives the reusable transfer theorem. Future stronger proof-complexity theorems must instantiate one of these local extraction routes with controlled overhead, add provenance, namespaces, controller-complete logs, learned-clause antecedent DAGs, RUP-complete support traces, canonical label-interval manifests, self-auditing proof syntax, append-only audit structure, public typed label syntax plus intrinsic support-locality, or semantic invariants, or explain why the proof system is genuinely nonlocal.

Normalization output	Applicable theorem	What must be proved about the proof system
Displayed formula atoms	Theorems 4.1 and 4.10	The proof exposes faithful formula labels and remains sound under all completions of unexposed formula labels.

Normalization output	Applicable theorem	What must be proved about the proof system
Explicit resolution assumptions	Theorems 4.1 and 4.13	A resolution refutation of an easy-completion CNF uses faithful formula-label unit assumptions, and those assumptions are part of the measured proof object.
Full-assignment resolution unit core	Theorems 4.1 and 4.15	A resolution refutation using the full hard labeling must use many distinct faithful formula-label unit leaves of each polarity.
Local resolution clauses	Theorems 4.1 and 4.16	A resolution refutation using faithful clauses over formula-label variables must expose many distinct labels of each polarity in the clause support.
Support-local auxiliary resolution	Theorems 4.1 and 4.18	A resolution refutation with auxiliary variables must have auxiliary constraints satisfiable under every completion preserving a fixed label support.
Support-local extension resolution	Theorems 4.1 and 4.20	Extension definitions may compute from a fixed formula-label support, and final extension clauses must remain satisfied whenever that support agrees with the hard labeling.
Dependency-cone extension normalizer	Theorems 4.1 and 4.23	A cone-annotated extension proof must expose a syntactic upper bound on the formula labels each final extension variable depends on.
Canonical syntactic-cone normalizer	Theorems 4.1 and 4.25	An acyclic extension proof has its final formula-label support extracted automatically from the extension-definition graph.
Final backward-slice lower bound	Theorems 4.1 and 4.27	The final backward extension slice must contain near-threshold many distinct formula-label inputs of each polarity.
Slice-explicit ER lower bound	Theorems 4.1, 4.29 and 4.30	A slice-explicit extension refutation must charge that final formula-label boundary in its proof-size measure.
Displayed acyclic ER length lower bound	Theorems 4.1, 4.27 and 4.32	If the acyclic extension definitions and final clauses are displayed, ordinary literal length already charges every formula-label input in the final backward slice.
Expansion-bounded compressed ER transfer	Theorems 4.32 and 4.34	If a compressed proof expands to displayed acyclic ER with expansion length $B(\|\rho\|)$, then $B(\|\rho\|)$ must be near-threshold large; polynomial expansion gives a polynomial-root lower bound on compressed length.
Macro-compressed ER reuse trade-off	Theorems 4.32 and 4.36	A displayed skeleton of length S, macro library of length D, and r invocations must satisfy $S + rD \geq 2n^{a-o(1)}$; ordinary macro accounting gives $L \geq n^{a/2-o(1)}$.
Support-declared macro ER normalizer	Theorems 4.18 and 4.38	A compressed macro proof with sound declared support S_ρ must satisfy $\|S_\rho\| \geq 2n^{a-o(1)}$ near threshold; any support budget $B(\|\rho\|)$ is therefore near-threshold large.
Support-audited LRAT trace normalizer	Theorems 4.18 and 4.40	An LRAT-style checked clausal trace with a sound support audit must expose near-threshold formula-label support; unaudited LRAT/DRAT remains outside the theorem.
Provenance-typed LRAT support audit	Theorems 4.40 and 4.42	A typed variable-role map plus block-local clause provenance and compatible auxiliary witnesses supplies the sound support audit; raw untyped LRAT/DRAT remains outside the theorem.
Namespace-typed LRAT antecedent closure	Theorems 4.40 and 4.44	A trusted formula-label variable namespace plus the LRAT antecedent closure of the empty clause supplies the used label support for witnessed auxiliary clauses; erasing the namespace returns to the raw-trace boundary.
Strict-prefix Qrep replay normalizer	Theorem 4.61	Quotient representative uses must replay from already charged same-theory certificate forests; delayed or hidden quotient certificates are private-row falsifiers.
Trace-complete DPLL decision-log normalizer	Theorems 6.14 and 6.15	A DPLL-style log must expose the complete finite decision controller, hard-tuple replay path, terminal value, and local clause or propagation witnesses certifying equivalence on the completion cylinder; partial transcripts and unaudited learned clauses are not normalized.
Antecedent-complete CDCL learned-clause normalizer	Theorems 6.15 and 6.16	A CDCL-style log must expose every learned clause used on the replay path through an acyclic antecedent DAG whose leaves are original local clauses, decisions, unit propagations, or lower-rank learned clauses; hidden learned-clause provenance is not normalized.
CDCL antecedent-erasure trichotomy	Theorems 4.46, 4.48 and 6.17	A CDCL log that erases learned-clause antecedents and role provenance must add equivalent provenance or a non-plain semantic invariant, pay conservative role-agnostic support, or become unsound on a same-skeleton semantic lift.
RUP-complete CDCL support normalizer	Theorems 6.14 and 6.18	A CDCL/DRUP-style log must expose checker-visible RUP propagation certificates over a known live clause database, parsed formula-label namespace or role information, and retained support for deleted-clause consequences; raw RUP/DRUP syntax without support parsing is not normalized.
Canonical-namespace RUP-CDCL standardizer	Theorems 6.18 and 6.19	A deterministic encoding manifest must reserve a formula-label variable interval, map it injectively to frontier coordinates, reconstruct the live clause database, and retain deleted-clause certificate supports; unmanifested DIMACS/RUP/DRUP syntax is not normalized.
Self-auditing RUP-CDCL trace	Theorems 4.56 and 4.57	The proof syntax itself contains the checked label decoder, live-database reconstruction, RUP propagation witnesses, and retained deleted-clause support; the support audit is computed from the proof artifact rather than a side manifest.
Append-only self-auditing RUP-CDCL trace	Theorems 4.56 to 4.58	If the self-auditing trace is deletion-free, the monotone clause-addition log and RUP certificates determine the touched formula-label support; no retained deleted-clause support records are needed.
Prefix-typed append-only RUP-CDCL trace	Theorems 4.56, 4.58 and 4.59	If formula-label variables are public syntax in an append-only RUP-CDCL system, the same monotone scan determines support without a proof-internal label decoder or retained deleted-clause support records.
Prefix-typed RUP support-witness necessity	Theorem 4.60	Public label syntax and append-only RUP checking do not certify completion-local auxiliary support by themselves; the system must provide a support-local witness or an equivalent intrinsic invariant.

Normalization output	Applicable theorem	What must be proved about the proof system
Support-audit necessity separator	Theorem 4.46	A linear-size checked clausal refutation can still require all fixed labels in its sound support audit; checked clausal unsatisfiability alone does not certify completion-local support.
Support-erasure indistinguishability separator	Theorem 4.48	The same untyped LRAT skeleton can be lifted with empty formula-label support or with full label support; role/provenance metadata is part of the support audit.
Plain LRAT extraction trichotomy	Theorem 4.50	A raw-trace extractor must add provenance, be role-agnostically conservative and pay the full support on the separator skeleton, or become unsound on a same-skeleton semantic lift.
Cone-budget extension lower bound	Theorem 4.52	The final-interface variable count times the maximum declared cone size must be near-threshold large.
Bounded-depth extension tradeoff	Theorem 4.53	For bounded-fanin extensions, the final-interface variable count times r^h must be near-threshold large.
Support-local normalizer	Theorems 4.1 and 4.55	A proof system supplies a support-local CNF and a support budget $B(\|\pi\|)$ for every accepted proof.
Carrier-independent audit extractor	Theorems 4.55 and 4.56	The proof-system artifact itself computes support, support-locality evidence, and budget without a carrier-specific namespace manifest, hidden role advice, or semantic oracle.
Private occurrence rows	Theorems 5.2 and 5.3	The proof exposes occurrence labels, and unexposed positions can be completed independently without invalidating soundness.
Equality quotient labels	Theorem 6.3	The proof's semantic equalities induce a sound quotient, and residual labels remain freely completable in the quotient affine space.
Nonlinear relation instance	Theorem 6.6	The proof's declared support contains a standard membership certificate for each finite semantic relation instance it imposes.
Acyclic OR-tree frontier	Theorem 6.7	After internal OR nodes are eliminated, root 0 forces the root and all frontier leaves, while root 1 requires the root plus one true leaf.
Quotiented OR-DAG frontier	Theorem 6.8	After internal OR nodes are eliminated and repeated reachable frontier formulas are soundly quotiented, root 0 forces the root and every distinct reachable frontier class, while root 1 requires the root plus one true reachable frontier class.
Acyclic Boolean frontier	Theorem 6.10	After an acyclic Boolean semantic frontier is exposed, support-locality is the root coordinate plus an ordinary pointwise certificate for the induced frontier function.
Visible adaptive frontier	Theorem 6.11	After a finite adaptive decision tree over frontier coordinates is exposed, support-locality is the root coordinate plus an ordinary pointwise certificate for the induced decision-tree function; the actual transcript path is sufficient support.
Visible adaptive branching-program frontier	Theorem 6.12	After a finite acyclic branching program over frontier coordinates is exposed, support-locality is the root coordinate plus an ordinary pointwise certificate for the induced branching-program function; the actual computation path is sufficient support.
Hidden adaptive transcript summary	Theorem 6.13	A small reported adaptive transcript is not a support normalizer unless completeness of the controller/path or enough dependency support is certified; otherwise the same transcript can hide all-frontier OR support.
Controller-complete adaptive audit	Theorem 6.14	A hidden adaptive transcript becomes a support normalizer when the proof object exposes a complete acyclic controller, replay path, and local equivalence certificate for the completion cylinder.
Nonlocal or compressed witness	None from this paper	The proof system escapes the present local completion models; a different argument is required.

9.10 Open problems

Open Problem 9.1 (Expected randomized and interactive footprints)**.** Theorem 4.7 extends the certificate lower bounds to the union of labels queried on accepting bounded-error executions. Prove, or refute, analogous lower bounds for expected footprint, transcript support, communication-query product, or information revealed by public-coin or interactive verifiers with bounded soundness error. Such a result would require a distributional or information-theoretic completion argument, not merely the deterministic certificate reduction used here.

Open Problem 9.2 (A concrete proof-system instantiation)**.** Identify a natural proof or certificate system for which every short proof of the carrier lower-bound statement can be normalized into one of the completion-sound models in Table 7. Theorem 4.10 handles the toy case where formula atoms are already displayed faithfully, while Theorems 4.13, 4.15, 4.16, 4.18, 4.20, 4.23, 4.52 and 4.53 handle restricted resolution cases where faithful label information is explicit, auxiliary constraints are support-local, extension definitions read only a fixed label support, or extension variables expose checked dependency cones with bounded final cone budget or bounded-fanin depth. Theorem 4.61 handles the quotient-compression case only when every representative use has a prior charged same-theory certificate forest. Theorem 4.46 shows why checked clausal unsatisfiability is not enough by itself: the normalizer must certify completion-local support, not merely a refutation.

Theorem 4.48 adds that untyped trace syntax does not determine the formula-label support without variable-role provenance. Theorem 4.50 turns this into the plain-trace alternative: add provenance, pay role-agnostic worst-case support, or lose soundness on an indistinguishable semantic lift. Theorem 6.13 adds the adaptive analogue: a reported transcript summary cannot replace a certified complete controller/path. Theorem 6.14 gives the matching positive audited-hidden-adaptive condition: expose the acyclic controller, replay path, and local equivalence certificate for the completion cylinder. Theorem 6.15 instantiates that condition for DPLL-style decision logs when the whole finite controller and the local clause or propagation witnesses are part of the proof object. Theorem 6.16 extends the instantiation to learned clauses only when every used learned clause carries an acyclic antecedent DAG down to audited local support. Theorem 4.42 proves the positive trusted-provenance case when role maps, block supports, and witness compatibility are checked. Theorem 4.55 shows that the lower bound follows automatically once such a normalizer is available. Theorem 4.56 turns the same open problem into an extractor target: compute support, support-locality evidence, and budget from the proof-system artifact itself, without a carrier-specific manifest or hidden role advice. Theorem 4.59 shows the current strongest RUP-style bridge: in an append-only trace, public typed label syntax replaces the proof-internal decoder. Theorem 4.60 shows why the remaining support-local auxiliary witness cannot be removed by that syntax alone. The open problem is now sharper: supply an intrinsic support-locality invariant for touched auxiliary clauses, handle deletion-bearing or untyped traces, go beyond typed RUP-CDCL to a stronger or less local proof system, or identify a non-plain LRAT/DRAT/CDCL invariant that supplies the missing semantic support without merely assuming the provenance object.

Open Problem 9.3 (Beyond local semantic-relation certificates)**.** Theorem 6.6 gives the local certificate cost of one nonlinear semantic relation, Theorem 6.7 gives the exact eliminated-frontier certificate for acyclic OR self-reduction trees, Theorem 6.8 gives the corresponding result for finite acyclic OR DAGs after sound frontier quotienting, Theorem 6.10 handles exposed finite acyclic Boolean frontier circuits by ordinary pointwise certificate complexity, Theorem 6.11 handles exposed finite adaptive decision trees, and Theorem 6.12 handles exposed finite acyclic adaptive branching programs. Theorem 6.13 shows that hidden adaptive summaries do not give such a theorem for free, while Theorem 6.14 records the sufficient audit when the hidden controller is locally certified against an exposed acyclic controller. Develop a residual-completion theorem for whole systems of nonlinear semantic constraints beyond exposed acyclic or visible-adaptive frontiers, beginning with bounded SAT self-reduction

$$Y(u) = Y(u|_{x_i=0}) \vee Y(u|_{x_i=1}).$$

The equality quotient gives an affine completion space. Self-reduction produces a constrained Boolean variety. The local certificate costs in Theorem 6.6 identify which labels an audit must expose one relation at a time, the OR-frontier theorems solve the collapsed pure-OR tree and shared-DAG relations exactly, the Boolean-frontier theorem reduces exposed acyclic semantics to standard certificate complexity, and the visible-adaptive theorems show that logged adaptive control is harmless only when the whole decision tree or branching program is audited. The hidden-transcript separator shows that a small reported transcript without this completeness certificate can hide all-frontier OR support; the controller-complete audit theorem identifies the positive certificate that would remove this ambiguity on a completion cylinder. The remaining problem is the global analogue of residual distance for hidden-adaptive, cyclic, learned-clause, or proof-system-hidden SAT self-reduction systems.

Open Problem 9.4 (An explicit hard SAT-labeled carrier)**.** Construct an explicit batched generator for which the actual SAT parity carrier has an unrestricted circuit lower bound above the substitution

threshold. Any such result must pass the formula-diversity, row-robustness, and quotient-distance tests proved here, as well as the reverse short-SAT burden in Theorem 7.4.

10 Conclusion

Label-induced source functions provide a simple laboratory for separating three kinds of local information. Distinct-formula certificates are limited by batched static lookup and therefore require near-threshold footprints of both polarities. Occurrence certificates are limited by sparse-row patching and therefore must force many rows; the aggregator's minimum forcing numbers determine the query cost, with parity requiring complete rows on both sides. Equality constraints replace private occurrences by an affine quotient, but completion soundness still demands large residual Hamming distance from the constant functions.

For polylogarithmic SAT parity carriers, these results sit exactly at a magnification threshold strong enough to imply SAT $\notin$ P/poly if the source lower bound could be proved. The paper does not supply that lower bound. It factors out a smaller, standard question: the nondeterministic query complexity and completion geometry of the induced circuit-hardness predicate. The displayed-atom, explicit-local resolution, support-local auxiliary-resolution, support-local extension-resolution, dependency-cone extension, canonical syntactic-cone, final backward-slice, slice-explicit ER, displayed acyclic ER, expansion-bounded compressed ER, macro-compressed ER, support-declared macro ER, support-audited LRAT, provenance-typed LRAT, namespace-typed LRAT, strict-prefix Qrep replay, trace-complete DPLL decision-log, antecedent-complete CDCL learned-clause, CDCL antecedent-erasure trichotomy, RUP-complete CDCL support normalizer, canonical-namespace RUP-CDCL standardizer, carrier-independent audit upgrade criterion, self-auditing RUP-CDCL instantiation, append-only self-auditing RUP-CDCL instantiation, prefix-typed append-only RUP-CDCL instantiation, prefix-typed RUP support-witness necessity, support-audit necessity, support-erasure indistinguishability, plain LRAT extraction trichotomy, semantic-relation support, acyclic OR-frontier, quotiented OR-DAG frontier, acyclic Boolean-frontier, visible adaptive frontier, visible adaptive branching-program frontier, hidden adaptive transcript insufficiency, controller-complete adaptive audit, cone-budget, bounded-depth, and normalization-transfer theorems show how this cohered package becomes a proof-size statement once local semantic atoms are faithfully extracted, explicitly used, support-locally represented, syntactically dependency-audited, expanded with controlled overhead, compressed only through visible macro reuse, summarized by a sound support declaration, certified by a checker-visible support audit, certified by typed block provenance or a trusted label-variable namespace plus LRAT antecedent closure, replayed from a strict-prefix quotient dictionary, logged as a trace-complete DPLL controller with local witnesses, logged as an antecedent-complete CDCL learned-clause DAG, accepted as a RUP-complete CDCL/DRUP trace whose propagation certificates expose parsed formula-label support over the live database, standardized by a canonical label-variable interval manifest and live-database log, internalized as a self-auditing trace whose proof object carries its decoder and support audit, simplified in the deletion-free case to an append-only audit with no retained deleted-clause support records, simplified further in the typed case to a public-label namespace with no proof-internal decoder, rejected when public label typing is asked to certify auxiliary support-locality by itself, lifted to a carrier-independent proof-system payoff by an intrinsic audit extractor, separated from unaudited clausal refutation soundness, protected against provenance-erased trace ambiguity, forced through the plain-trace trichotomy, checked against local nonlinear semantic relations, collapsed to a pure OR frontier with exact residual certificate cost, exposed as an acyclic Boolean frontier whose support cost is ordinary certificate complexity, logged as a visible adaptive frontier whose transcript path is

a certificate, logged as a visible adaptive branching program whose computation path is a certificate, rejected as an incomplete hidden adaptive transcript that can mask all-frontier support, or accepted as an audited hidden-adaptive controller only when the proof object pays for controller completeness and local equivalence. The next decisive step is to supply an intrinsic support-locality invariant for touched auxiliary clauses, handle deletion-bearing or untyped traces, or supply an explicit SAT-labeled source lower bound.

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